DOI. No. 10.1109/MAES.2016.150029

Geometry of Complex Data

K. James Sangston, **Georgia Tech Research Institute, Atlanta, GA USA**

Geometric algebra has been called a "unified language for mathematics and physics." Sometimes known as Clifford algebra, it is based on the notion of an invertible product of vectors that captures the geometric relationship between two vectors, i.e., their relative magnitudes and the angle between them. This seemingly simple concept leads to a rich system of algebra and calculus that encompasses the diverse areas of complex numbers, quaternions, vectors, tensors, spinors, and differential forms. This tutorial provides a basic introduction to geometric algebra and presents formulations of known electrical engineering and signal processing concepts to illustrate some inherent advantages of geometric algebra for formulating and solving problems involving vectors. Being introductory, the goal of the tutorial is to introduce this emerging area that, although old as a mathematics discipline, has only recently started to garner significant attention in engineering communities. Geometric algebra should give another potentially powerful tool for pursuing research in any area that uses vectors.

I. INTRODUCTION

> *David [Hestenes] said that physicists had not learned properly how to multiply vectors and, as a result of attempts to overcome this, had evolved a variety of mathematical systems and notations that has come to resemble Babel.*
>
> (Gull et al., 1993)

When the author first encountered this statement, he was both intrigued and amused. Although the statement is directed at physicists, for purposes of working with vectors, this statement may be directed at engineers as well. But we know how to multiply vectors, right? After all, we have the inner (or dot) product, and in three dimensions, we have the cross product. So not only do we know how to multiply vectors, but in some cases we also have more than one way to do so.

Yet a closer look at the Gull et al. (1993) article quickly convinced the author that Hestenes is correct—as part of our general education, we engineers do not normally learn how to multiply vectors properly, if by multiplication we want to encompass division as well. In this article, we review some basic ideas of geometric algebra—which explain how to multiply and divide vectors—and then examine some basic electrical engineering notions from this geometric perspective to see how these ideas may be applied.

Engineers routinely work with scalars and vectors. Geometrically, one may think of a scalar as a real number attached to a point. In this sense, it has no dimension. A vector, however, may be thought of as a real number attached to a direction. In particular, a vector may be thought of as comprising three things: 1) a line along which the vector is directed, 2) a magnitude, and 3) an orientation, i.e., the sign of the vector. Thus, geometrically, a vector may be thought of as an oriented length; as such, such a vector is a one-dimensional quantity. These notions are illustrated in Fig. 1.

It would only seem natural, then, that the extension from the zero-dimensional scalar to the one-dimensional vector could (and perhaps should) be carried to higher dimensions. For example, following this line of thinking, it seems one should be able to define a quantity that has a magnitude and an orientation in a plane rather than simply along a line. Indeed, one can define such a quantity—it is called a bivector, which we discuss further later. As such, a bivector is an extension of the idea of a vector and represents an oriented area. Just as a vector is associated with a given line, a bivector is associated with a given plane. Having made the conceptual leap from a vector to a bivector, it should not then take a great conceptual leap to continue this process to higher dimensions and define an oriented volume, an oriented 4-volume, etc. This is precisely what William Clifford's algebra—also known as geometric algebra—does. It builds on the exterior algebra of Hermann Grassmann, who introduced the notion of an oriented k-volume for any positive integer k. These notions are illustrated in Fig. 2 up through $k = 3$.

This kind of extension is interesting in its own right, but a question naturally arises: Why would electrical engineers want to take the time to learn geometric algebra? One answer arises

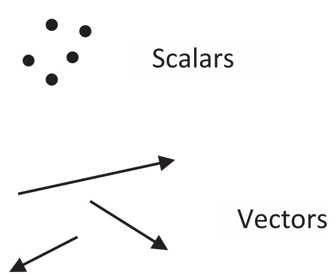
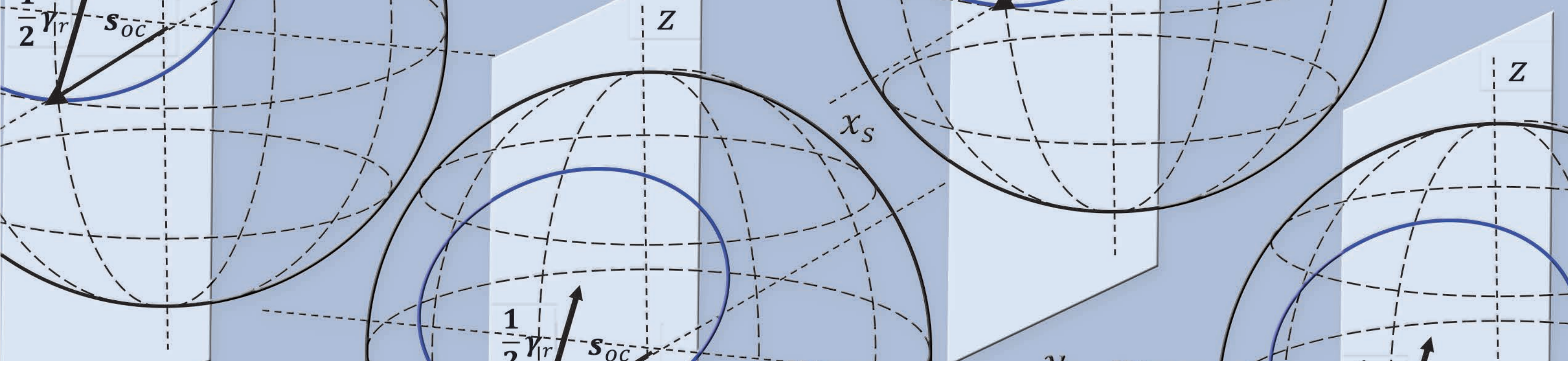


Fig. 1.
Geometric interpretation of scalars and vectors.

Author's address: GTRI, SEAL, 7220 Richardson Rd, Smyrna, GA 30080, E-mail: (jim.sangston@gtri.gatech.ed).
Manuscript received July 6, 2015; revised September 14, 2015; released for publication November 8, 2015.
Refereeing of this contribution was handled by W. Blair.
0885-8985/$26.00



from electrical engineers' routine work with complex quantities. For example, because of Euler's theorem, it is convenient to represent sinusoidal signals as complex phasors, which are given a geometric interpretation as quantities in a plane. They are sometimes described as vectors that rotate in a plane, but they also have an operational aspect, because they are routinely used to describe processes such as rotations and dilations.

Despite their convenience, the use of complex phasors can lead to conceptual difficulties in learning various important physical concepts. For example, what should beginning students make of the notion of a physical quantity such as reactance being described as the imaginary part of the impedance? Given the preceding brief discussion of directed quantities, we might suspect that complex numbers, which lead to the complex plane, are somehow be related to bivectors. As we discuss later, geometric algebra shows clearly that complex numbers may indeed be understood through the use of bivectors and as a result gives new ways of looking at physical quantities represented by complex numbers.

In addition to encountering many quantities that are represented by complex numbers, engineers often encounter complex vectors. For example, an engineer working on signal processing problems often encounters a collection of data samples represented by a complex vector:

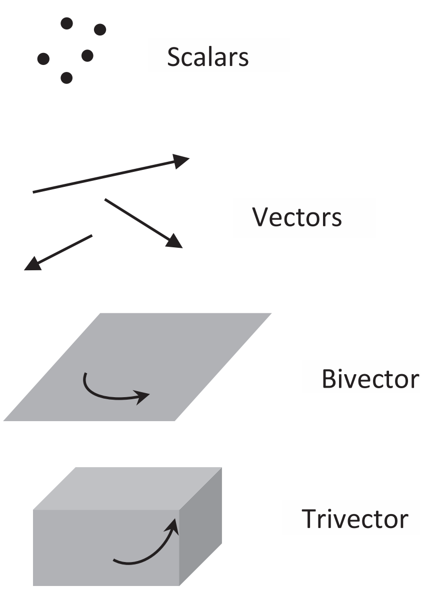


Fig. 2.
Geometric interpretation of directed quantities.

$$\begin{bmatrix} \mathcal{I}_1 + i\mathcal{Q}_1 \\ \vdots \\ \mathcal{I}_N + i\mathcal{Q}_N \end{bmatrix}$$

where $\mathcal{I}_1, \cdots, \mathcal{I}_N$ are the in-phase samples and $\mathcal{Q}_1, \cdots, \mathcal{Q}_N$ are the quadrature samples. While this representation captures the observed data and is convenient to use algebraically, it is not clear how to interpret this vector geometrically. For example, if a complex number is a vector, then is a complex vector a vector of vectors? If so, how should we interpret such a quantity geometrically? A real vector has the geometric interpretation of a magnitude and a direction. For example, let $\boldsymbol{x}$ be a real vector in two dimensions:

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \tag{1.1}$$

where x_1 and x_2 are real numbers representing the components of this vector. In writing $\boldsymbol{x}$ in this component form, we generally mean

$$\boldsymbol{x} = x_1\hat{\boldsymbol{e}}_1 + x_2\hat{\boldsymbol{e}}_2 \tag{1.2}$$

where $\hat{\boldsymbol{e}}_1$ and $\hat{\boldsymbol{e}}_2$ are orthogonal unit vectors in a two-dimensional space. In this view, $x_1\hat{\boldsymbol{e}}_1$ and $x_2\hat{\boldsymbol{e}}_2$ are each vectors, and their sum is a vector, with the geometric interpretation shown in Fig. 3. The vector $\boldsymbol{x}$ has a length and a direction in this two-dimensional space and hence a clear geometric interpretation.

Examine a complex vector:

$$\boldsymbol{c} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} c_{1,r} + ic_{1,i} \\ c_{2,r} + ic_{2,i} \end{bmatrix} \tag{1.3}$$

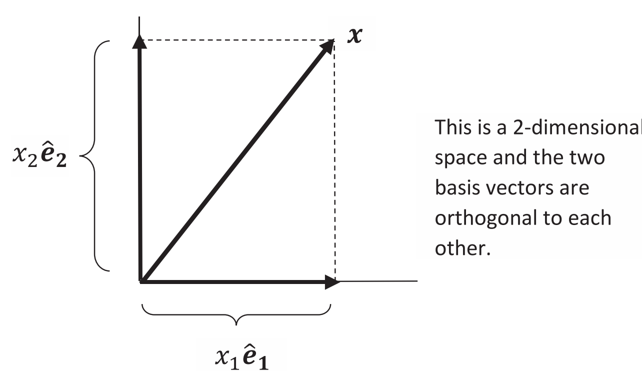


Fig. 3.
Geometric interpretation of a real vector.

where c_1 and c_2 are complex numbers representing the components of the complex vector $\boldsymbol{c}$. What do we actually mean by writing the complex vector in this component form? By analogy with real vectors, do we mean

$$\boldsymbol{c} = c_1\hat{\boldsymbol{e}}_1 + c_2\hat{\boldsymbol{e}}_2 \tag{1.4}$$

where $\hat{\boldsymbol{e}}_1$ and $\hat{\boldsymbol{e}}_2$ are again real orthogonal unit vectors? At first glance, this probably looks like gibberish to most engineers, because we are multiplying a real unit vector by a complex number. However, as we discuss later, geometric algebra gives a meaningful interpretation to precisely this formulation.

This article begins by presenting the basic principles of geometric algebra. These principles lead to the geometric product of two vectors, which differs from both the inner product and the cross product that most engineers learn as undergraduates. In particular, it is a new[1] product that permits division by nonzero vectors. These principles are then used to show how a complex number may be related to the geometric product (and the bivector mentioned earlier). With these ideas in place, the article then presents several electrical engineering notions rewritten in the language of geometric algebra:

1. Power concepts such as instantaneous active and reactive power
2. Basic signal processing notions such as the discrete Fourier transform (DFT) and signal detection
3. Rotations in a hyperbolic plane
4. A three-dimensional Smith chart in microwave engineering

Many other applications are of interest, and the paper presents an extensive set of references to the existing literature.

Geometric algebra is a system of mathematics. It does not lead to new engineering results any more than complex numbers do or linear algebra does. However, geometric algebra facilitates and promotes a geometric, as well as an algebraic, way of thinking and thus may lead to insights that suggest new solutions to problems faced by engineers. Accordingly, the focus of this article is on presenting a geometric view of a selection of basic electrical engineering notions to foster new directions for research, rather than on presenting empirical or analytical results from the assessment of new techniques. Such a geometric perspective offers a potentially powerful approach to various problems that arise in electrical engineering.

II. BASIC IDEAS OF GEOMETRIC ALGEBRA

This section gives an introduction to geometric algebra. Several excellent references teach the basics of geometric algebra, as well as many advanced topics on both geometric algebra and its related geometric calculus, and are included at the end of this paper for more information. In this overview, we do little more than introduce a small part of the basic ideas sufficient to motivate the rest of the paper.

[1] The author uses the word "new" here in the belief that the geometric product is not known to many engineers. It actually is not new, having been introduced by William Clifford in 1878.

Consider two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ in a Euclidean space (Fig. 4). Regardless of the number of dimensions, we have the inner product

$$\boldsymbol{a}\cdot\boldsymbol{b} = \boldsymbol{b}\cdot\boldsymbol{a} = |\boldsymbol{a}||\boldsymbol{b}|\cos\theta_{ab}. \tag{2.1}$$

We have not assigned axes to represent the vectors, because they are irrelevant to the inner product, which depends only on the magnitudes of the vectors and the angle between them. Observe now that the inner product is not invertible. Geometrically, this is illustrated in Fig. 5. In particular, given the inner product $\boldsymbol{a}\cdot\boldsymbol{b}$ and one of the vectors, e.g., $\boldsymbol{a}$, we cannot obtain the other vector $\boldsymbol{b}$ uniquely. Thus, we cannot use the inner product by itself to define multiplication and division of vectors.

Let us temporarily limit our considerations to a three-dimensional Euclidean space (Fig. 6). In three dimensions, we again have the inner product, which is defined for any number of dimensions. In addition, we have the cross product:

$$\boldsymbol{a}\times\boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|\sin\theta_{ab}\,\hat{\boldsymbol{n}} \tag{2.2}$$

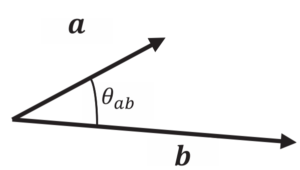


Fig. 4.
Arbitrary vectors in N-dimensional Euclidean space.

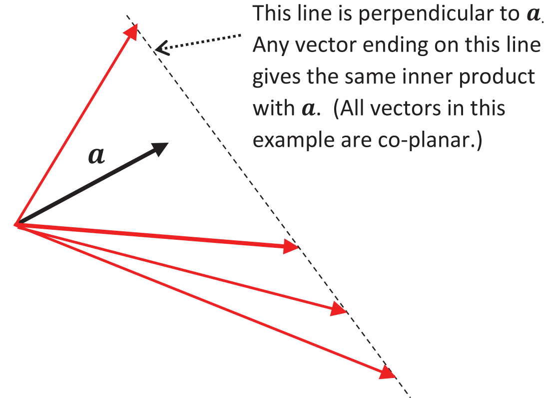


Fig. 5.
Noninvertibility of the inner product.

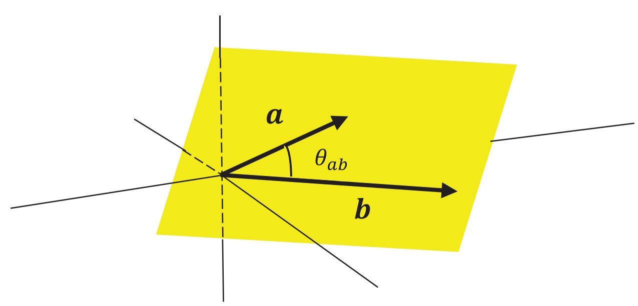


Fig. 6.
Arbitrary vectors in three-dimensional Euclidean space.

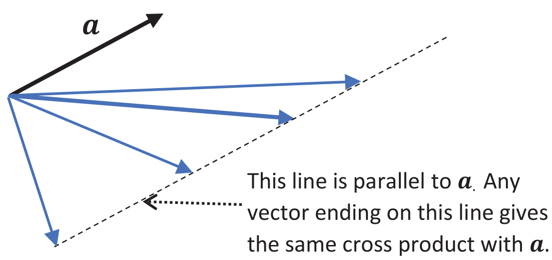


Fig. 7.
Noninvertibility of the cross product.

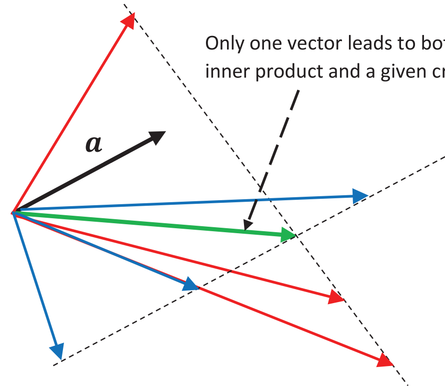


Fig. 8.
Invertibility of combination of inner product and cross product.

where $\hat{\boldsymbol{n}}$ is a unit vector pointing normal to the plane defined by $\boldsymbol{a}$ and $\boldsymbol{b}$ in accordance with a right-handed coordinate system.

Observe that, like the inner product, the cross product is not invertible, as illustrated in Fig. 7. However, knowing both the inner product and the cross product gives us information about the relative lengths of $\boldsymbol{a}$ and $\boldsymbol{b}$, as well as the angle $\theta_{\boldsymbol{ab}}$ between them. This follows from the basic relations

$$\boldsymbol{a}\cdot\boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|\cos\theta_{ab} \tag{2.3}$$

and

$$|\boldsymbol{a}\times\boldsymbol{b}| = |\boldsymbol{a}||\boldsymbol{b}|\sin\theta_{ab} \tag{2.4}$$

where $|\boldsymbol{a}\times\boldsymbol{b}|$ defines the length of a vector and is a nonnegative number; it implicitly assumes $\sin\theta_{ab} \geq 0$. The sign—and thus orientation—of $\boldsymbol{a}\times\boldsymbol{b}$ is embodied in the assumed right-handed system giving rise to the unit normal vector arising from $\boldsymbol{a}\times\boldsymbol{b}$.

The invertibility arising from using both $\boldsymbol{a}\cdot\boldsymbol{b}$ and $\boldsymbol{a}\times\boldsymbol{b}$ is suggested in Fig. 8, formed from Figs. 5 and 7. Algebraically, (2.3) and (2.4) suggest a relationship between these two products (in three dimensions) and complex numbers. In particular, if we represent the vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ by complex numbers in the plane that is common to both vectors, e.g.,

$$\boldsymbol{a} \to \mathfrak{z}_a = |\boldsymbol{a}|\cos\varphi_a + i|\boldsymbol{a}|\sin\varphi_a = |\boldsymbol{a}|e^{i\varphi_a} \tag{2.5}$$

$$\boldsymbol{b} \to \mathfrak{z}_b = |\boldsymbol{b}|\cos\varphi_b + i|\boldsymbol{b}|\sin\varphi_b = |\boldsymbol{b}|e^{i\varphi_b} \tag{2.6}$$

then we have

$$\mathfrak{z}_a^*\mathfrak{z}_b = |\boldsymbol{a}||\boldsymbol{b}|e^{i\theta_{ab}} = |\boldsymbol{a}||\boldsymbol{b}|\cos\theta_{ab} + i|\boldsymbol{a}||\boldsymbol{b}|\sin\theta_{ab} \tag{2.7}$$

with

$$\theta_{ab} = \varphi_b - \varphi_a. \tag{2.8}$$

These considerations suggest that algebraically, we may be able to use the inner and cross products to formulate a complex number to represent a product of vectors $\boldsymbol{ab}$ as

$$\boldsymbol{ab} = \boldsymbol{a}\cdot\boldsymbol{b} + I_{\boldsymbol{ab}}|\boldsymbol{a}\times\boldsymbol{b}| = |\boldsymbol{a}||\boldsymbol{b}|\cos\theta_{ab} + I_{\boldsymbol{ab}}|\boldsymbol{a}||\boldsymbol{b}|\sin\theta_{ab} \tag{2.9}$$

where $I_{\boldsymbol{ab}}$ is a quantity that behaves in some sense as $i = \sqrt{-1}$ but also must restrict this complex number to the plane defined by $\boldsymbol{a}$ and $\boldsymbol{b}$. Again, $|\boldsymbol{a}\times\boldsymbol{b}|$ is nonnegative and implicitly assumes $\sin\theta_{ab} \geq 0$, so $I_{\boldsymbol{ab}}$ must also provide an orientation (e.g., of a right-handed system).

In this way, we can view the product of two three-dimensional vectors as a complex number in the two-dimensional plane they form. We could then seek to write

$$\boldsymbol{ab} = |\boldsymbol{a}||\boldsymbol{b}|e^{I_{\boldsymbol{ab}}\theta_{ab}} \tag{2.10}$$

in an analogy with multiplication of complex numbers. However, $I_{\boldsymbol{ab}}$ cannot simply be $i = \sqrt{-1}$, because at a minimum, it must also define the plane of $\boldsymbol{a}$ and $\boldsymbol{b}$ and provide an orientation in that plane.

With these background considerations as motivation, we look more generally in the next couple of sections at how to multiply two vectors. We begin with the outer product, which is more general than the cross product.

A. THE OUTER PRODUCT

Let $\hat{\boldsymbol{e}}_1$ and $\hat{\boldsymbol{e}}_2$ be an orthonormal basis for a plane, and write

$$\boldsymbol{a} = a_1\hat{\boldsymbol{e}}_1 + a_2\hat{\boldsymbol{e}}_2 \tag{2.11}$$

$$\boldsymbol{b} = b_1\hat{\boldsymbol{e}}_1 + b_2\hat{\boldsymbol{e}}_2 \tag{2.12}$$

to represent two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ in this plane. The quantity

$$\boldsymbol{a}\wedge\boldsymbol{b} = (a_1b_2 - a_2b_1)\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 \tag{2.13}$$

with $\hat{e}_1\hat{e}_2 = -\hat{e}_2\hat{e}_1$ is defined as the outer product (sometimes known as the exterior product or the wedge product) of the vectors $\boldsymbol{a}$ and $\boldsymbol{b}$. In three dimensions, the quantity

$$\boldsymbol{a}\times\boldsymbol{b} = \left(a_1b_2 - a_2b_1\right)\hat{e}_3 \tag{2.14}$$

gives the cross product, which is a vector. The outer product, however, is not limited to three dimensions but more generally applies in any number of dimensions. Evidently, it defines a quantity that is neither a scalar nor a vector. In particular, the outer product $\boldsymbol{a}\wedge\boldsymbol{b}$ defines an oriented area and, as discussed in the introduction, extends the notion of a vector, which is an oriented length, to yield a bivector.

To see why the outer product is said to define an oriented area, we observe that the quantity

$$\left|a_1b_2 - a_2b_1\right| \tag{2.15}$$

is the area of the parallelogram defined by the two vectors (Fig. 9). As a result, we may think of $\boldsymbol{a}\wedge\boldsymbol{b}$ as a kind of product of the vectors $\boldsymbol{a}$ and $\boldsymbol{b}$. It also has an orientation defined by the order in which the vectors appear in the outer product (Fig. 10).

Like the cross-product, the outer product is antisymmetric, i.e.,

$$\boldsymbol{a}\wedge\boldsymbol{b} = -\boldsymbol{b}\wedge\boldsymbol{a} \tag{2.16}$$

and distributive, i.e.,

$$\boldsymbol{a}\wedge\left(\boldsymbol{b}+\boldsymbol{c}\right) = \boldsymbol{a}\wedge\boldsymbol{b}+\boldsymbol{a}\wedge\boldsymbol{c} \tag{2.17}$$

$$\left(\boldsymbol{a}+\boldsymbol{b}\right)\wedge\boldsymbol{c} = \boldsymbol{a}\wedge\boldsymbol{c}+\boldsymbol{b}\wedge\boldsymbol{c}. \tag{2.18}$$

Unlike the cross product, however, the outer product is associative, i.e.,

$$\left(\boldsymbol{a}\wedge\boldsymbol{b}\right)\wedge\boldsymbol{c} = \boldsymbol{a}\wedge\left(\boldsymbol{b}\wedge\boldsymbol{c}\right) = \boldsymbol{a}\wedge\boldsymbol{b}\wedge\boldsymbol{c}. \tag{2.19}$$

It immediately follows from the preceding definitions that

$$\boldsymbol{a}\wedge\boldsymbol{a} = 0 \tag{2.20}$$

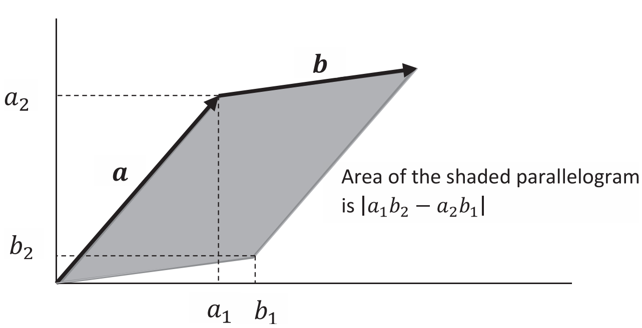


Fig. 9.
Area of parallelogram.

and

$$\hat{e}_1\wedge\hat{e}_2 = \hat{e}_1\hat{e}_2. \tag{2.21}$$

We also find that the area associated with $\hat{e}_1\wedge\hat{e}_2$ is equal to 1. Thus, $\hat{e}_1\hat{e}_2$ is a unit bivector in the $\hat{e}_1,\hat{e}_2$ plane, and geometrically, it represents an oriented unit area within a plane. We discuss its algebraic properties later.

As an extension of the notion of a vector, a bivector comprises the following:

1. A plane
2. An area
3. An orientation within the plane

It thus extends the one-dimensional concept of an oriented length to the two-dimensional concept of an oriented area. However, the bivector does not define a shape. Although it is convenient to represent a bivector by a parallelogram, as was done in the preceding figures, the bivector is an oriented number in a plane—not a specific shape. It could be equally well represented by a circle or any other planar figure with the given area. This is because the bivector only embodies information about two vectors taken as a pair, whereas defining a specific parallelogram in a plane, for example, requires information about the individual vectors. This perhaps may be most easily seen by forming the outer product of the vectors $\boldsymbol{a}+\alpha\boldsymbol{b}$ and $\boldsymbol{b}$, where α is a scalar. Because $\boldsymbol{b}\wedge\boldsymbol{b} = 0$, this gives the same outer product as $\boldsymbol{a}\wedge\boldsymbol{b}$. However, the actual parallelogram formed by $\boldsymbol{a}+\alpha\boldsymbol{b}$ and $\boldsymbol{b}$ differs from the parallelogram formed by $\boldsymbol{a}$ and $\boldsymbol{b}$ whenever $\alpha \neq 0$ (Fig. 11).

With the notion of the outer product in hand, we are in a position to explore multiplication of vectors.

B. MULTIPLICATION OF VECTORS

Let us consider how we might define multiplication of two distinct nonzero real N-dimensional vectors $\boldsymbol{a}$ and $\boldsymbol{b}$:

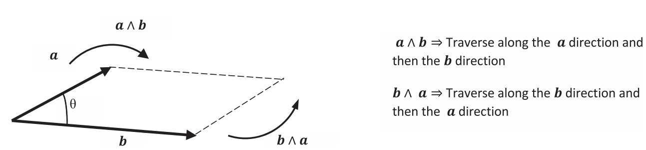


Fig. 10.
Bivector as outer product of two vectors.

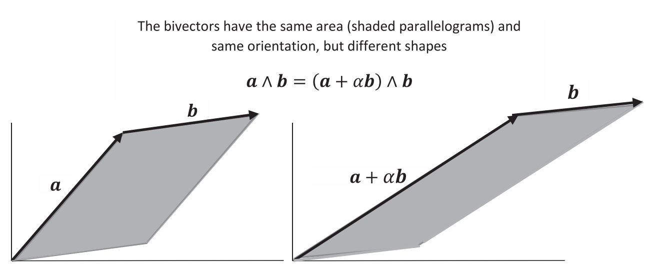


Fig. 11.
Bivector has no specific shape.

Consider the product of two vectors

$$\boldsymbol{ab}. \tag{2.22}$$

In defining multiplication of vectors, we would also like to be able to divide by nonzero vectors. When we say "divide by a vector $\boldsymbol{a}$," what we mean is we want to be able to multiply by a vector $\boldsymbol{a}^{-1}$ that has the property

$$\boldsymbol{a}\boldsymbol{a}^{-1} = \boldsymbol{a}^{-1}\boldsymbol{a} = 1. \tag{2.23}$$

The vector $\boldsymbol{a}^{-1}$ (if it exists) is called the multiplicative inverse of the vector $\boldsymbol{a}$. This will permit us to divide by a nonzero vector $\boldsymbol{a}$ as follows. Assume we have a product of vectors $\boldsymbol{ab}$ (which we have not yet defined) and the inverse $\boldsymbol{a}^{-1}$ (which we also have not yet defined), and we want to divide $\boldsymbol{ab}$ by $\boldsymbol{a}$ to recover $\boldsymbol{b}$. Let us represent this division as

$$\boldsymbol{a}^{-1}\left(\boldsymbol{ab}\right).$$

If we require multiplication of vectors to be associative, i.e.,

$$\boldsymbol{a}^{-1}\left(\boldsymbol{ab}\right) = \left(\boldsymbol{a}^{-1}\boldsymbol{a}\right)\boldsymbol{b} \tag{2.24}$$

then we may use $\boldsymbol{a}^{-1}\boldsymbol{a} = 1$ to find

$$\boldsymbol{a}^{-1}\left(\boldsymbol{ab}\right) = \left(\boldsymbol{a}^{-1}\boldsymbol{a}\right)\boldsymbol{b} = \boldsymbol{b}. \tag{2.25}$$

Thus, we have divided by the nonzero vector $\boldsymbol{a}$. We could also have examined

$$\left(\boldsymbol{ab}\right)\boldsymbol{a}^{-1}.$$

However, in general, $(\boldsymbol{ab})\boldsymbol{a}^{-1}$ does not equal $\boldsymbol{a}^{-1}(\boldsymbol{ab})$. Thus, the order of multiplication is important, and we do not require multiplication of vectors to be commutative.

The challenge is to define a product of vectors that permits us to use this approach. To proceed, we adopt a rule—called the contraction rule—such that the square of any vector is given by a scalar:

$$\boldsymbol{a}^2 = \boldsymbol{aa} = \boldsymbol{a}\cdot\boldsymbol{a} \tag{2.26}$$

where $\boldsymbol{a}\cdot\boldsymbol{a}$ is the inner product (sometimes known as the scalar product or dot product) of $\boldsymbol{a}$ with itself. We are not restricting ourselves to Euclidean spaces. The concept of the inner product is well defined in non-Euclidean spaces and thus can be used here to define the square of a vector. It is a scalar; thus, the contraction rule causes a vector—i.e., a one-dimensional quantity—to contract into a scalar—i.e., a zero-dimensional quantity. Provided $\boldsymbol{a}\cdot\boldsymbol{a}$ is nonzero, we can define

$$\boldsymbol{a}^{-1} \triangleq \frac{\boldsymbol{a}}{\boldsymbol{a}\cdot\boldsymbol{a}}. \tag{2.27}$$

In making this definition, we are requiring multiplication of vectors and scalars to be permissible. With this definition, we have

$$\boldsymbol{a}^{-1}\boldsymbol{a} = \frac{\boldsymbol{a}}{\boldsymbol{a}\cdot\boldsymbol{a}}\boldsymbol{a} = \frac{\boldsymbol{a}^2}{\boldsymbol{a}\cdot\boldsymbol{a}} = 1. \tag{2.28}$$

By the same approach, we also find $\boldsymbol{a}\boldsymbol{a}^{-1} = 1$. Thus, $\boldsymbol{a}^{-1}$ is the multiplicative inverse of $\boldsymbol{a}$. With these assumptions, our definition of $\boldsymbol{a}^{-1}$ leads to

$$\boldsymbol{a}^{-1}\boldsymbol{ab} = \frac{\boldsymbol{a}}{\boldsymbol{a}\cdot\boldsymbol{a}}\boldsymbol{ab} = \frac{\boldsymbol{a}^2}{\boldsymbol{a}\cdot\boldsymbol{a}}\boldsymbol{b} = \boldsymbol{b} \tag{2.29}$$

Similarly, we obtain

$$\boldsymbol{b}\boldsymbol{a}\boldsymbol{a}^{-1} = \boldsymbol{b}. \tag{2.30}$$

The adoption of the contraction rule, along with associativity of the product of vectors, has led to a meaningful way to perform division with nonzero vectors—provided we can define a product of vectors.

We now examine how the contraction rule allows us to define a product of two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$. Regardless of the number of dimensions N, these two vectors share a two-dimensional subspace, which we call a plane. Let $\hat{\boldsymbol{e}}_1$ and $\hat{\boldsymbol{e}}_2$ be an orthonormal basis for this plane, and write $\boldsymbol{a}$ and $\boldsymbol{b}$ as in (2.11) and (2.12). Then, the product of these two vectors may be written as follows:

$$\boldsymbol{ab} = \left(a_1\hat{\boldsymbol{e}}_1 + a_2\hat{\boldsymbol{e}}_2\right)\left(b_1\hat{\boldsymbol{e}}_1 + b_2\hat{\boldsymbol{e}}_2\right) \tag{2.31}$$

It is convenient to require the product to be distributive, i.e.:

$$\left(\boldsymbol{a}+\boldsymbol{b}\right)\boldsymbol{c} = \boldsymbol{ac} + \boldsymbol{bc} \tag{2.32}$$

$$\boldsymbol{a}\left(\boldsymbol{b}+\boldsymbol{c}\right) = \boldsymbol{ab} + \boldsymbol{ac} \tag{2.33}$$

With this property, we can write

$$\boldsymbol{ab} = \left(a_1\hat{\boldsymbol{e}}_1\right)\left(b_1\hat{\boldsymbol{e}}_1\right) + \left(a_1\hat{\boldsymbol{e}}_1\right)\left(b_2\hat{\boldsymbol{e}}_2\right) + \left(a_2\hat{\boldsymbol{e}}_2\right)\left(b_1\hat{\boldsymbol{e}}_1\right) + \left(a_2\hat{\boldsymbol{e}}_2\right)\left(b_2\hat{\boldsymbol{e}}_2\right). \tag{2.34}$$

It is also convenient to assume multiplication of vectors with scalars is commutative, i.e.,

$$\lambda\boldsymbol{a} = \boldsymbol{a}\lambda \tag{2.35}$$

where λ is a scalar. We have already assumed that multiplication of vectors is associative, i.e.,

$$\left(\boldsymbol{ab}\right)\boldsymbol{c} = \boldsymbol{a}\left(\boldsymbol{bc}\right) = \boldsymbol{abc}. \tag{2.36}$$

With these two properties, we can remove the parentheses from our product and write

$$\boldsymbol{ab} = a_1b_1\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_1 + a_2b_2\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_2 + a_1b_2\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 + a_2b_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_1. \tag{2.37}$$

Thus far, the properties we have required are properties shared by real and complex numbers. However, unlike the real and complex numbers, the product of vectors is not commutative.

To proceed we use the contraction rule to obtain

$$\hat{\boldsymbol{e}}_1^2 = \epsilon_1 \tag{2.38}$$

$$\hat{e}_2^2 = \epsilon_2 \tag{2.39}$$

where each ϵ_1 and ϵ_2 is individually equal to 1 or –1. This follows from $\hat{e}_1$ and $\hat{e}_2$ being unit vectors.[2] Our product becomes

$$ab = a \cdot b + \left(a_1 b_2 \hat{e}_1 \hat{e}_2 + a_2 b_1 \hat{e}_2 \hat{e}_1\right) \tag{2.40}$$

where we have incorporated the inner product

$$a \cdot b = \epsilon_1 a_1 b_1 + \epsilon_2 a_2 b_2 \tag{2.41}$$

into the right-hand side of this result. It is apparent the inner product arose from the contraction rule.

When $\epsilon_1 = \epsilon_2 = 1$, the expression in (2.41) is clearly the inner product in a two-dimensional Euclidean space. When ϵ_1 and ϵ_2 differ, the space is said to be pseudo-Euclidean. More generally, if we have an N-dimensional space in which p basis vectors square to +1 and q basis vectors square to –1, where $N = p + q$, the resulting pseudo-Euclidean space is denoted $R^{p,q}$ and the inner product is given by

$$a \cdot b = \sum_{k=1}^{p} a_k b_k - \sum_{k=p+1}^{N} a_k b_k. \tag{2.42}$$

N-dimensional Euclidean space is denoted by $R^{N,0} = R^N$. As we know from special relativity, $R^{1,3}$ is the pseudo-Euclidean space for describing Minkowski space-time.

To further explore the implications of the contraction rule, let us examine the product $a^2 = aa$, where $a = a_1\hat{e}_1 + a_2\hat{e}_2$ and both a_1 and a_2 are nonzero scalars. We find from our general approach earlier that

$$a^2 = a \cdot a + a_1 a_2 \left(\hat{e}_1 \hat{e}_2 + \hat{e}_2 \hat{e}_1\right). \tag{2.43}$$

From the contraction rule, we have $a^2 = a \cdot a = \epsilon_1 a_1^2 + \epsilon_2 a_2^2$. Therefore, because neither a_1 nor a_2 is zero, it follows that we necessarily have

$$\hat{e}_1 \hat{e}_2 = -\hat{e}_2 \hat{e}_1. \tag{2.44}$$

Going back to our general product, we therefore have

$$ab = a \cdot b + \left(a_1 b_2 - a_2 b_1\right) \hat{e}_1 \hat{e}_2. \tag{2.45}$$

The second term is the outer product of a and b. Thus, through use of the contraction rule, we have arrived at a product of two vectors given by

$$ab = a \cdot b + a \wedge b. \tag{2.46}$$

This product is called the geometric product and is a fundamental result in geometric algebra.

Examine

$$ba = b \cdot a + b \wedge a. \tag{2.47}$$

From the facts

$$b \cdot a = a \cdot b \tag{2.48}$$

$$b \wedge a = -a \wedge b \tag{2.49}$$

we may write

$$ba = a \cdot b - a \wedge b. \tag{2.50}$$

We may then combine this result with the product ab to obtain

$$a \cdot b = \frac{ab + ba}{2} \tag{2.51}$$

$$a \wedge b = \frac{ab - ba}{2}. \tag{2.52}$$

Thus, we see that the geometric product is more basic than either of the inner or outer products, because each of them can be obtained from the geometric product.[3] The inner product is the symmetric part of the geometric product, and the outer product is the antisymmetric part. It follows that

$$ab = \begin{cases} ba & \text{if } a \text{ and } b \text{ are parallel} \\ -ba & \text{if } a \text{ and } b \text{ are orthogonal.} \end{cases} \tag{2.53}$$

Thus, the symmetry or antisymmetry of the geometric product captures information about the relative directions of a and b in their shared plane. In general, however, the geometric product is neither symmetric nor antisymmetric.

Using the geometric product defined earlier, we may compute the product of three vectors that share the same plane[4]:

abc.

To see this, let us set $c = c_1\hat{e}_1 + c_2\hat{e}_2$ and use our earlier result for ab to write

$$\begin{aligned} abc &= \left(a \cdot b + \left(a_1 b_2 - a_2 b_1\right)\hat{e}_1\hat{e}_2\right)c \\ &= \left(a \cdot b\right)c + \left(a_1 b_2 - a_2 b_1\right)\hat{e}_1\hat{e}_2\left(c_1\hat{e}_1 + c_2\hat{e}_2\right) \\ &= \left(a \cdot b\right)c + c_1\left(a_1 b_2 - a_2 b_1\right)\hat{e}_1\hat{e}_2\hat{e}_1 + c_2\left(a_1 b_2 - a_2 b_1\right)\hat{e}_1\hat{e}_2\hat{e}_2. \end{aligned} \tag{2.54}$$

If we use the associativity of the product and the antisymmetry of the bivector, we find the following:

[2] Although in Euclidean space the inner product of a unit vector $\hat{e}$ with itself is the positive scalar $|\hat{e}|^2 = 1$, more generally in non-Euclidean spaces it can be –1. For example, the four-dimensional space where the squares of three basis vectors are negative and the square of the remaining basis vector is positive is particularly important and leads to the hyperbolic geometry of Minkowski space.

[3] We could have obtained the geometric product axiomatically without assuming a basis and then derived the inner and outer products from the geometric product. This is how a mathematician would approach the geometric product. However, it seems clearer for engineers to approach it (at least initially) through the use of a basis, as was done here.

[4] In a later section, we extend these ideas to multiplication of non-coplanar vectors.

$$\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_1 = -\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 = -\epsilon_1\hat{\boldsymbol{e}}_2 \tag{2.55}$$

$$\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_2 = \epsilon_2\hat{\boldsymbol{e}}_1 \tag{2.56}$$

Hence,

$$\boldsymbol{abc} = (\boldsymbol{a}\cdot\boldsymbol{b})\boldsymbol{c} + (a_1b_2 - a_2b_1)\left[\epsilon_2 c_2\hat{\boldsymbol{e}}_1 - \epsilon_1 c_1\hat{\boldsymbol{e}}_2\right]. \tag{2.57}$$

This is another vector in the same $\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2$ plane. Therefore, we see that the product of three coplanar vectors is another vector in the same plane. As a result, we may think of the product $\boldsymbol{ab}$ as an operator that converts a vector $\boldsymbol{c}$ in the $\boldsymbol{a},\boldsymbol{b}$ plane into another vector in the same plane. The product $\boldsymbol{ab}$ is an example of a spinor.

In this fashion, we can also divide the product $\boldsymbol{ab}$ by the non-zero vector $\boldsymbol{c}$:

$$\frac{\boldsymbol{ab}}{\boldsymbol{c}}$$

However, as already discussed, we must be careful to ascertain what we mean by this expression. In particular, because multiplication is not commutative, we have two possible interpretations:

$$\frac{\boldsymbol{ab}}{\boldsymbol{c}} = \begin{cases} \boldsymbol{abc}^{-1} = \dfrac{\boldsymbol{abc}}{|\boldsymbol{c}|^2} \\ \boldsymbol{c}^{-1}\boldsymbol{ab} = \dfrac{\boldsymbol{cab}}{|\boldsymbol{c}|^2} \end{cases} \tag{2.58}$$

In general, these are different vectors, and which interpretation is appropriate would be determined by the context of a particular problem. In either case, dividing the product of two vectors by a third coplanar vector results in a vector in the plane.

Recall that $\hat{\boldsymbol{e}}_1$ and $\hat{\boldsymbol{e}}_2$ are orthonormal basis vectors in the $\boldsymbol{a},\boldsymbol{b}$ plane and define

$$I_{\hat{e}_1\hat{e}_2} \triangleq \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2. \tag{2.59}$$

We are interested in the algebraic properties of $I_{\hat{e}_1\hat{e}_2}$—particularly $I^2_{\hat{e}_1\hat{e}_2}$. We have

$$\begin{aligned} I^2_{\hat{e}_1\hat{e}_2} &= (\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2)^2 = \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 \\ &= -\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_1 && (\text{since } \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 = -\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_1) \\ &= -\epsilon_2\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_1 && (\text{since } \hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_2 = |\hat{\boldsymbol{e}}_2|^2 = \epsilon_2) \\ &= -\epsilon_1\epsilon_2 && (\text{since } \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_1 = |\hat{\boldsymbol{e}}_1|^2 = \epsilon_1) \end{aligned} \tag{2.60}$$

This is an interesting result. Consider first the case in which $\epsilon_1 = \epsilon_2 = 1$. This choice gives rise to a Euclidean plane, for which we see

$$I^2_{\hat{e}_1\hat{e}_2} = -1. \tag{2.61}$$

In a Euclidean plane, the unit bivector $I_{\hat{e}_1\hat{e}_2}$ may be viewed as being analogous to $i = \sqrt{-1}$. As we discuss later, this property gives rise to a connection between the complex numbers and the geometric product of two vectors in a Euclidean plane.

On the other hand, consider the case in which $\epsilon_1 = 1$ and $\epsilon_2 = -1$. This choice gives rise to the hyperbolic plane. For the hyperbolic plane, we see

$$I^2_{\hat{e}_1\hat{e}_2} = 1. \tag{2.62}$$

As we discuss later, this property gives rise to the Lorentz transformations that arise in special relativity.

Because algebraically we have $I^2_{\hat{e}_1\hat{e}_2} = 1$ or -1, we may define the following function of a scalar x:

$$e^{I_{\hat{e}_1\hat{e}_2}x} \triangleq \sum_{k=0}^{\infty}\frac{\left(I_{\hat{e}_1\hat{e}_2}x\right)^k}{k!} \tag{2.63}$$

Using the associativity of the product of vectors, we may write this summation as follows:

$$\begin{aligned} e^{I_{\hat{e}_1\hat{e}_2}x} &= \sum_{m=0}^{\infty}\frac{\left(I^2_{\hat{e}_1\hat{e}_2}\right)^m x^{2m}}{2m!} + I_{\hat{e}_1\hat{e}_2}\sum_{m=0}^{\infty}\frac{\left(I^2_{\hat{e}_1\hat{e}_2}\right)^m x^{2m+1}}{(2m+1)!} \\ &= \begin{cases} \cos x + I_{\hat{e}_1\hat{e}_2}\sin x, & I^2_{\hat{e}_1\hat{e}_2} = -1 \\ \cosh x + I_{\hat{e}_1\hat{e}_2}\sinh x, & I^2_{\hat{e}_1\hat{e}_2} = 1 \end{cases} \end{aligned} \tag{2.64}$$

This result is a version of Euler's formula that incorporates the unit bivector $I_{\hat{e}_1\hat{e}_2}$ and is applicable in both the Euclidean and the hyperbolic planes.

In the remainder of this tutorial, we limit our attention to a Euclidean space, except as explicitly stated otherwise in section VI. The primary consequence is that we can write the following for an arbitrary vector $\boldsymbol{a}$:

$$\boldsymbol{a}\cdot\boldsymbol{a} = |\boldsymbol{a}|^2 \geq 0$$

The meaning of this expression, as giving the magnitude squared of the vector $\boldsymbol{a}$, should be familiar. Although much of what we discuss is valid for more general pseudo-Euclidean spaces, this tutorial emphasizes geometric ideas and relies on intuition in Euclidean geometry. Thus, to avoid abstraction for its own sake, the discussion is limited to Euclidean geometry. Section VI relaxes this assumption and uses geometric algebra to study the hyperbolic plane, which has extensive uses in electrical engineering.

C. ROTATION OF VECTORS VIA SPINORS

Our first task is to examine how spinors—i.e., a product of two vectors—may be used to define rotations in a Euclidean space. To do so, it is convenient to define the reverse of the geometric product as follows:

$$(\boldsymbol{ab})^{\dagger} \triangleq \boldsymbol{ba} \tag{2.65}$$

Examine the following quantity:

$$(\boldsymbol{ab})^{\dagger}\,\boldsymbol{ab} = \boldsymbol{baab} \tag{2.66}$$

From the contraction rule, this becomes

$$\boldsymbol{baab} = (\boldsymbol{a}\cdot\boldsymbol{a})(\boldsymbol{b}\cdot\boldsymbol{b}). \tag{2.67}$$

Moreover, because we are in a Euclidean space, this may be written as follows:

$$\boldsymbol{baab} = |\boldsymbol{a}|^2|\boldsymbol{b}|^2 \tag{2.68}$$

However, from the geometric product, we also may write

$$\begin{aligned}(\boldsymbol{ab})^\dagger\ \boldsymbol{ab} &= (\boldsymbol{b}\cdot\boldsymbol{a}+\boldsymbol{b}\wedge\boldsymbol{a})(\boldsymbol{a}\cdot\boldsymbol{b}+\boldsymbol{a}\wedge\boldsymbol{b})\\ &= (\boldsymbol{b}\cdot\boldsymbol{a})(\boldsymbol{a}\cdot\boldsymbol{b})+(\boldsymbol{b}\wedge\boldsymbol{a})(\boldsymbol{a}\wedge\boldsymbol{b})+(\boldsymbol{b}\cdot\boldsymbol{a})(\boldsymbol{a}\wedge\boldsymbol{b})+(\boldsymbol{b}\wedge\boldsymbol{a})(\boldsymbol{a}\cdot\boldsymbol{b})\\ &= (\boldsymbol{a}\cdot\boldsymbol{b})^2+(\boldsymbol{b}\wedge\boldsymbol{a})(\boldsymbol{a}\wedge\boldsymbol{b})+(\boldsymbol{a}\cdot\boldsymbol{b})[\boldsymbol{a}\wedge\boldsymbol{b}+\boldsymbol{b}\wedge\boldsymbol{a}]\\ &= (\boldsymbol{a}\cdot\boldsymbol{b})^2+(\boldsymbol{b}\wedge\boldsymbol{a})(\boldsymbol{a}\wedge\boldsymbol{b})\end{aligned} \tag{2.69}$$

where we have used $\boldsymbol{b}\cdot\boldsymbol{a} = \boldsymbol{a}\cdot\boldsymbol{b}$ and $\boldsymbol{a}\wedge\boldsymbol{b} = -\boldsymbol{b}\wedge\mathbf{a}$. Thus, we have

$$|\boldsymbol{a}|^2|\boldsymbol{b}|^2 = (\boldsymbol{a}\cdot\boldsymbol{b})^2+(\boldsymbol{b}\wedge\boldsymbol{a})(\boldsymbol{a}\wedge\boldsymbol{b}). \tag{2.70}$$

Dividing both sides by the scalar $|\boldsymbol{a}|^2|\boldsymbol{b}|^2$ yields

$$1 = \frac{(\boldsymbol{a}\cdot\boldsymbol{b})^2}{|\boldsymbol{a}|^2|\boldsymbol{b}|^2}+\frac{(\boldsymbol{b}\wedge\boldsymbol{a})(\boldsymbol{a}\wedge\boldsymbol{b})}{|\boldsymbol{a}|^2|\boldsymbol{b}|^2}. \tag{2.71}$$

Let us write

$$\boldsymbol{a}\wedge\boldsymbol{b} = (a_1b_2-a_2b_1)I_{\hat{e}_1\hat{e}_2} \tag{2.72}$$

$$\boldsymbol{b}\wedge\boldsymbol{a} = -(a_1b_2-a_2b_1)I_{\hat{e}_1\hat{e}_2} \tag{2.73}$$

$$|\boldsymbol{a}\wedge\boldsymbol{b}|^2 \triangleq (a_1b_2-a_2b_1)^2. \tag{2.74}$$

Then, we may write

$$(\boldsymbol{b}\wedge\boldsymbol{a})(\boldsymbol{a}\wedge\boldsymbol{b}) = -|\boldsymbol{a}\wedge\boldsymbol{b}|^2 I^2_{\hat{e}_1\hat{e}_2} = |\boldsymbol{a}\wedge\boldsymbol{b}|^2. \tag{2.75}$$

Thus, we have

$$1 = \frac{(\boldsymbol{a}\cdot\boldsymbol{b})^2}{|\boldsymbol{a}|^2|\boldsymbol{b}|^2}+\frac{|\boldsymbol{a}\wedge\boldsymbol{b}|^2}{|\boldsymbol{a}|^2|\boldsymbol{b}|^2}. \tag{2.76}$$

We may now define an angle $\theta\boldsymbol{ab}$ by

$$\cos^2\theta_{\boldsymbol{ab}} \triangleq \frac{(\boldsymbol{a}\cdot\boldsymbol{b})^2}{|\boldsymbol{a}|^2|\boldsymbol{b}|^2} \tag{2.77}$$

$$\sin^2\theta_{\boldsymbol{ab}} \triangleq \frac{|\boldsymbol{a}\wedge\boldsymbol{b}|^2}{|\boldsymbol{a}|^2|\boldsymbol{b}|^2}. \tag{2.78}$$

With these definitions, we may write

$$\boldsymbol{ab} = |\boldsymbol{a}||\boldsymbol{b}|\cos\theta_{\boldsymbol{ab}} \pm |\boldsymbol{a}||\boldsymbol{b}|\sin\theta_{\boldsymbol{ab}} I_{\hat{e}_1\hat{e}_2} = |\boldsymbol{a}||\boldsymbol{b}|e^{\pm I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}. \tag{2.79}$$

The sign is determined by the sign of $(a_1b_1 - a_2b_1)$. We find similarly that

$$\boldsymbol{ba} = (\boldsymbol{ab})^\dagger = |\boldsymbol{a}||\boldsymbol{b}|e^{\mp I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}. \tag{2.80}$$

We may use this spinor formulation to explore rotations of vectors.

Previously, we saw that the product of three coplanar vectors is another coplanar vector and that, as a result, we may view the product of two vectors as a spinor that operates on vectors in the same plane as the two vectors. In this section, we examine the operation of a spinor on a vector that may have non-coplanar components and show how this may be used to define an analytically convenient approach to rotations. Thus, let us again consider the product

abc

where now, however, the vector $\boldsymbol{c}$ has both coplanar and non-coplanar components:

$$\boldsymbol{c} = c_\parallel\hat{\boldsymbol{e}}_\parallel + c_\perp\hat{\boldsymbol{e}}_\perp \tag{2.81}$$

In this expression, $c_\parallel\hat{\boldsymbol{e}}_\parallel = c_1\hat{\boldsymbol{e}}_1 + c_2\hat{\boldsymbol{e}}_2$ is the part of $\boldsymbol{c}$ in the $\hat{\boldsymbol{e}}_1,\hat{\boldsymbol{e}}_2$ plane and $\hat{\boldsymbol{e}}_\perp$ is orthogonal to both $\hat{\boldsymbol{e}}_1$ and $\hat{\boldsymbol{e}}_2$:

$$\hat{\boldsymbol{e}}_\perp\cdot\hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{e}}_\perp\cdot\hat{\boldsymbol{e}}_2 = 0 \tag{2.82}$$

We have

$$\boldsymbol{abc} = c_\parallel\boldsymbol{ab}\hat{\boldsymbol{e}}_\parallel + c_\perp\boldsymbol{ab}\hat{\boldsymbol{e}}_\perp. \tag{2.83}$$

We already know that $\boldsymbol{ab}(c_\parallel\hat{\boldsymbol{e}}_\parallel)$ is a vector in the $\hat{\boldsymbol{e}}_1,\hat{\boldsymbol{e}}_2$ plane. To consider this particular vector further, we use Euler's formula to write

$$\begin{aligned}c_\parallel\boldsymbol{ab}\hat{\boldsymbol{e}}_\parallel &= \boldsymbol{ab}(c_1\hat{\boldsymbol{e}}_1+c_2\hat{\boldsymbol{e}}_2) = |\boldsymbol{a}||\boldsymbol{b}|e^{\pm I_{\hat{e}_1\hat{e}_2}}(c_1\hat{\boldsymbol{e}}_1+c_2\hat{\boldsymbol{e}}_2)\\ &= |\boldsymbol{a}||\boldsymbol{b}|\{\cos\theta_{\boldsymbol{ab}} \pm \sin\theta_{\boldsymbol{ab}}I_{\hat{e}_1\hat{e}_2}\}(c_1\hat{\boldsymbol{e}}_1+c_2\hat{\boldsymbol{e}}_2).\end{aligned} \tag{2.84}$$

To carry out this computation, we first observe the following from the properties of the geometric product:

$$\hat{\boldsymbol{e}}_1 I_{\hat{e}_1\hat{e}_2} = \hat{\boldsymbol{e}}_1(\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2) = \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{e}}_2 \tag{2.85}$$

$$\hat{\boldsymbol{e}}_2 I_{\hat{e}_1\hat{e}_2} = \hat{\boldsymbol{e}}_2(\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2) = -\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_1 = -\hat{\boldsymbol{e}}_1 \tag{2.86}$$

$$I_{\hat{e}_1\hat{e}_2}\hat{\boldsymbol{e}}_1 = (\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2)\hat{\boldsymbol{e}}_1 = -\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 = -\hat{\boldsymbol{e}}_2 \tag{2.87}$$

$$I_{\hat{e}_1\hat{e}_2}\hat{\boldsymbol{e}}_2 = (\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2)\hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{e}}_1 \tag{2.88}$$

Using these results leads to

$$\begin{aligned}c_\parallel\boldsymbol{ab}\hat{\boldsymbol{e}}_\parallel &= |\boldsymbol{a}||\boldsymbol{b}|[c_1\cos\theta_{\boldsymbol{ab}} \pm c_2\sin\theta_{\boldsymbol{ab}}]\hat{\boldsymbol{e}}_1 + [\mp c_1\sin\theta_{\boldsymbol{ab}} + c_2\cos\theta_{\boldsymbol{ab}}]\hat{\boldsymbol{e}}_2\\ &= |\boldsymbol{a}||\boldsymbol{b}|(c_{rotated,1}\hat{\boldsymbol{e}}_1 + c_{rotated,2}\hat{\boldsymbol{e}}_2).\end{aligned} \tag{2.89}$$

This vector is also in the plane and is simply a dilated (by $|\boldsymbol{a}||\boldsymbol{b}|$) and rotated (by the angle $\theta_{\boldsymbol{ab}}$) version of the vector $c_1\hat{\boldsymbol{e}}_1 + c_2\hat{\boldsymbol{e}}_2$. Thus, we have dilated and rotated the part of $\boldsymbol{c}$ that is in the plane.

We now examine $c_\perp \boldsymbol{ab}\hat{\boldsymbol{e}}_\perp$. We find

$$\begin{aligned} c_\perp \boldsymbol{ab}\hat{\boldsymbol{e}}_\perp &= (\boldsymbol{a}\cdot\boldsymbol{b})c_\perp\hat{\boldsymbol{e}}_\perp + c_\perp(a_1b_2 - a_2b_1)I_{\hat{e}_1\hat{e}_2}\hat{\boldsymbol{e}}_\perp \\ &= (\boldsymbol{a}\cdot\boldsymbol{b})c_\perp\hat{\boldsymbol{e}}_\perp + c_\perp(a_1b_2 - a_2b_1)\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_\perp. \end{aligned} \tag{2.90}$$

This is the sum of a vector and a new kind of object formed from the product of three orthogonal unit vectors. This object is called a trivector, which we consider further later. Thus, the product $\boldsymbol{abc}$, where $\boldsymbol{c}$ has components that are orthogonal to the $\boldsymbol{a},\boldsymbol{b}$ plane, results in the sum of a vector and a trivector.

Because the three unit vectors are mutually orthogonal, we may sift $\hat{\boldsymbol{e}}_\perp$ through the product, changing signs as we go, to write the following:

$$I_{\hat{e}_1\hat{e}_2}\hat{\boldsymbol{e}}_\perp = \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_\perp = -\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_\perp\hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{e}}_\perp\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{e}}_\perp I_{\hat{e}_1\hat{e}_2} \tag{2.91}$$

Hence, the unit bivector $I_{\hat{e}_1\hat{e}_2}$ commutes with all vectors that are orthogonal to the $\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2$ plane.

We use this property of $I_{\hat{e}_1\hat{e}_2}$ to further explore the operation of a spinor. In particular, let us examine a two-sided spinor operation

$$\boldsymbol{abcba} = |\boldsymbol{a}|^2|\boldsymbol{b}|^2 e^{I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}\boldsymbol{c}e^{-I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}} \tag{2.92}$$

where we have used the Euler representations of the products $\boldsymbol{ab}$ and $\boldsymbol{ba}$ that we obtained earlier and for simplicity have chosen the sign corresponding to $\boldsymbol{a}\wedge\boldsymbol{b} > 0$; the opposite orientation simply interchanges the signs of $I_{\hat{e}_1\hat{e}_2}$ in (2.92). Because the magnitudes of the vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ simply scale the right-hand side, we set them equal to 1 and examine the operation

$$\hat{\boldsymbol{a}}\hat{\boldsymbol{b}}\boldsymbol{c}\hat{\boldsymbol{b}}\hat{\boldsymbol{a}} = e^{I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}\boldsymbol{c}e^{-I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}} \tag{2.93}$$

where $\hat{\boldsymbol{a}}$ and $\hat{\boldsymbol{b}}$ are unit-length vectors. This may be written as follows:

$$\hat{\boldsymbol{a}}\hat{\boldsymbol{b}}\boldsymbol{c}\hat{\boldsymbol{b}}\hat{\boldsymbol{a}} = e^{I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}\left\{c_\parallel\hat{\boldsymbol{e}}_\parallel\right\}e^{-I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}} + e^{I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}\left\{c_\perp\hat{\boldsymbol{e}}_\perp\right\}e^{-I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}} \tag{2.94}$$

We look at each of these terms separately.

First let us look at the part of $\boldsymbol{c}$ that is in the plane:

$$e^{I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}\left\{c_\parallel\hat{\boldsymbol{e}}_\parallel\right\}e^{-I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}} = e^{I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}\left\{c_1\hat{\boldsymbol{e}}_1 + c_2\hat{\boldsymbol{e}}_2\right\}e^{-I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}} \tag{2.95}$$

We already know that

$$e^{I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}\left\{c_\parallel\hat{\boldsymbol{e}}_\parallel\right\} = c_{rotated,1}\hat{\boldsymbol{e}}_1 + c_{rotated,2}\hat{\boldsymbol{e}}_2. \tag{2.96}$$

Continuing with our computation of the two-sided operation, we find the following:

$$\begin{aligned} e^{I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}\left\{c_\parallel\hat{\boldsymbol{e}}_\parallel\right\}e^{-I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}} &= \left\{c_{rotated,1}\hat{\boldsymbol{e}}_1 + c_{rotated,2}\hat{\boldsymbol{e}}_2\right\}e^{-I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}} \\ &= \left\{c_{rotated,1}\hat{\boldsymbol{e}}_1 + c_{rotated,2}\hat{\boldsymbol{e}}_2\right\}\left\{\cos\theta_{\boldsymbol{ab}} - \sin\theta_{\boldsymbol{ab}}I_{\hat{e}_1\hat{e}_2}\right\} \\ &= \left[c_{rotated,1}\cos\theta_{\boldsymbol{ab}} + c_{rotated,2}\sin\theta_{\boldsymbol{ab}}\right]\hat{\boldsymbol{e}}_1 \\ &\quad + \left[-c_{rotated,1}\sin\theta_{\boldsymbol{ab}} + c_{rotated,2}\cos\theta_{\boldsymbol{ab}}\right]\hat{\boldsymbol{e}}_2 \end{aligned} \tag{2.97}$$

This again is simply a planar rotation of the vector $c_{rotated,1}\hat{\boldsymbol{e}}_1 + c_{rotated,2}\hat{\boldsymbol{e}}_2$, which already incorporates a rotation of the planar part of c, by an additional angle $\theta_{\boldsymbol{ab}}$. Hence, we see that the two-sided operation

$$e^{I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}\left\{c_\parallel\hat{\boldsymbol{e}}_\parallel\right\}e^{-I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}$$

rotates the coplanar vector $c_1\hat{\boldsymbol{e}}_1 + c_2\hat{\boldsymbol{e}}_2$ in the plane by the angle $2\theta_{\boldsymbol{ab}}$.

It may seem strange to implement a two-sided operation to obtain a rotation of $2\theta_{\boldsymbol{ab}}$ when we can implement a one-sided operation and obtain a rotation of $\theta_{\boldsymbol{ab}}$. However, the effect of the two-sided operation on the non-coplanar part of $\boldsymbol{c}$ yields great benefit. Examine the following:

$$\begin{aligned} e^{I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}\left\{c_\perp\hat{\boldsymbol{e}}_\perp\right\}e^{-I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}} &= c_\perp\left\{\cos\theta_{\boldsymbol{ab}} + \sin\theta_{\boldsymbol{ab}}I_{\hat{e}_1\hat{e}_2}\right\} \\ &\quad \times\hat{\boldsymbol{e}}_\perp\left\{\cos\theta_{\boldsymbol{ab}} - \sin\theta_{\boldsymbol{ab}}I_{\hat{e}_1\hat{e}_2}\right\} \\ &= c_\perp\left\{\cos^2\theta_{\boldsymbol{ab}}\hat{\boldsymbol{e}}_\perp + \sin\theta_{\boldsymbol{ab}}\cos\theta_{\boldsymbol{ab}}I_{\hat{e}_1\hat{e}_2}\hat{\boldsymbol{e}}_\perp\right. \\ &\quad \left. -\cos\theta_{\boldsymbol{ab}}\sin\theta_{\boldsymbol{ab}}\hat{\boldsymbol{e}}_\perp I_{\hat{e}_1\hat{e}_2} - \sin^2\theta_{\boldsymbol{ab}}I_{\hat{e}_1\hat{e}_2}\hat{\boldsymbol{e}}_\perp I_{\hat{e}_1\hat{e}_2}\right\} \end{aligned} \tag{2.98}$$

As we discussed earlier, $I_{\hat{e}_1\hat{e}_2}$ commutes with any vector that is orthogonal to the $\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2$ plane—hence, it commutes with $\hat{\boldsymbol{e}}_\perp$. Therefore,

$$\sin\theta_{\boldsymbol{ab}}\cos\theta_{\boldsymbol{ab}}I_{\hat{e}_1\hat{e}_2}\hat{\boldsymbol{e}}_\perp - \cos\theta_{\boldsymbol{ab}}\sin\theta_{\boldsymbol{ab}}\hat{\boldsymbol{e}}_\perp I_{\hat{e}_1\hat{e}_2} = 0 \tag{2.99}$$

$$-\sin^2\theta_{\boldsymbol{ab}}I_{\hat{e}_1\hat{e}_2}\hat{\boldsymbol{e}}_\perp I_{\hat{e}_1\hat{e}_2} = -\sin^2\theta_{\boldsymbol{ab}}I^2_{\hat{e}_1\hat{e}_2}\hat{\boldsymbol{e}}_\perp = \sin^2\theta_{\boldsymbol{ab}}\hat{\boldsymbol{e}}_\perp. \tag{2.100}$$

Our result becomes

$$e^{I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}}\left\{c_\perp\hat{\boldsymbol{e}}_\perp\right\}e^{-I_{\hat{e}_1\hat{e}_2}\theta_{\boldsymbol{ab}}} = c_\perp\left\{\cos^2\theta_{\boldsymbol{ab}} + \sin^2\theta_{\boldsymbol{ab}}\right\}\hat{\boldsymbol{e}}_\perp = c_\perp\hat{\boldsymbol{e}}_\perp. \tag{2.101}$$

The two-sided operation has no effect on vectors that are orthogonal to the $\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2$ plane. Therefore, by implementing the two-sided operation instead of the one-sided operation, we have implemented a rotation of the part of the vector that is in the plane, at the same time avoiding the trivector that arises from the one-side operation. This leads to a convenient approach to defining rotations.

Define the following rotors (which are unit spinors obtained as the product of two unit vectors):

$$R_{\theta_{ab}} \triangleq e^{I_{ab}\theta_{ab}} = \hat{\boldsymbol{a}}\hat{\boldsymbol{b}} \tag{2.102}$$

$$R^\dagger_{\theta_{ab}} \triangleq e^{-I_{ab}\theta_{ab}} = \hat{\boldsymbol{b}}\hat{\boldsymbol{a}} \tag{2.103}$$

In these expressions, $I_{\boldsymbol{ab}}$ is the unit bivector in the $\boldsymbol{a},\boldsymbol{b}$ plane with the same orientation as $\boldsymbol{a} \wedge \boldsymbol{b}$. The two-sided spinor operation is now expressed as follows:

$$R_{\theta_{ab}} \boldsymbol{c} R_{\theta_{ab}}^{\dagger}$$

This operation rotates the part of the vector $\boldsymbol{c}$ in the plane containing $\hat{\boldsymbol{a}}$ and $\hat{\boldsymbol{b}}$ by the angle $2\theta_{\boldsymbol{ab}}$, but leaves untouched the part of the vector $\boldsymbol{c}$ orthogonal to the plane. Therefore, we see that as long as we operate only on coplanar vectors, we can define a rotation via a one-sided operation. However, the one-sided operation no longer yields just a vector when applied to vectors with components outside the plane of interest. In this more general case, rotation by an angle θ in a given plane can be implemented conveniently via a two-sided operation with rotors that each implement half of the desired rotation. This two-sided operation gives the desired rotation of coplanar vectors and leaves the orthogonal vectors unaffected (Fig. 12).

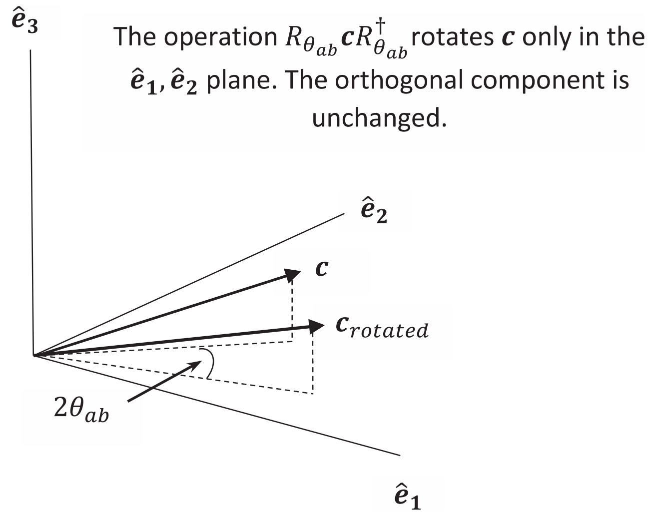


Fig. 12.
Rotation using rotors.

D. VECTOR PRODUCTS IN THE EUCLIDEAN PLANE AND COMPLEX NUMBERS

As discussed earlier, in a Euclidean space, the individual inner and outer products may be described geometrically by

$$\boldsymbol{a} \cdot \boldsymbol{b} = |\boldsymbol{a}||\boldsymbol{b}|\cos\theta_{ab} \tag{2.104}$$

$$|\boldsymbol{a} \wedge \boldsymbol{b}| = |\boldsymbol{a}||\boldsymbol{b}|\sin\theta_{ab} \tag{2.105}$$

Here, $\sin\theta_{\boldsymbol{ab}} \geq 0$ (hence, $0 \leq \theta_{\boldsymbol{ab}} \leq \pi$). From this, we see explicitly that the geometric product $\boldsymbol{ab}$ embodies information about the following:

1. The magnitudes $|\boldsymbol{a}|$ and $|\boldsymbol{b}|$ of the vectors $\boldsymbol{a}$ and $\boldsymbol{b}$
2. The angle $\theta_{\boldsymbol{ab}}$ between them
3. The plane that they share (captured by the product $\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2$)

In terms of the magnitudes of the vectors and the angles between them, we have

$$\boldsymbol{ab} = |\boldsymbol{a}||\boldsymbol{b}|\cos\theta_{ab} + I_{\boldsymbol{ab}}|\boldsymbol{a}||\boldsymbol{b}|\sin\theta_{ab} = |\boldsymbol{a}||\boldsymbol{b}|e^{I_{\boldsymbol{ab}}\theta_{ab}}. \tag{2.106}$$

Here, the notation $I_{\boldsymbol{ab}}$ represents a unit bivector in the $\boldsymbol{a},\boldsymbol{b}$ plane and hence represents either $I_{\hat{e}_1\hat{e}_2}$ or $-I_{\hat{e}_1\hat{e}_2} = I_{\hat{e}_2\hat{e}_1}$ in the earlier notation—the specific choice is determined by the sign of $(a_1b_2 - a_2b_1)$. Because $I_{\boldsymbol{ab}}^2 = -1$, (2.106) gives the result we conjectured in (2.9). However, I_{ab} is not an imaginary number—it is a unit bivector. In addition to satisfying $I_{\boldsymbol{ab}}^2 = -1$, a unit bivector gives rise to rotations. To see this, consider the unit bivector $I_{\hat{e}_1\hat{e}_2}$ and examine

$$\hat{\boldsymbol{e}}_1 I_{\hat{e}_1\hat{e}_2} = \hat{\boldsymbol{e}}_2 \tag{2.107}$$

$$\hat{\boldsymbol{e}}_2 I_{\hat{e}_1\hat{e}_2} = -\hat{\boldsymbol{e}}_1. \tag{2.108}$$

If $\hat{\boldsymbol{e}}_1$ represents the real axis and $\hat{\boldsymbol{e}}_2$ represents the imaginary axis, then right multiplication of a vector by the unit bivector $I_{\hat{e}_1\hat{e}_2}$ causes the same 90-deg rotation as multiplication by $+i$ does in the complex plane. Similarly, left multiplication by $I_{\hat{e}_1\hat{e}_2}$ is like multiplication by $-i$ in the complex plane.

The relationship between a vector and a corresponding complex number follows from

$$\hat{\boldsymbol{e}}_1\boldsymbol{a} = \hat{\boldsymbol{e}}_1 a_1 \hat{\boldsymbol{e}}_1 + \hat{\boldsymbol{e}}_1 a_2 \hat{\boldsymbol{e}}_2 = a_1 \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_1 + a_2 \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 = a_1 + I_{\hat{e}_1\hat{e}_2} a_2 \tag{2.109}$$

(scalars commute with everything in geometric algebra, so $a_2\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 a_2 = I_{\hat{e}_1\hat{e}_2} a_2$) and

$$(\hat{\boldsymbol{e}}_1\boldsymbol{a})^{\dagger} = \boldsymbol{a}\hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{e}}_1 a_1 \hat{\boldsymbol{e}}_1 + a_2 \hat{\boldsymbol{e}}_2 \hat{\boldsymbol{e}}_1 = a_1 - I_{\hat{e}_1\hat{e}_2} a_2. \tag{2.110}$$

Thus, if the product $\hat{\boldsymbol{e}}_1\boldsymbol{a}$ is the complex number, then the reverse $(\hat{\boldsymbol{e}}_1\boldsymbol{a})^{\dagger}$ is its complex conjugate. In this way, we can easily switch between two-dimensional vectors and complex numbers but nonetheless maintain a distinct view of these two kinds of quantities.

To compare this formulation with our earlier attempt to define multiplication of vectors by representing the vectors as complex numbers, we write the following:

$$\mathfrak{z}_a^* = (\hat{\boldsymbol{e}}_1\boldsymbol{a})^{\dagger} = \boldsymbol{a}\hat{\boldsymbol{e}}_1 = a_1 - I_{\hat{e}_1\hat{e}_2} a_2 \tag{2.111}$$

$$\mathfrak{z}_b = \hat{\boldsymbol{e}}_1\boldsymbol{b} = b_1 + I_{\hat{e}_1\hat{e}_2} b_2 \tag{2.112}$$

Hence,

$$\begin{aligned} \mathfrak{z}_a^*\mathfrak{z}_b = \boldsymbol{a}\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_1\boldsymbol{b} &= \left(a_1 - I_{\hat{e}_1\hat{e}_2} a_2\right)\left(b_1 + I_{\hat{e}_1\hat{e}_2} b_2\right) \\ &= a_1b_1 - I_{\hat{e}_1\hat{e}_2}^2 a_2b_2 + I_{\hat{e}_1\hat{e}_2}\left(a_1b_2 - a_2b_1\right) \\ &= a_1b_1 + a_2b_2 + I_{\hat{e}_1\hat{e}_2}\left(a_1b_2 - a_2b_1\right) = \boldsymbol{ab}. \end{aligned} \tag{2.113}$$

We could have obtained $\mathfrak{z}_a^*\mathfrak{z}_b = \boldsymbol{ab}$ by simply using $\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_1 = 1$.

In addition to restricting our "complex numbers" to the appropriate plane, a unit bivector determines an orientation for the product $\boldsymbol{ab}$. Thus, I_{ab} is generally different from the unit imaginary number $i = \sqrt{-1}$, and the geometric product $\boldsymbol{ab}$ is generally different from the product of two complex numbers.

E. THE OUTER PRODUCT AND *K* VECTORS

The introduction suggested that we should be able to extend the notion of an oriented length to the more general notion of an oriented k-volume. Equipped with the outer product, we can do just that. Let $\boldsymbol{a}$, $\boldsymbol{b}$, and $\boldsymbol{c}$ be three vectors and examine the following:

$$A_3 = \boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c} \tag{2.114}$$

The outer product is associative, so this expression is well defined without parentheses. It follows from the preceding results that if one of the vectors is equal to another, then this triple product is equal to zero. For example, let $\boldsymbol{a} = \boldsymbol{b}$. Then, we can write

$$A_3 = \left(\boldsymbol{a} \wedge \boldsymbol{a}\right) \wedge \boldsymbol{c} \tag{2.115}$$

and because $\boldsymbol{a} \wedge \boldsymbol{a} = 0$, we find $A_3 = 0$. From this, we can conclude that if $\boldsymbol{a}$, $\boldsymbol{b}$, and $\boldsymbol{c}$ are linearly independent, then

$$\boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c} \neq 0. \tag{2.116}$$

This is a convenient and succinct characterization of the notion of linear independence. Thus, the triple outer product of linearly independent vectors forms a nonzero quantity called a trivector and represents an oriented volume (Fig. 13).

The orientation of a trivector (like a bivector) is determined by the order in which the vectors appear in the product. Because the exterior product is antisymmetric, we find that there are two possible orientations (i.e., two possible signs—just like for a vector or a bivector). Thus, reversing the order of two of the vectors changes the orientation. If $\hat{\boldsymbol{e}}_1$, $\hat{\boldsymbol{e}}_2$, and $\hat{\boldsymbol{e}}_3$ form an orthonormal set, then we find

$$\hat{\boldsymbol{e}}_1 \wedge \hat{\boldsymbol{e}}_2 \wedge \hat{\boldsymbol{e}}_3 = \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_3 \triangleq I_3. \tag{2.117}$$

Such a quantity is a unit trivector I_3. Examine the following:

$$I_3^2 = \left(\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_3\right)^2 = \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_3\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_3 \tag{2.118}$$

Because the unit vectors in this expression are orthogonal and the geometric product of two orthogonal vectors is antisymmetric, we may reorder this expression to obtain

$$I_3^2 = -\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_3\hat{\boldsymbol{e}}_3. \tag{2.119}$$

Because we have unit vectors in this expression, we find (in Euclidean space)

$$I_3^2 = -\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_3\hat{\boldsymbol{e}}_3 = -1. \tag{2.120}$$

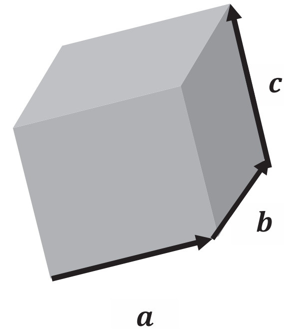


Fig. 13.
Example of trivector.

Thus, we have found another quantity that squares to –1. It is an attribute of geometric algebra that several geometrically distinct quantities can square to –1.

This process can be continued to form a k-blade for any $k = 1,2,\cdots,N$ (where N is the dimension of the space we are working in) by forming

$$A_k = \boldsymbol{a}_1 \wedge \boldsymbol{a}_2 \wedge \cdots \wedge \boldsymbol{a}_k \tag{2.121}$$

from k linearly independent vectors. If the vectors forming the k-blade are orthonormal vectors, we obtain a unit k-blade given by the following:

$$I_k = \hat{\boldsymbol{e}}_1 \wedge \hat{\boldsymbol{e}}_2 \wedge \cdots \wedge \hat{\boldsymbol{e}}_{\boldsymbol{k}} = \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 \cdots \hat{\boldsymbol{e}}_{\boldsymbol{k}} \tag{2.122}$$

The highest-grade unit k-blade in the space—i.e., the unit N-blade I_N in an N-dimensional space—is called the pseudoscalar. Thus, for example, in two dimensions, the pseudoscalar is the unit bivector in the plane. A general k-vector can then be obtained as a linear combination of k-blades. For each k-vector, the number k denotes the grade of the element.

Following the same steps we followed earlier to compute I_2^2 and I_3^2, we may show in a Euclidean space that

$$I_k^2 = \left(-1\right)^{\frac{k(k-1)}{2}}, \qquad k = 1,2,\cdots,N. \tag{2.123}$$

From this, we see that the pseudoscalar squares to either 1 or –1 as a function of the dimension.

F. THE GEOMETRIC ALGEBRA ASSOCIATED WITH $R^{p,q}$

We can form linear combinations of k-blades of varying grades to form a general multivector. For example, as we have seen in a two-dimensional space (i.e., a plane), we have four basic elements from which we can form a multivector:

- The unit scalar 1
- Two unit vectors $\hat{\boldsymbol{e}}_1$ and $\hat{\boldsymbol{e}}_2$
- The unit pseudoscalar $I_2 = \hat{\boldsymbol{e}}_1 \wedge \hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2$

In a plane, a general multivector A_2 may be written as follows:

$$A_2 = \underbrace{a_0}_{\text{Scalar}} + \underbrace{a_{11}\hat{\boldsymbol{e}}_1 + a_{12}\hat{\boldsymbol{e}}_2}_{\text{Vectors}} + \underbrace{a_2 I_2}_{\text{Pseudoscalar}} \tag{2.124}$$

Because the square of a pseudoscalar is a scalar, the product of any two multivectors formed from the vectors of a two-dimensional space is another multivector in the same space. Thus, the multivectors form a closed system with respect to the geometric product.

Similarly, in three dimensions, we have eight basic elements for forming a multivector:

- A unit scalar 1
- Three unit vectors $\hat{\boldsymbol{e}}_1$, $\hat{\boldsymbol{e}}_2$, and $\hat{\boldsymbol{e}}_3$
- Three unit bivectors $\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2$, $\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_3$, and $\hat{\boldsymbol{e}}_3\hat{\boldsymbol{e}}_1$
- The unit pseudoscalar $I_3 = \hat{\boldsymbol{e}}_1 \wedge \hat{\boldsymbol{e}}_2 \wedge \hat{\boldsymbol{e}}_3 = \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_3$

In three dimensions, a general multivector A_3 may be written as follows:

$$A_3 = \underbrace{a_0}_{\text{Scalar}} + \underbrace{a_{11}\hat{\boldsymbol{e}}_1 + a_{12}\hat{\boldsymbol{e}}_2 + a_{13}\hat{\boldsymbol{e}}_3}_{\text{Vectors}} + \underbrace{a_{21}\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 + a_{22}\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_3 + a_{23}\hat{\boldsymbol{e}}_3\hat{\boldsymbol{e}}_1}_{\text{Bivectors}} + \underbrace{a_3 I_3}_{\text{Pseudoscalar}} \tag{2.125}$$

As in the two-dimensional case, it follows that because the square of a pseudoscalar is a scalar, this system of multivectors is closed under the geometric product.

More generally, this process can be continued to any dimension N, and the set of multivectors for an N-dimensional vector space forms a geometric algebra with the dimension 2^N. The geometric algebra associated with the space $R^{p,q}$ ($N = p + q$)) is denoted $G_{p,q}$. To see why the dimension is 2^N, we consider that we can form $\begin{pmatrix} N \\ k \end{pmatrix}$ unit k-blades by choosing the orthonormal basis vectors of the vector space k at a time. We can therefore form

$$\sum_{k=0}^{N} \begin{pmatrix} N \\ k \end{pmatrix} = 2^N \tag{2.126}$$

such unit blades, which then act as a basis for the geometric algebra.

From the perspective of geometric algebra, we have the following observations:

1. The product of two vectors is an example of a more general multivector.
2. An imaginary number is a bivector in a Euclidean plane.
3. A complex number is the geometric product of two vectors in a Euclidean plane, i.e., the multivector obtained by taking the geometric product of a vector with whatever unit basis axis one wants to designate as the real axis. In geometric algebra, a complex number may be viewed as a spinor operating in the shared plane of the given vector and unit vector.

The earlier discussion about multiplying vectors in two dimensions applies generally to multiplying two $2N$-dimensional vectors where the unit bivector lies in the plane defined by the two vectors. Moreover, as discussed, the outer product is defined in any number of dimensions. As a result, the geometric product defined earlier applies in any number of dimensions. Here, we have discussed its application in two dimensions to introduce the notion that the geometric product leads to algebraic manipulations—i.e., multiplication and division, as well as addition and subtraction—through the introduction of a new quantity called a bivector. More generally, the geometric product leads in arbitrary dimensions to algebraic manipulations operating on multivectors, which as we have seen are generalizations of vectors and bivectors. We do not explore these ideas further in this tutorial, but the references give an excellent introduction to these considerations, as well as many other fascinating results.

III. APPLICATION: A GEOMETRIC INTERPRETATION OF BASIC POWER CONCEPTS

Traditionally, in circuit theory, voltage and current are considered complex phasors that arise from considering sinusoidal signals. Complex phasors are introduced because they render many problems analytically tractable and thus prove convenient. However, the introduction of complex quantities into a problem involving real quantities can be confusing and can sometimes obscure a geometric understanding that may be helpful in learning new concepts. In this section, we revisit some basic electrical engineering concepts from a geometric perspective, but in doing so, we avoid the use of complex quantities.

To begin, consider a sinusoidal current signal:

$$i(t) = I\cos(\omega_i t + \theta_i) \tag{3.1}$$

From a simple trigonometric identity, we may write

$$i(t) = I[\cos\omega_i t\cos\theta_i - \sin\omega_i t\sin\theta_i] = \mathcal{I} \cdot \mathcal{B} \tag{3.2}$$

where we have introduced

$$\mathcal{I} = I\begin{bmatrix} \cos\omega_i t \\ \sin\omega_i t \end{bmatrix} = I\cos\omega_i t\hat{\boldsymbol{\varphi}}_1 + I\sin\omega_i t\hat{\boldsymbol{\varphi}}_2 \tag{3.3}$$

$$\mathcal{B}_i = \begin{bmatrix} \cos\theta_i \\ -\sin\theta_i \end{bmatrix} = \cos\theta_i\hat{\boldsymbol{\varphi}}_1 - \sin\theta_i\hat{\boldsymbol{\varphi}}_2 \tag{3.4}$$

$$\hat{\boldsymbol{\varphi}}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{3.5}$$

$$\hat{\boldsymbol{\varphi}}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{3.6}$$

Observe that $\hat{\boldsymbol{\varphi}}_1$ and $\hat{\boldsymbol{\varphi}}_2$ form an orthonormal set of vectors and therefore represent a set of basis vectors in a plane. Thus, we can represent the current $i(t)$ by a real vector $\mathcal{I}$ that rotates in a plane, as shown in Fig. 14. Given the value of the phase θ_i, we can recover the actual value of $i(t)$ from $\mathcal{I}$ through the dot product with $\mathcal{B}_i$.

Similarly, let us introduce a sinusoidal voltage:

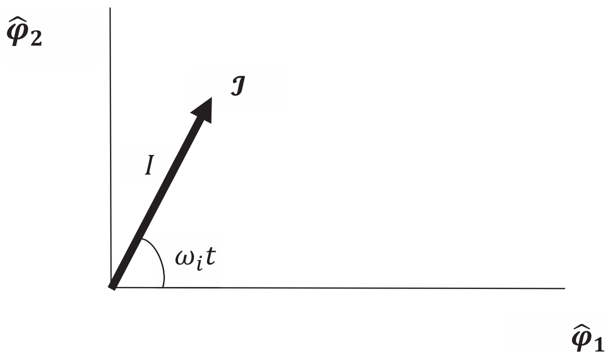


Fig. 14.
Vector representation of sinusoidal signal.

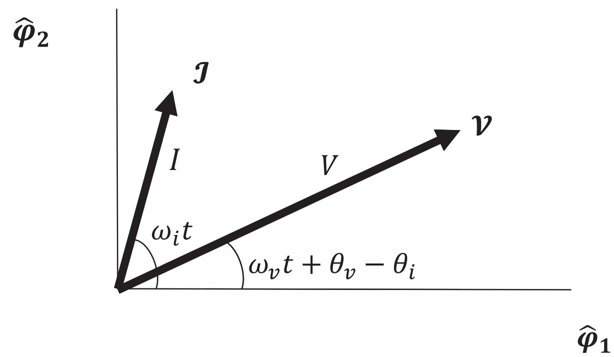


Fig. 15.
Rotating current and voltage vectors.

$$v(t) = V\cos(\omega_v t + \theta_v) \tag{3.7}$$

To formulate this quantity as a vector in the same basis as the current vector, let us rewrite it as

$$\begin{aligned} v(t) &= V\cos(\omega_v t + \theta_v - \theta_i + \theta_i) \\ &= V\left[\cos(\omega_v t + \theta_v - \theta_i)\cos\theta_i - \sin(\omega_v t + \theta_v - \theta_i)\sin\theta_i\right] \\ &= \mathcal{V}\cdot\mathcal{B}_i \end{aligned} \tag{3.8}$$

where $\mathcal{B}_i$ is as given earlier and

$$\begin{aligned} \mathcal{V} &= V\begin{bmatrix}\cos(\omega_v t + \theta_v - \theta_i) \\ \sin(\omega_v t + \theta_v - \theta_i)\end{bmatrix} \\ &= V\cos(\omega_v t + \theta_v - \theta_i)\hat{\varphi}_1 + V\sin(\omega_v t + \theta_v - \theta_i)\hat{\varphi}_2. \end{aligned} \tag{3.9}$$

We have the diagram in Fig. 15 for our two rotating vectors.

With the preceding formulation in mind, let us define voltage and current simply as vectors in the plane with basis vectors $\hat{\varphi}_1$ and $\hat{\varphi}_2$:

$$\mathcal{V} = v_1\hat{\varphi}_1 + v_2\hat{\varphi}_2; \quad \mathcal{I} = i_1\hat{\varphi}_1 + i_2\hat{\varphi}_2 \tag{3.10}$$

Examine the geometric product of these two vectors:

$$\mathcal{VI} = \mathcal{V}\cdot\mathcal{I} + \mathcal{V}\wedge\mathcal{I} = v_1 i_1 + v_2 i_2 + (v_1 i_2 - v_2 i_1)\hat{\varphi}_1\hat{\varphi}_2 \tag{3.11}$$

Writing $v_1 = V\cos(\omega_v t + \theta_v - \theta_i)$, etc., from earlier yields

$$\begin{aligned} \mathcal{V}\cdot\mathcal{I} &= v_1 i_1 + v_2 i_2 \\ &= VI\cos(\omega_v t + \theta_v - \theta_i)\cos\omega_i t + VI\sin(\omega_v t + \theta_v - \theta_i)\sin\omega_i t \\ &= VI\cos((\omega_v - \omega_i)t + \theta_v - \theta_i). \end{aligned} \tag{3.12}$$

This quantity is called the active power. It is given by a scalar.

We also have

$$\begin{aligned} v_1 i_2 - v_2 i_1 &= VI\cos(\omega_v t + \theta_v - \theta_i)\sin\omega_i t \\ &\quad - VI\sin(\omega_v t + \theta_v - \theta_i)\cos\omega_i t \\ &= VI\sin((\omega_v - \omega_i)t + \theta_v - \theta_i). \end{aligned} \tag{3.13}$$

This quantity is called the reactive power. As we see, the reactive power can be meaningfully defined by a reactive power bivector with the geometric interpretation of an oriented area in the plane shared by the current and voltage vectors.

The entire quantity $\mathcal{P} = \mathcal{VI}$ may be deemed the apparent power spinor:

$$\begin{aligned} \mathcal{P} &= \mathcal{V}\cdot\mathcal{I} + \mathcal{V}\wedge\mathcal{I} \\ &= VI\cos((\omega_v - \omega_i)t + \theta_v - \theta_i) \\ &\quad + VI\sin((\omega_v - \omega_i)t + \theta_v - \theta_i)\hat{\varphi}_1\hat{\varphi}_2 \\ &= VIe^{\hat{\varphi}_1\hat{\varphi}_2((\omega_v - \omega_i)t + \theta_v - \theta_i)} \end{aligned} \tag{3.14}$$

There was no need to introduce the notion of the complex conjugate of the current phasor to define the apparent power.

Generally, in simple circuits $\omega_v = \omega_i$, in which case the geometric formulation reduces to

$$\mathcal{V}\cdot\mathcal{I} = VI\cos(\theta_v - \theta_i) \tag{3.15}$$

$$\mathcal{V}\wedge\mathcal{I} = VI\sin(\theta_v - \theta_i)\hat{\varphi}_1\hat{\varphi}_2 \tag{3.16}$$

$$\mathcal{P} = VIe^{\hat{\varphi}_1\hat{\varphi}_2(\theta_v - \theta_i)}. \tag{3.17}$$

It is immediately clear from this approach that active power and reactive power are fundamentally different concepts represented by two different geometric objects—active power is given by an unoriented scalar, whereas reactive power is represented by an oriented area. Recognizing that these two concepts describe fundamentally different geometric objects may facilitate understanding for beginning students. Furthermore (as expected), it is clear that their actual values depend on the phase difference between the voltage and the current signals, not on the values of the phases themselves. The geometric approach taken here encompasses the same results as the complex phasor approach but embodies a further geometric distinction between active and reactive power and thus potentially conveys more information than the traditional approach.

To formulate these same concepts in terms of impedance, let us define

$$\mathcal{Z} \triangleq \mathcal{VI}^{-1}. \tag{3.18}$$

As we discussed earlier, the inverse of a nonzero vector is a vector. Thus, we see that $\mathcal{Z}$ is given by the geometric product of two vectors and hence is a spinor. We call $\mathcal{Z}$ the impedance spinor. From our definition of the inverse of a vector, we obtain

$$\mathcal{Z} = \frac{\mathcal{V}\mathcal{I}}{|\mathcal{I}|^2} = \frac{\mathcal{P}}{I^2}. \tag{3.19}$$

From this, we see the apparent power spinor is a scaled version of the impedance spinor.

Using our earlier results, we find

$$\begin{aligned}\mathcal{Z} &= \frac{V}{I} e^{\hat{\varphi}_1\hat{\varphi}_2((\omega_v-\omega_i)t+\theta_v-\theta_i)} \\ &= \frac{V}{I}\cos\left((\omega_v-\omega_i)t+\theta_v-\theta_i\right) \\ &\quad+\frac{V}{I}\sin\left((\omega_v-\omega_i)t+\theta_v-\theta_i\right)\hat{\varphi}_1\hat{\varphi}_2.\end{aligned} \tag{3.20}$$

When $\omega_v = \omega_i$, this formulation reduces to

$$\mathcal{Z} = \frac{V}{I}\cos(\theta_v-\theta_i) + \frac{V}{I}\sin(\theta_v-\theta_i)\hat{\varphi}_1\hat{\varphi}_2. \tag{3.21}$$

The quantity $\frac{V}{I}\cos(\theta_v-\theta_i)$ is a scalar called the resistance. We call the quantity $\frac{V}{I}\sin(\theta_v-\theta_i)\hat{\varphi}_1\hat{\varphi}_2$ the reactance bivector. We see that the reactance $\frac{V}{I}\sin(\theta_v-\theta_i)$ is simply the area of the reactance bivector; it is not imaginary.

Let us consider the reverse of $\mathcal{Z}$, obtained by reversing the order of the vectors appearing in the product that defines $\mathcal{Z}$:

$$\mathcal{Z}^\dagger = \mathcal{I}^{-1}\mathcal{V} \tag{3.22}$$

From this relation, we find

$$\begin{aligned}\mathcal{Z}^\dagger\mathcal{Z} &= \mathcal{I}^{-1}\mathcal{V}\mathcal{V}\mathcal{I}^{-1} = \frac{\mathcal{I}\mathcal{V}\mathcal{V}\mathcal{I}}{|\mathcal{I}|^4} = \frac{|\mathcal{V}|^2}{|\mathcal{I}|^2} \\ &= \frac{\mathcal{V}\mathcal{I}\mathcal{I}\mathcal{V}}{|\mathcal{I}|^4} = \mathcal{V}\mathcal{I}^{-1}\mathcal{I}^{-1}\mathcal{V} = \mathcal{Z}\mathcal{Z}^\dagger.\end{aligned} \tag{3.23}$$

Hence, we may define

$$\mathcal{Z}^{-1} \triangleq \frac{|\mathcal{I}|^2}{|\mathcal{V}|^2}\mathcal{Z}^\dagger. \tag{3.24}$$

This quantity is also a spinor, which we call the admittance spinor. The relations among voltage, current, and impedance are therefore given by

$$\mathcal{V} = \mathcal{Z}\mathcal{I} \tag{3.25}$$

$$\mathcal{I} = \mathcal{Z}^{-1}\mathcal{V}. \tag{3.26}$$

These relations show that the impedance spinor rotates and dilates the current vector into the voltage vector, and the admittance spinor rotates and dilates the voltage vector back into the current vector. This approach makes clear geometric distinctions between vectors—i.e., the current and the voltage—and spinors—i.e., the impedance and the admittance. These geometric distinctions are lost when all quantities are represented by complex numbers, as is traditionally done.

It is of some historical interest to observe that as early as 1893, MacFarlane proposed the use of plane algebra to address problems involving alternating currents:

> In plane algebra, the quantities considered can be represented by vectors lying in a common plane, and the relation considered is that which may exist between one vector and another. Let ***A*** and ***R*** be two such vectors lying in the plane of the paper; the axis of this plane is constant and may be denoted by *a*. In the more general algebra of space, *a* varies, hence must always be stated explicitly. The ratio of ***R*** to ***A*** consists of a change of length which may be denoted by *r*, and a change of direction which may be denoted by a^θ, that is, a turning of θ radians around the axis *a*. Thus $\boldsymbol{R} = ra^\theta\boldsymbol{A}$ and reciprocally $\boldsymbol{A} = \frac{1}{r}a^{-\theta}\boldsymbol{R}$. The ratio or rotator raθ is a directed quantity but it is not a vector.… Each of the vectors in the plane may be specified by such a rotator, a given initial vector being understood. The confusion of such rotator with the rotated line has been the principal cause of the obscurity which has clouded the algebra of complex numbers. (MacFarlane, 1893)

This is a remarkable passage—with the substitution of "geometric algebra in a Euclidean plane" for "plane algebra" and "spinor" for "rotator," this passage is consistent with geometric algebra and the material we have discussed herein. MacFarlane distinguishes between vectors and rotators and teaches that one vector is obtained from another by the operation of a rotator, which changes both the length and the angle of the vector on which it operates. Moreover, he insists that the rotator and the "rotated line"—i.e., the vector—must not be confused. Hence, this passage can be interpreted as a suggestion that geometric algebra, i.e., "plane algebra," be adopted as the proper mathematical basis for analyzing alternating currents. MacFarlane also refers to the "more general algebra of space," which appears to be a reference to geometric algebra in R^3. Historically it is quite intriguing that the next paper presented at the conference at which MacFarlane presented the quoted paper was Steinmetz (1893). The work of Steinmetz ultimately led to the adoption of the complex phasor approach that is ubiquitous in electrical engineering. We are left to wonder how electrical engineering would have developed if Clifford's algebra, set forth in 1878, had been more widely known and MacFarlane's use of plane algebra had been adopted in electrical engineering.

These elementary considerations have been intended only to address the simplest ideas arising from representing power and impedance in the context of geometric algebra. They have been presented as a reminder that a sinusoidal signal naturally can be represented by a real vector in a plane and that complex phasors are used primarily because of their analytic convenience. However, their use is not necessary and can foster confusion by discarding fundamental geometric differences in quantities of interest. Use of real vectors in the context of geometric algebra may lead to a better, more intuitive understanding of such basic concepts as power and impedance. Several researchers have applied the ideas of geometric algebra to such power considerations; see Willems (1996), Menti et al. (2007), Castilla and Bravo (2009), Lev-Ari and Stankovic (2009), and Petroianu (2014) for further investigations along these lines.

IV. APPLICATION: A GEOMETRIC INTERPRETATION OF THE DFT

In this section, we use the concepts of geometric algebra to explore a geometric interpretation of the DFT. Because the DFT is most generally formulated in terms of complex quantities, we begin by exploring the notion of a complex vector.

A. GEOMETRIC INTERPRETATION OF A COMPLEX DATA VECTOR AS A SPINOR EXPANSION

If $\boldsymbol{r}$ is a real vector in R^N and $\{\hat{\boldsymbol{e}}_{\boldsymbol{k}}, k = 1, \cdots, N\}$ is an orthonormal basis in R^N, then as is well known, we may write

$$\boldsymbol{r} = \sum_{k=1}^{N} r_k \hat{\boldsymbol{e}}_{\boldsymbol{k}} \tag{4.1}$$

where

$$r_k = \boldsymbol{r} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{k}}. \tag{4.2}$$

This representation of the vector $\boldsymbol{r}$ encompasses well-known geometric interpretations for both r_k and $r_k\hat{\boldsymbol{e}}_{\boldsymbol{k}}$. In particular, r_k is a scalar representing the magnitude and sign of the vector obtained by projecting $\boldsymbol{r}$ into the subspace spanned by $\hat{\boldsymbol{e}}_{\boldsymbol{k}}$, and $r_k\hat{\boldsymbol{e}}_{\boldsymbol{k}}$ is the projected vector so obtained.

We now develop an analogous expansion in terms of two-dimensional subspaces. We begin with an arbitrary real vector $\boldsymbol{d}$ in R^{2N} and assume an orthonormal basis $\{\hat{\boldsymbol{e}}_{\boldsymbol{l}}, l = 1, \cdots, 2N\}$ for this space. We may then write

$$\boldsymbol{d} = \sum_{k=1}^{N} \boldsymbol{d}_k \tag{4.3}$$

with

$$\boldsymbol{d} = d_{1,k}\hat{\boldsymbol{e}}_{\boldsymbol{k}} + d_{2,k}\hat{\boldsymbol{e}}_{\boldsymbol{k+N}} \tag{4.4}$$

$$d_{1,k} = \boldsymbol{d} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{k}} \tag{4.5}$$

$$d_{2,k} = \boldsymbol{d} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{k+N}}. \tag{4.6}$$

This is simply a straightforward expansion of $\boldsymbol{d}$ by projecting it into N orthogonal two-dimensional subspaces. We use the geometric product to define

$$c_k = \boldsymbol{d}_k \hat{\boldsymbol{e}}_{\boldsymbol{k}} = d_{1,k} + i_k d_{2,k} \tag{4.7}$$

where

$$i_k = \hat{\boldsymbol{e}}_{\boldsymbol{k+N}} \hat{\boldsymbol{e}}_{\boldsymbol{k}} \tag{4.8}$$

is a unit bivector in the $\hat{\boldsymbol{e}}_{\boldsymbol{k+N}}, \hat{\boldsymbol{e}}_{\boldsymbol{k}}$ plane and we have used $\hat{\boldsymbol{e}}_{\boldsymbol{k}}\hat{\boldsymbol{e}}_{\boldsymbol{k}} = 1$. The quantity c_k is a spinor and, as discussed earlier, may be thought of as being analogous to a complex number. However, it is restricted to the $\hat{\boldsymbol{e}}_{\boldsymbol{k+N}}, \hat{\boldsymbol{e}}_{\boldsymbol{k}}$ plane, where the bivector i_k replaces $i = \sqrt{-1}$. With this definition, we easily find that by multiplying both sides of (4.7) by $\hat{\boldsymbol{e}}_{\boldsymbol{k}}$,

$$\boldsymbol{d}_k = c_k \hat{\boldsymbol{e}}_{\boldsymbol{k}}. \tag{4.9}$$

In one sense, the relationship in (4.9) is remarkable. It multiplies a real vector by a complex number to obtain a real vector. This is generally not done in electrical engineering.

From this result, we see that the spinor c_k may be viewed as an operator that transforms $\hat{\boldsymbol{e}}_{\boldsymbol{k}}$ into $\boldsymbol{d}_k$. We may now use (4.9) in (4.3) to write

$$\boldsymbol{d} = \sum_{k=1}^{N} c_k \hat{\boldsymbol{e}}_{\boldsymbol{k}}. \tag{4.10}$$

We may also use the techniques of geometric algebra (see, for example, Doran and Lasenby, 2003) to find

$$c_k = -(\boldsymbol{d} \cdot i_k)\hat{\boldsymbol{e}}_{\boldsymbol{k+N}}. \tag{4.11}$$

The right-hand side of (4.10) represents the $2N$-dimensional vector $\boldsymbol{d}$ as an N-dimensional complex vector with a definite geometric interpretation. To explore this assertion, first

$$|\boldsymbol{d}|^2 = \sum_{k=1}^{N} |\boldsymbol{d}_k|^2 \tag{4.12}$$

and then

$$|\boldsymbol{d}_k|^2 = d_{1,k}^2 + d_{2,k}^2 = |c_k|^2. \tag{4.13}$$

Thus,

$$|\boldsymbol{d}|^2 = \sum_{k=1}^{N} |c_k|^2. \tag{4.14}$$

This relation holds for any choice of the orthonormal basis $\{\hat{\boldsymbol{e}}_{\boldsymbol{l}}, l = 1, \cdots, 2N\}$. In addition, we may show in a straightforward manner

$$|\boldsymbol{d}_k| = |\boldsymbol{d}| \cos\psi_k \tag{4.15}$$

for some angle ψ_k. Finally, using $i_k^2 = -1$, we may rewrite c_k as

$$c_k = |c_k| e^{i_k\theta_k} = |\boldsymbol{d}| \cos\psi_k e^{i_k\theta_k}. \tag{4.16}$$

Thus, we have the diagram in Fig. 16.

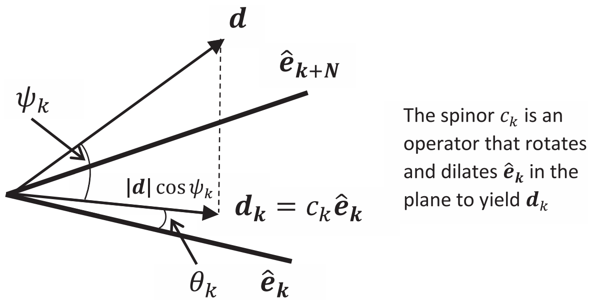


Fig. 16.
Spinor interpretation.

Because c_k, $k = 1,\cdots,N$ may be interpreted as complex numbers in their respective two-dimensional subspaces (i.e., planes), we see that the complex vector expansion of the vector $\boldsymbol{d}$ in (4.3) and (4.9)–(4.11) is analogous to the real vector expansion of the vector $\boldsymbol{r}$ in (4.1) and (4.2). First, (4.1) projects $\boldsymbol{r}$ into N one-dimensional subspaces represented by the unit vectors $\hat{\boldsymbol{e}}_{\boldsymbol{k}}$, whereas (4.3) projects $\boldsymbol{d}$ into N two-dimensional subspaces represented by the unit bivectors $\hat{\boldsymbol{e}}_{\boldsymbol{k+N}}\,\hat{\boldsymbol{e}}_{\boldsymbol{k}}$. In addition, the inner product of $\boldsymbol{r}$ with the unit vector $\hat{\boldsymbol{e}}_{\boldsymbol{k}}$ leads to the scalar r_k as in (4.2), whereas the inner product of $\boldsymbol{d}$ with the unit bivector $\hat{\boldsymbol{e}}_{\boldsymbol{k+N}}\,\hat{\boldsymbol{e}}_{\boldsymbol{k}}$ leads to the spinor c_k as in (4.11). Finally, the operation of the scalar r_k on the unit vector $\hat{\boldsymbol{e}}_{\boldsymbol{k}}$ yields the projected vector r_k, whereas the operation of the spinor c_k on the unit vector $\hat{\boldsymbol{e}}_{\boldsymbol{k}}$ yields the projected vector $\boldsymbol{d}_k$. Thus, the expansion of $\boldsymbol{d}$ in (4.10) is analogous to the expansion of $\boldsymbol{r}$ in (4.1) but encompasses complex numbers as represented in (4.7).

To further understand the earlier spinor expansion, we consider a specific orthonormal basis. We reserve the notation $\hat{\boldsymbol{e}}_{\boldsymbol{l}}$ and c_l for general unspecified bases and their corresponding spinors and assign different notations for specific choices of the basis and spinor. In particular, let

$$\hat{\boldsymbol{\varphi}}_l = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \qquad l = 1,\cdots,2N \tag{4.17}$$

where the nonzero entry is in the lth position. We call this the observed data basis. It is clearly orthonormal. If we have a sequence of N observed data samples, for example, in-phase and quadrature samples $\mathcal{I}_k, \mathcal{Q}_k\; k = 1,\cdots,N$, then in the observed data basis, we set

$$\boldsymbol{d} = \begin{bmatrix} \mathcal{I}_1 \\ \vdots \\ \mathcal{I}_N \\ \mathcal{Q}_1 \\ \vdots \\ \mathcal{Q}_N \end{bmatrix} \tag{4.18}$$

and find

$$f_k = \mathcal{I}_k + i_{k,\varphi}\mathcal{Q}_k, \qquad k = 1,\cdots,N \tag{4.19}$$

with $i_{k,\varphi} = \hat{\boldsymbol{\varphi}}_{k+N}\hat{\boldsymbol{\varphi}}_k$. Our expansion becomes

$$\boldsymbol{d} = \sum_{k=1}^{N} f_k \hat{\boldsymbol{\varphi}}_k. \tag{4.20}$$

We may interpret the right-hand side of (4.20) as giving rise to the complex vector

$$\boldsymbol{f} = \begin{bmatrix} \mathcal{I}_1 + i_{1,\varphi}\mathcal{Q}_1 \\ \vdots \\ \mathcal{I}_N + i_{N,\varphi}\mathcal{Q}_N \end{bmatrix}. \tag{4.21}$$

In this interpretation, each observed complex sample $f_k = \mathcal{I}_k + i_{k,\varphi}\mathcal{Q}_k$ is a spinor that operates in its respective $\hat{\boldsymbol{\varphi}}_{k+N}, \hat{\boldsymbol{\varphi}}_k$ plane to transform the unit vector $\hat{\boldsymbol{\varphi}}_{\boldsymbol{k}}$ into the real vector

$$\boldsymbol{d}_k = Re(f_k)\hat{\boldsymbol{\varphi}}_k + Im(f_k)\hat{\boldsymbol{\varphi}}_{k+N} = \mathcal{I}_k\hat{\boldsymbol{\varphi}}_k + \mathcal{Q}_k\hat{\boldsymbol{\varphi}}_{k+N} \tag{4.22}$$

which also lies in the $\hat{\boldsymbol{\varphi}}_{k+N}, \hat{\boldsymbol{\varphi}}_k$ plane. Thus, we may interpret $\mathcal{I}$ and $\mathcal{Q}$ data, which traditionally are represented as complex vectors, as either a $2N$-dimensional real vector or a collection of N operators (spinors) that operate on unit vectors in N two-dimensional subspaces of a $2N$-dimensional space.

B. A GENERAL DISCRETE TRANSFORM

We begin by assuming a sinusoidal signal

$$s(t) = A\cos\big((\omega_0 + \omega_\Delta)t + \theta\big). \tag{4.23}$$

We take the nominal frequency ω_0 as known and treat ω_Δ as being offset from this known nominal frequency. Using simple sum and difference identities from trigonometry, we may rewrite this signal as

$$s(t) = \mathcal{I}_{\Delta,\theta}(t)\cos\omega_0 t - \mathcal{Q}_{\Delta,\theta}(t)\sin\omega_0 t \tag{4.24}$$

where

$$\mathcal{I}_{\Delta,\theta}(t) = A\big[\cos\theta\cos\omega_\Delta t - \sin\theta\sin\omega_\Delta t\big] \tag{4.25}$$

$$\mathcal{Q}_{\Delta,\theta}(t) = A\big[\cos\theta\sin\omega_\Delta t + \sin\theta\cos\omega_\Delta t\big]. \tag{4.26}$$

Because we have assumed ω_0 is known, it follows from this representation that if we know $\mathcal{I}_{\Delta,\theta}(t)$ and $\mathcal{Q}_{\Delta,\theta}(t)$ then we also know $s(t)$. Thus, we may view the pair $\big(\mathcal{I}_{\Delta,\theta}(t), \mathcal{Q}_{\Delta,\theta}(t)\big)$ as a representation of the signal $s(t)$. As is well known, in many applications, it is more convenient to work with the pair $\big(\mathcal{I}_{\Delta,\theta}(t), \mathcal{Q}_{\Delta,\theta}(t)\big)$ than directly with the signal $s(t)$.

Let us sample the signal at times $t = t_i$ to obtain two sets of samples,

$$\begin{aligned}\mathcal{I}_{\Delta,\theta,i} &= \mathcal{I}_{\Delta,\theta}(t_i) \\ \mathcal{Q}_{\Delta,\theta,i} &= \mathcal{Q}_{\Delta,\theta}(t_i)\end{aligned} \qquad i = 1,\cdots,N. \tag{4.27}$$

We may assume $t_1 = 0$; otherwise, we may absorb the value of $(\omega_0 + \omega_\Delta)t_1$ into the value of the phase offset θ and then measure all subsequent times from the time t_1. Examine the following vectors:

$$\boldsymbol{d}_{\Delta,\theta} = \begin{bmatrix} \mathcal{I}_{\Delta,\theta,1} \\ \vdots \\ \mathcal{I}_{\Delta,\theta,N} \\ \mathcal{Q}_{\Delta,\theta,1} \\ \vdots \\ \mathcal{Q}_{\Delta,\theta,N} \end{bmatrix} \tag{4.28}$$

$$\boldsymbol{b}_\theta = \begin{bmatrix} \cos\omega_0 t_1 \\ \vdots \\ \cos\omega_0 t_N \\ -\sin\omega_0 t_1 \\ \vdots \\ -\sin\omega_0 t_N \end{bmatrix} \tag{4.29}$$

With these definitions, we may recover the signal sample $s(t_i)$ as

$$s(t_i) = d_{\Delta,\theta,i} b_{0,i} + d_{\Delta,\theta,i+N} b_{0,i+N}. \tag{4.30}$$

In this way, we see it is meaningful to represent the set of N signal samples $s(t_i)$, $i = 1,\cdots,N$, by a $2N$-dimensional real vector $\boldsymbol{d}_{\Delta,\theta}$, where each frequency offset ω_Δ and each phase offset θ gives rise to a different vector $\boldsymbol{d}_{\Delta,\theta}$.

Let us also define

$$\hat{\boldsymbol{p}}_\Delta = \frac{1}{\sqrt{N}} \begin{bmatrix} \cos\omega_\Delta t_1 \\ \vdots \\ \cos\omega_\Delta t_N \\ \sin\omega_\Delta t_1 \\ \vdots \\ \sin\omega_\Delta t_N \end{bmatrix}; \qquad \hat{\boldsymbol{p}}'_\Delta = \frac{1}{\sqrt{N}} \begin{bmatrix} -\sin\omega_\Delta t_N \\ \vdots \\ -\sin\omega_\Delta t_N \\ \cos\omega_\Delta t_1 \\ \vdots \\ \cos\omega_\Delta t_N \end{bmatrix}. \tag{4.31}$$

One may show that these two $2N$-dimensional vectors are orthonormal. Therefore, we may write

$$\boldsymbol{d}_{\Delta,\theta} = A\cos\theta\,\hat{\boldsymbol{p}}_\Delta + A\sin\theta\,\hat{\boldsymbol{p}}'_\Delta. \tag{4.32}$$

Thus, we have the diagram in Fig. 17.

The unit vectors $\hat{\boldsymbol{p}}_\Delta$ and $\hat{\boldsymbol{p}}'_\Delta$ define a two-dimensional subspace, i.e., a plane, of the larger $2N$-dimensional space, and the vector $\boldsymbol{d}_{\Delta,\theta}$ resides in this subspace. The amplitude A of the signal samples is embodied in the magnitude of the vector $\boldsymbol{d}_{\Delta,\theta}$, the phase offset θ is embodied in the amount by which $\boldsymbol{d}_{\Delta,\theta}$ is rotated from the vector $\hat{\boldsymbol{p}}_\Delta$, and the frequency offset is embodied in the unit vectors $\hat{\boldsymbol{p}}_\Delta$ and $\hat{\boldsymbol{p}}'_\Delta$. Hence, each frequency offset ω_Δ gives rise to a different two-dimensional subspace.

Motivated by these observations, we represent a more general sampled signal $r(t_i)$, $i=1,\cdots,N$, by the $2N$-dimensional real vector

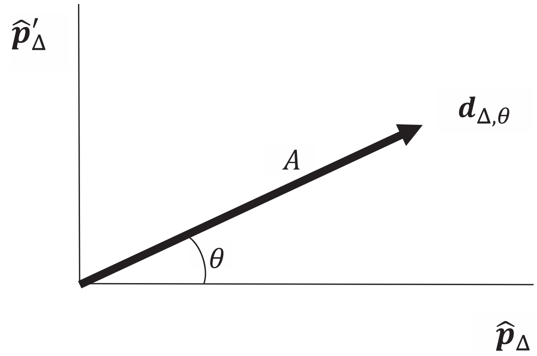


Fig. 17.
Vector representation of N signal samples.

$$\boldsymbol{d} = \begin{bmatrix} \mathcal{I}_1 \\ \vdots \\ \mathcal{I}_N \\ \mathcal{Q}_1 \\ \vdots \\ \mathcal{Q}_N \end{bmatrix}. \tag{4.33}$$

As should be clear, if the received signal $r(t)$ is a sinusoidal signal of the form $s(t) = A\cos\left((\omega_0 + \omega_\Delta)t + \theta\right)$, then the vector $\boldsymbol{d} = \boldsymbol{d}_{\Delta,\theta}$ will lie in the two-dimensional planar subspace specified by the unit vectors $\hat{\boldsymbol{p}}_\Delta$ and $\hat{\boldsymbol{p}}'_\Delta$, which in turn depend on the offset frequency ω_Δ. If the received signal comprises this sinusoidal signal $s(t)$ plus noise, then the effect of the noise will be twofold. It will rotate and dilate the vector $\boldsymbol{d}_{\Delta,\theta}$ within the plane, and it will cause the vector $\boldsymbol{d}_{\Delta,\theta}$ to potentially move out of the plane. Thus, in the absence of noise, the vector $\boldsymbol{d}_{\Delta,\theta}$ will lie in a subspace determined by the offset frequency ω_Δ and will rotate within this specific subspace as θ varies. Deviation outside of this subspace is caused by noise.

Let us assume two arbitrary orthonormal bases given by $\{\hat{\boldsymbol{p}}_k, \hat{\boldsymbol{p}}_{k+N}, k = 1,\cdots,N\}$ and $\{\hat{\boldsymbol{\varphi}}_k, \hat{\boldsymbol{\varphi}}_{k+N}, k = 1,\cdots,N\}$ (not necessarily the bases defined earlier). Generally, for the vector $\boldsymbol{d}$, we may write

$$F_k = (\boldsymbol{d}\cdot\hat{\boldsymbol{p}}_k) + i_{k,p}(\boldsymbol{d}\cdot\hat{\boldsymbol{p}}_{k+N}) \tag{4.34}$$

$$f_k = (\boldsymbol{d}\cdot\hat{\boldsymbol{\varphi}}_k) + i_{k,\varphi}(\boldsymbol{d}\cdot\hat{\boldsymbol{\varphi}}_{k+N}) \tag{4.35}$$

where

$$i_{k,p} = \hat{\boldsymbol{p}}_{k+N}\hat{\boldsymbol{p}}_k \tag{4.36}$$

$$i_{k,\varphi} = \hat{\boldsymbol{\varphi}}_{k+N}\hat{\boldsymbol{\varphi}}_k. \tag{4.37}$$

The quantities F_k and f_k are the spinors (4.7) associated with expansions of $\boldsymbol{d}$ in the two different bases. Let us expand $\boldsymbol{d}$ in the $\hat{\boldsymbol{\varphi}}_k$ basis, whose spinors are f_k, $k=1,\cdots,N$:

$$\boldsymbol{d} = \sum_{i=1}^{N} \boldsymbol{d}_i \tag{4.38}$$

$$\boldsymbol{d}_i = \left(\boldsymbol{d}\cdot\hat{\boldsymbol{\varphi}}_i\right)\hat{\boldsymbol{\varphi}}_i + \left(\boldsymbol{d}\cdot\hat{\boldsymbol{\varphi}}_{i+N}\right)\hat{\boldsymbol{\varphi}}_{i+N} \tag{4.39}$$

If we substitute this expansion for $\boldsymbol{d}$ in the spinor expression for F_k in the other $\hat{\boldsymbol{p}}_k$ basis, we obtain the following:

$$\begin{aligned} F_k &= \sum_{i=1}^{N}\left\{\left(\boldsymbol{d}_i\cdot\hat{\boldsymbol{p}}_k\right) + i_{k,p}\left(\boldsymbol{d}_i\cdot\hat{\boldsymbol{p}}_{k+N}\right)\right\} \\ &= \sum_{i=1}^{N}\left\{\left(\boldsymbol{d}\cdot\hat{\boldsymbol{\varphi}}_i\right)\left(\hat{\boldsymbol{\varphi}}_i\cdot\hat{\boldsymbol{p}}_k\right) + \left(\boldsymbol{d}\cdot\hat{\boldsymbol{\varphi}}_{i+N}\right)\left(\hat{\boldsymbol{\varphi}}_{i+N}\cdot\hat{\boldsymbol{p}}_k\right)\right\} \\ &\quad + i_{k,p}\sum_{i=1}^{N}\left\{\left(\boldsymbol{d}\cdot\hat{\boldsymbol{\varphi}}_i\right)\left(\hat{\boldsymbol{\varphi}}_i\cdot\hat{\boldsymbol{p}}_{k+N}\right) + \left(\boldsymbol{d}\cdot\hat{\boldsymbol{\varphi}}_{i+N}\right)\left(\hat{\boldsymbol{\varphi}}_{i+N}\cdot\hat{\boldsymbol{p}}_{k+N}\right)\right\} \end{aligned} \tag{4.40}$$

Let us write in the suggestive notation:

$$Re\left(f_i\right) = \left(\boldsymbol{d}\cdot\hat{\boldsymbol{\varphi}}_i\right) \tag{4.41}$$

$$Im\left(f_i\right) = \left(\boldsymbol{d}\cdot\hat{\boldsymbol{\varphi}}_{i+N}\right) \tag{4.42}$$

$$Re\left(F_i\right) = \left(\boldsymbol{d}\cdot\hat{\boldsymbol{p}}_i\right) \tag{4.43}$$

$$Im\left(F_i\right) = \left(\boldsymbol{d}\cdot\hat{\boldsymbol{p}}_{i+N}\right) \tag{4.44}$$

Then, we may write

$$\begin{aligned} F_k &= \sum_{i=1}^{N}\Big\{Re\left(f_i\right)\left[\hat{\boldsymbol{\varphi}}_i\cdot\hat{\boldsymbol{p}}_k + i_{k,p}\left(\hat{\boldsymbol{\varphi}}_i\cdot\hat{\boldsymbol{p}}_{k+N}\right)\right] \\ &\quad + Im\left(f_i\right)\left[\hat{\boldsymbol{\varphi}}_{i+N}\cdot\hat{\boldsymbol{p}}_k + i_{k,p}\left(\hat{\boldsymbol{\varphi}}_{i+N}\cdot\hat{\boldsymbol{p}}_{k+N}\right)\right]\Big\} \end{aligned} \tag{4.45}$$

$$\begin{aligned} f_k &= \sum_{i=1}^{N}\Big\{Re\left(F_i\right)\left[\hat{\boldsymbol{p}}_i\cdot\hat{\boldsymbol{\varphi}}_k + i_{k,\varphi}\left(\hat{\boldsymbol{p}}_i\cdot\hat{\boldsymbol{\varphi}}_{k+N}\right)\right] \\ &\quad + Im\left(F_i\right)\left[\hat{\boldsymbol{p}}_{i+N}\cdot\hat{\boldsymbol{\varphi}}_k + i_{k,\varphi}\left(\hat{\boldsymbol{p}}_{i+N}\cdot\hat{\boldsymbol{\varphi}}_{k+N}\right)\right]\Big\}. \end{aligned} \tag{4.46}$$

We obtained the second result by simply reversing the roles of the bases and spinors in the preceding development. These two equations form a general discrete transform pair between spinors associated with different orthonormal bases for a given vector $\boldsymbol{d}$ in the $2N$-dimensional space.

C. THE DFT

As an example, let $\left\{\hat{\boldsymbol{\varphi}}_k, \hat{\boldsymbol{\varphi}}_{k+N},\ k = 1,\cdots,N\right\}$ be the observed data basis discussed earlier—where f_k, $k = 1,\cdots, N$, comprise the $\mathcal{I}$ and $\mathcal{Q}$ data samples—and let $\left\{\hat{\boldsymbol{p}}_k, \hat{\boldsymbol{p}}_{k+N},\ k = 1,\cdots,N\right\}$ be the Fourier basis given by the following:

$$\hat{\boldsymbol{p}}_k = \frac{1}{\sqrt{N}}\begin{bmatrix} \cos\left(2\pi\left(k-1\right)\frac{0}{N}\right) \\ \vdots \\ \cos\left(2\pi\left(k-1\right)\frac{N-1}{N}\right) \\ \sin\left(2\pi\left(k-1\right)\frac{0}{N}\right) \\ \vdots \\ \sin\left(2\pi\left(k-1\right)\frac{N-1}{N}\right) \end{bmatrix} \tag{4.47}$$

$$\hat{\boldsymbol{p}}_{k+N} = \frac{1}{\sqrt{N}}\begin{bmatrix} -\sin\left(2\pi\left(k-1\right)\frac{0}{N}\right) \\ \vdots \\ -\sin\left(2\pi\left(k-1\right)\frac{N-1}{N}\right) \\ \cos\left(2\pi\left(k-1\right)\frac{0}{N}\right) \\ \vdots \\ \cos\left(2\pi\left(k-1\right)\frac{N-1}{N}\right) \end{bmatrix} \qquad k = 1,\cdots,N \tag{4.48}$$

Then, we find

$$\hat{\boldsymbol{\varphi}}_i\cdot\hat{\boldsymbol{p}}_k = \frac{1}{\sqrt{N}}\cos\left(2\pi\frac{\left(i-1\right)\left(k-1\right)}{N}\right) = \hat{\boldsymbol{p}}_i\cdot\hat{\boldsymbol{\varphi}}_k \tag{4.49}$$

$$\hat{\boldsymbol{\varphi}}_{i+N}\cdot\hat{\boldsymbol{p}}_k = \frac{1}{\sqrt{N}}\sin\left(2\pi\frac{\left(i-1\right)\left(k-1\right)}{N}\right) = \hat{\boldsymbol{p}}_i\cdot\hat{\boldsymbol{\varphi}}_{k+N} \tag{4.50}$$

$$\hat{\boldsymbol{\varphi}}_i\cdot\hat{\boldsymbol{p}}_{k+N} = -\frac{1}{\sqrt{N}}\sin\left(2\pi\frac{\left(i-1\right)\left(k-1\right)}{N}\right) = \hat{\boldsymbol{p}}_{i+N}\cdot\hat{\boldsymbol{\varphi}}_k \tag{4.51}$$

$$\hat{\boldsymbol{\varphi}}_{i+N}\cdot\hat{\boldsymbol{p}}_{k+N} = \frac{1}{\sqrt{N}}\cos\left(2\pi\frac{\left(i-1\right)\left(k-1\right)}{N}\right) = \hat{\boldsymbol{p}}_{i+N}\cdot\hat{\boldsymbol{\varphi}}_{k+N}. \tag{4.52}$$

In these particular bases, we find

$$\begin{aligned} F_k &= \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\left\{Re\left(f_i\right)\cos\left(2\pi\frac{\left(i-1\right)\left(k-1\right)}{N}\right)\right. \\ &\quad \left. + Im\left(f_i\right)\sin\left(2\pi\frac{\left(i-1\right)\left(k-1\right)}{N}\right)\right\} \\ &\quad + i_{k,p}\frac{1}{\sqrt{N}}\sum_{i=1}^{N}\left\{Im\left(f_i\right)\cos\left(2\pi\frac{\left(i-1\right)\left(k-1\right)}{N}\right)\right. \\ &\quad \left. - Re\left(f_i\right)\sin\left(2\pi\frac{\left(i-1\right)\left(k-1\right)}{N}\right)\right\} \end{aligned} \tag{4.53}$$

$$
\begin{aligned}
f_k = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \Bigg\{ & Re(F_i) \cos\left(2\pi \frac{(i-1)(k-1)}{N} \right) \\
& - Im(F_i) \sin\left(2\pi \frac{(i-1)(k-1)}{N} \right) \Bigg\} \\
& + i_{k,\varphi} \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \Bigg\{ Im(F_i) \cos\left(2\pi \frac{(i-1)(k-1)}{N} \right) \\
& + Re(F_i) \sin\left(2\pi \frac{(i-1)(k-1)}{N} \right) \Bigg\}.
\end{aligned}
\tag{4.54}
$$

These two equations form a DFT–inverse discrete Fourier transform (IDFT) pair, where $i = \sqrt{-1}$ is replaced by $i_{k,p}$ in the DFT and by $i_{k,\varphi}$ in the IDFT. They represent a specific example of the more general transform pair given earlier.

In the geometric interpretation suggested by geometric algebra,

$$F_k = |\boldsymbol{d}| \cos\psi_k e^{i_{k,p}\theta} \tag{4.55}$$

is a spinor that transforms the unit vector $\hat{\boldsymbol{p}}_k$ into the vector

$$
\begin{aligned}
\boldsymbol{d}_k &= F_k \hat{\boldsymbol{p}}_k = Re(F_k)\,\hat{\boldsymbol{p}}_k + Im(F_k)\,\hat{\boldsymbol{p}}_{k+N} \\
&= |\boldsymbol{d}| \cos\psi_k \left\{ \cos\theta_k \hat{\boldsymbol{p}}_k + \sin\theta_k \hat{\boldsymbol{p}}_{k+N} \right\}.
\end{aligned}
\tag{4.56}
$$

This interpretation shows that each component of the DFT represents a phase rotation and magnitude dilation. In addition, from these considerations, we see that $\{f_k,\ k = 1,\cdots,N\}$ and $\{F_k,\ k = 1,\cdots,N\}$ are spinor components corresponding to expansions of $\boldsymbol{d}$ in two different bases. In particular, the complex data representation in (4.33) arises from projecting the vector $\boldsymbol{d}$ onto N planes determined by the observed data basis, whereas the DFT representation in (4.53) arises from projecting this same vector $\boldsymbol{d}$ onto a different set of N planes, namely, those determined from the Fourier basis. Thus, the complex vector and the DFT are merely different representations of the same underlying data vector $\boldsymbol{d}$ (Fig. 18).

Not surprisingly, we find for these two specific bases

$$|\boldsymbol{d}|^2 = \sum_{k=1}^{N} |f_k|^2 = \sum_{k=1}^{N} |F_k|^2. \tag{4.57}$$

This relation expresses Parseval's theorem and simply states that the magnitude of the vector $\boldsymbol{d}$ is independent of the basis chosen to represent it.

Using (4.55), we may rewrite our DFT expansion as

$$\boldsymbol{d} = |\boldsymbol{d}| \sum_{k=1}^{N} \cos\psi_k e^{i_{k,p}\theta_k} \hat{\boldsymbol{p}}_k. \tag{4.58}$$

This spinor viewpoint suggests a new way of looking at the DFT. Instead of viewing it in terms of the traditional magnitude and phase pairs $(|F_k|,\theta_k)$, the geometric perspective suggests we view it as comprising the triplets $(|\boldsymbol{d}|,\psi_k,\theta_k)$, where

$$\psi_k = \cos^{-1}\left\{ \frac{|F_k|}{|\boldsymbol{d}|} \right\} = \tan^{-1}\left\{ \frac{\sqrt{|\boldsymbol{d}|^2 - |F_k|^2}}{|F_k|} \right\}. \tag{4.59}$$

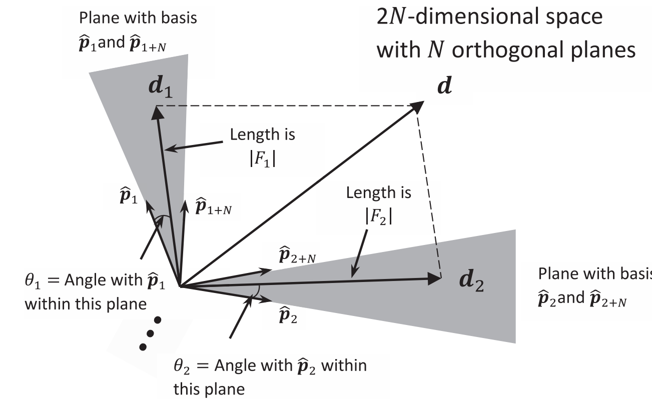


Fig. 18.
Geometry of DFT.

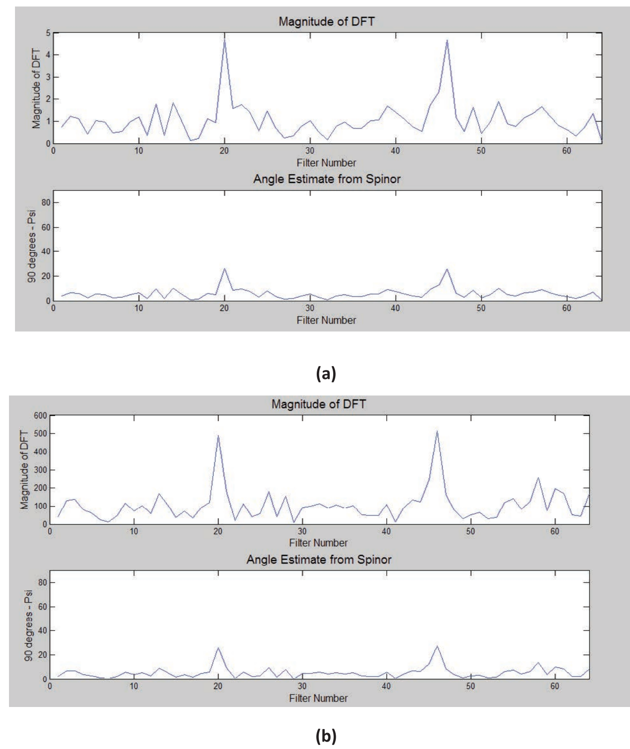


Fig. 19.
Comparison of magnitude of DFT and angle of spinor. (a) Noise power = 1 (arbitrary units). (b) Noise power = 10 000 (arbitrary units).

This view isolates the angles ψ_k and θ_k from the size of the data, as reflected in $|\boldsymbol{d}|$.

As an example of the potential benefit of reinterpreting the DFT in this manner, the plots in Fig. 19 show the output of a DFT for two sinusoidal signals embedded in white Gaussian noise. For each signal, the single-pulse signal-to-noise ratio (SNR) is −3 dB, and the number of pulses used in computing the DFT (via the fast Fourier transform) is 64. The top plot (Fig. 19a) shows the magnitude $|F_k|$, and the bottom plot (Fig. 19b) shows the angle $\frac{\pi}{2} - \psi_k$ (in degrees; we plot this quantity instead of ψ_k for ease of comparison).

In each case, the angle plot is a scaled version of the magnitude plot. This behavior is expected, because $|F_k| = |\boldsymbol{d}| \cos\psi_k$, and thus the size of the data as reflected in $|\boldsymbol{d}|$, simply scales the magnitude.

However, note the differences in scale of the magnitude plots vis-à-vis the unchanging scale of the angle plots. For each of these plots, the same sinusoidal signals were input to the DFT, but the noise power increased (in arbitrary units) from 1 to 10 000 and the signal power increased to maintain the SNR at −3 dB. As expected, the scale of the magnitude response $|F_k|$ changes accordingly. However, the scale of the angle response $\frac{\pi}{2} - \psi_k$ does not change with the level of input noise power. This behavior is expected from (4.55), which shows that ψ_k is independent of $|\boldsymbol{d}|$. This is an interesting property and suggests that in some applications, the spinor angle response can be used in place of $|F_k|$ to obtain a response from the DFT that is independent of the size of the noise power. As one example of the utility of this property, it may be possible to detect signals in nonstationary noise without needing to adjust a detection threshold to follow the constantly varying noise power.

For example, consider a detector formed from the output of the DFT as follows:

$$\frac{\pi}{2} - \psi_k \underset{<}{\overset{>}{}} T \tag{4.60}$$

Because the left-hand side is independent of $|\boldsymbol{d}|$, the threshold T is independent of the noise power level. This can potentially simplify implementation of a detector. For instance, when the magnitude of the DFT is compared to a threshold to make a detection decision, the threshold must be adjusted in accordance with the known or estimated power level of the noise. If the noise power level changes, the threshold must change accordingly to control false alarms. With an approach that uses $\frac{\pi}{2} - \psi_k$ in place of $|F_k|$, however, there is no need to change the threshold as the noise power level changes.

To explore the efficacy of this kind of detector in, for example, nonstationary noise, consider the following plots. The first plot (Fig. 20) shows the power level of white Gaussian noise as a function of time. Each point on this plot represents a variance value that was used to generate 64 samples of white, complex, Gaussian noise for input into a DFT, and this was repeated for a sequence of 1000 DFTs, each DFT comprising 64 pulses. Thus, for each successive DFT, the noise power level of the white Gaussian noise at the input was chosen randomly from a uniform distribution on [0,20]. Noise with this nonstationary behavior in the power level will be very spiky noise, and a traditional detector that compares the output of the DFT to a fixed threshold would have to set a very high threshold to control false alarms.

The next plots show scenarios in which two sinusoidal signals at different Doppler frequencies have been added to the nonstationary noise. The average single-pulse SNR for each signal is −13 dB. The first two plots (Figs. 21a and 21b) show the angle $\frac{\pi}{2} - \psi_k$ (in degrees) of the spinor DFT and the magnitude $|F_k|$ of the DFT, respectively. The number of time samples is limited to 100 for display purposes.

In both of these plots, the signals are clearly visible in filter numbers 20 and 47. To make a detection decision, however, the output of the detectors must be compared to a threshold. Thus, in the next two plots (Figs. 22a and 22b), the data of the previous two plots have been thresholded. The respective thresholds were chosen after inspecting the data so as to control false alarms.

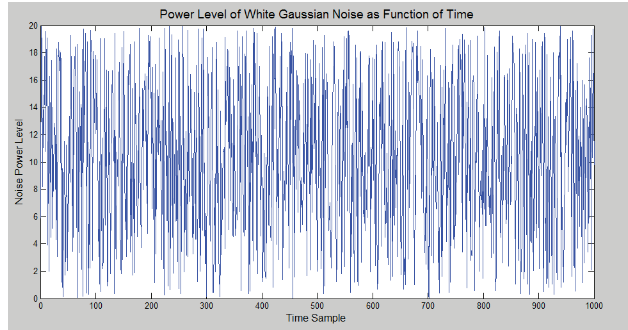
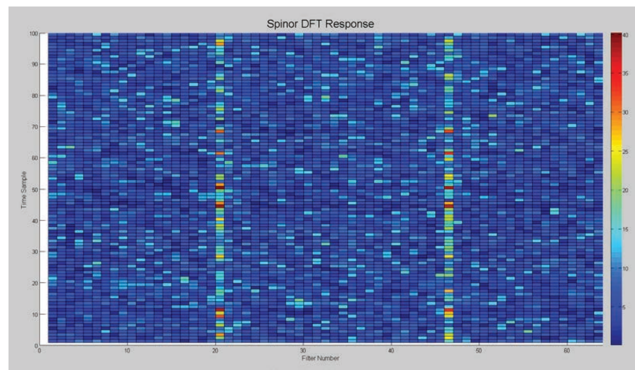
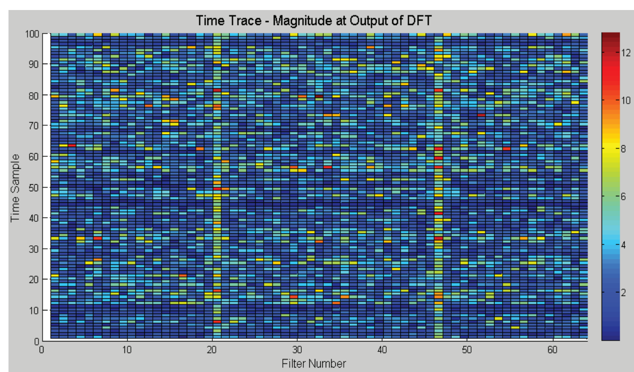


Fig. 20.
Noise power level of nonstationary noise.



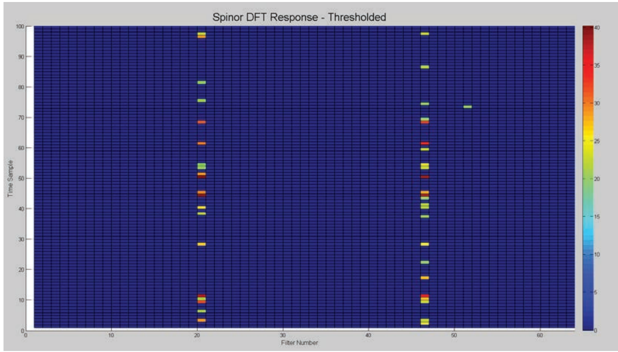
(a)



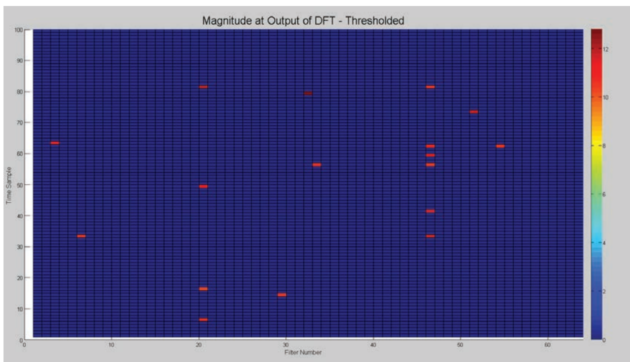
(b)

Fig. 21.
(a) Spinor DFT. (b) Magnitude of DFT. Nonstationary noise, average SNR = −13 dB.

It is evident that despite the severe changes in the noise power level as a function of time, an acceptable threshold can be chosen for the spinor DFT detector to control false alarms. In the spinor DFT detector, the signals in filters 20 and 47 are often detectable, and the false alarms have been controlled (in this example, there is only one). With the magnitude DFT detector, however, it is not possible to detect these targets yet maintain an acceptable false-alarm level. As the example in Fig. 21 shows, a significant number of false alarms occur even though the threshold has been raised sufficiently high to prevent most detections of the targets. Raising

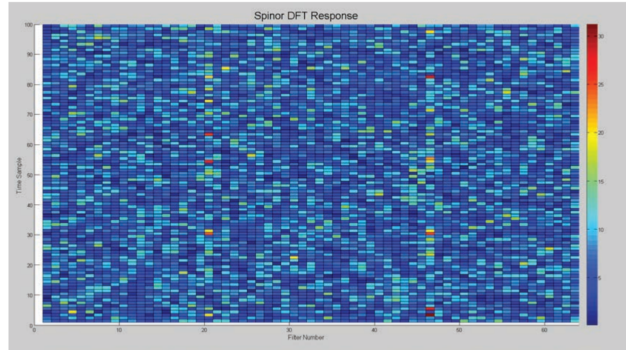


(a)

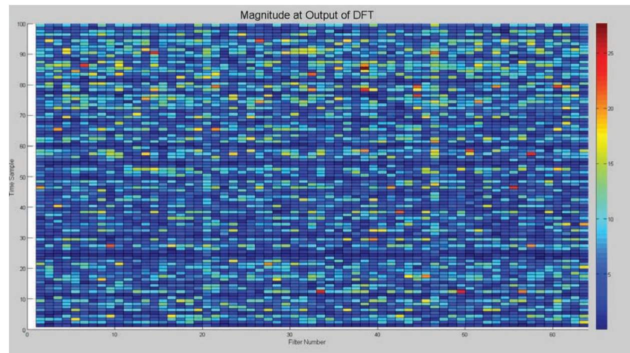


(b)

Fig. 22.
(a) Spinor DFT. (b) Magnitude of DFT after thresholding.



(a)



(b)

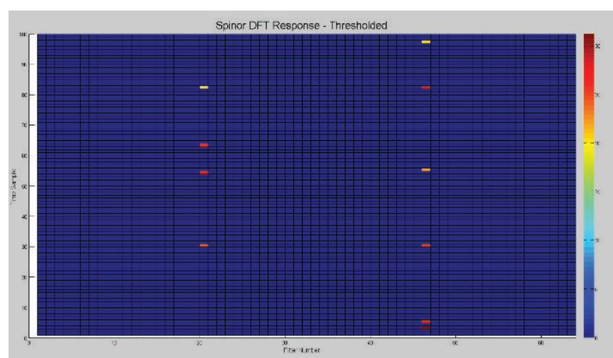
Fig. 23.
(a) Spinor DFT. (b) Magnitude of DFT. Nonstationary noise, average SNR = −20 dB.

the threshold even higher to eliminate the false alarms will only further prevent detection of the targets.

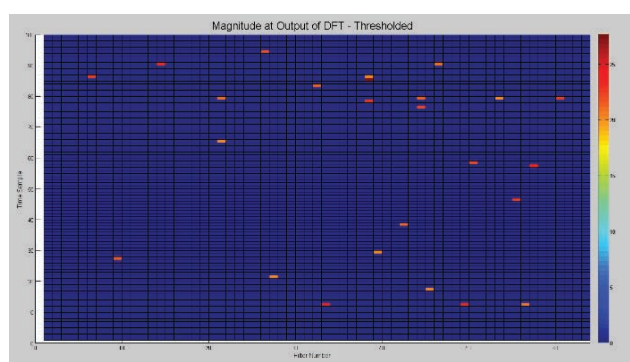
The next plots (Figs. 23a and 23b) repeat the preceding scenario except that now the average single-pulse SNR is −20 dB. This scenario was created by keeping the same signal levels as earlier but raising the average noise power by randomly choosing the noise power level from a uniform distribution on [0,100]. In the thresholded plots that follow (Figs. 24a and 24b), the threshold for the magnitude DFT detector was increased to account for the increase in average noise power. However, the threshold used with the spinor DFT detector was the same as the threshold used earlier.

Despite the significant increase in the input noise power level, the spinor DFT detector with the same threshold as shown earlier controls false alarms and occasionally detects the very small target. These detections evidently occur when the nonstationary randomly varying noise power level at the input to the DFT is low compared to the signal power. The magnitude DFT detector, however, has a large number of false alarms, although the threshold was raised sufficiently high to eliminate almost all detections of the actual targets. Unlike the spinor DFT detector, it does not detect targets when the noise power level at the input to the DFT is low. Thus, the detector based on reinterpreting the DFT as a spinor shows some promise as an effective detector and merits further investigation. In particular, it remains to explore in more depth how to set the thresholds for use with this detector.

We may object that it would be fairer to compare the spinor DFT detector to a constant false alarm rate (CFAR) detector that



(a)



(b)

Fig. 24.
(a) Spinor DFT. (b) Magnitude of DFT after thresholding.

estimates the noise level at each detection decision. While perhaps true—and worthy of further exploration—the primary observation here is that the spinor DFT uses a fixed threshold and thus offers a potentially simpler implementation than the traditional CFAR detector. More work should be done to investigate this issue.

In addition to exploring the implications of a geometric reinterpretation of the DFT, it should be of interest to explore the implications of using other bases besides the Fourier basis in the general discrete transform pair. For example, an interesting question is whether a discrete form of the fractional Fourier transform can be obtained through appropriate choice of the basis in the general discrete transform pair. More research in this area is suggested.

V. APPLICATION: A GEOMETRIC APPROACH TO SIGNAL DETECTION

In this section, we use the geometric perspective gained from geometric algebra to examine a simple signal detection problem. A common approach to developing detection structures is to start with two separate models for the statistics of the observed data under the assumptions that 1) a signal is present or 2) a signal is not present and then derive the likelihood ratio to obtain the detector. Often, however, the models for the statistics under the two hypotheses—i.e., that the signal is or is not present—are not available, especially in the multivariate case in which the decision is made based on multiple data samples. In such cases, we show that we can use a geometric approach to obtain detection structures that are reasonable heuristically and that encompass and expand upon some well-known detectors.

We begin with a discussion of the Hermitian inner product, which arises often in detection processing.

A. HERMITIAN INNER PRODUCT

Consider the Hermitian inner product

$$\boldsymbol{s}^* \cdot \boldsymbol{z} \tag{5.1}$$

where $\boldsymbol{s}$ and $\boldsymbol{z}$ are complex vectors obtained as follows. We assume we have complex samples represented by in-phase and quadrature components

$$\mathcal{I}_k + i\mathcal{Q}_k, \qquad k = 0,\cdots,N-1 \tag{5.2}$$

from which we obtain an N-dimensional complex vector

$$\boldsymbol{z} = \begin{bmatrix} \mathcal{I}_0 + i\mathcal{Q}_0 \\ \vdots \\ \mathcal{I}_{N-1} + i\mathcal{Q}_{N-1} \end{bmatrix}. \tag{5.3}$$

A steering vector $\boldsymbol{s}$ is similarly defined by

$$\boldsymbol{s} = \begin{bmatrix} \mathcal{I}_{p,0} + i\mathcal{Q}_{p,0} \\ \vdots \\ \mathcal{I}_{p,N-1} + i\mathcal{Q}_{p,N-1} \end{bmatrix} \tag{5.4}$$

for suitable choices of the complex steering samples $\mathcal{I}_{p,k} + i\mathcal{Q}_{p,k}$, $k = 0,\cdots,N-1$. This steering vector can be general and can represent such quantities as a Doppler vector, whose components capture the pulse-to-pulse Doppler rotation of a frequency-shifted signal, or an array steering vector, whose components define a beam or null steering direction used in array processing.

Consider the following real vectors defined in R^{2N}:

$$\boldsymbol{p} = \begin{bmatrix} \mathcal{I}_{p,0} \\ \vdots \\ \mathcal{I}_{p,N-1} \\ \mathcal{Q}_{p,0} \\ \vdots \\ \mathcal{Q}_{p,N-1} \end{bmatrix} \tag{5.5}$$

$$\boldsymbol{p}' = \begin{bmatrix} -\mathcal{Q}_{p,0} \\ \vdots \\ -\mathcal{Q}_{p,N-1} \\ \mathcal{I}_{p,0} \\ \vdots \\ \mathcal{I}_{p,N-1} \end{bmatrix} \tag{5.6}$$

$$\boldsymbol{d} = \begin{bmatrix} \mathcal{I}_0 \\ \vdots \\ \mathcal{I}_{N-1} \\ \mathcal{Q}_0 \\ \vdots \\ \mathcal{Q}_{N-1} \end{bmatrix} \tag{5.7}$$

We assume a normalization such that $|\boldsymbol{s}| = |\boldsymbol{p}| = |\boldsymbol{p}'| = 1$. We also have

$$\boldsymbol{p} \cdot \boldsymbol{p}' = 0. \tag{5.8}$$

Thus, we see that $\boldsymbol{p}$ and $\boldsymbol{p}'$ are orthonormal vectors that, taken together, form a two-dimensional subspace (i.e., a plane) in our $2N$-dimensional space. We find in a straightforward manner that

$$\boldsymbol{s}^* \cdot \boldsymbol{z} = \boldsymbol{p} \cdot \boldsymbol{d} - i\, \boldsymbol{p}' \cdot \boldsymbol{d}. \tag{5.9}$$

These observations suggest we study the real vector

$$\boldsymbol{d}_p = (\boldsymbol{p} \cdot \boldsymbol{d})\,\boldsymbol{p} + (\boldsymbol{p}' \cdot \boldsymbol{d})\,\boldsymbol{p}' \tag{5.10}$$

which is the projection of the vector $\boldsymbol{d}$ into the subspace (i.e., plane) defined by $\boldsymbol{p}$ and $\boldsymbol{p}'$. To relate $\boldsymbol{d}_p$ to $\boldsymbol{s}^* \cdot \boldsymbol{z}$, we can multiply $\boldsymbol{d}_p$ from the left by $\boldsymbol{p}$:

$$pd_p = (p \cdot d)\,pp + (p' \cdot d)\,pp' = (p \cdot d)\,pp - (p' \cdot d)\,p'p. \tag{5.11}$$

From the geometric product, we find

$$pp = 1 \tag{5.12}$$

$$p'p' = p' \wedge p = i_p. \tag{5.13}$$

Here, $i_p = p' \wedge p$ is a unit bivector in the p,p' plane. With these observations, we have

$$pd_p = p \cdot d - i_p p' \cdot d = Re\left[s^* \cdot z\right] - i_p\, Im\left[s^* \cdot z\right]. \tag{5.14}$$

We can therefore give the Hermitian inner product $s^* \cdot z$ a geometric interpretation as the geometric product of two $2N$-dimensional real vectors representing the steering signal and the data:

$$s^* \cdot z = pd_p. \tag{5.15}$$

Thus, $s^* \cdot z$ is a measure of the relative magnitudes of the real vectors p and d_p and the angle between them in the p,p' plane.

We have the diagram in Fig. 25 in our $2N$-dimensional space showing the orthogonal projection of the vector d down onto the plane shared by p and p'. In this diagram, the vector d_p is in the p,p' plane and is obtained by orthogonally projecting d through the angle ψ_p onto the p,p' plane, i.e., onto i_p. Thus, we have

$$p \cdot d = p \cdot d_p = |p||d_p|\cos\theta_p \tag{5.16}$$

$$p' \cdot d = p' \cdot d_p = |p||d_p|\sin\theta_p. \tag{5.17}$$

Hence, it is evident that

$$s^* \cdot z = |p||d|\cos\psi_p\left\{\cos\theta_p - i_p\sin\theta_p\right\} = |d|\cos\psi_p e^{-i_p\theta_p} \tag{5.18}$$

where we have used our assumption $|p| = 1$ and have recognized that $|d_p| = |d|\cos\psi_p$. Therefore, we see that the Hermitian inner product of two complex vectors gives rise to a spinor in geometric algebra. The magnitude of the vector d_p—and thus the magnitude of the spinor $s^* \cdot z$—is independent of how we choose p and p' as axes in the plane (they may be rotated arbitrarily). The angle θ_p, however, is a function of how these axes are chosen. These axes in turn are a function of how we choose to split our steering vector into in-phase and quadrature parts.

In computing $s^* \cdot z$, we see that the first step is to obtain d_p by projecting d onto i_p. If we know the initial phase offset ϕ, then in the absence of noise we can project d onto the plane in such a way as to obtain the scalar $Re[s^* \cdot z]$ directly. This is because with knowledge of ϕ, we can define the in-phase and quadrature components of our steering vector—and thus p and p'—in such a way that in the absence of noise, we would have

$$d = d_p = (p \cdot d)\,p = Ap. \tag{5.19}$$

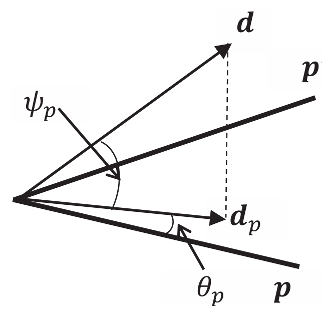


Fig. 25.
Projection of data vector onto plane defined by steering vector.

For example, we can simply rotate the in-phase and quadrature components of the steering vector to obtain

$$e^{i\phi}\left(\mathcal{I}_{p,i} + i\mathcal{Q}_{p,i}\right). \tag{5.20}$$

In this case, the vector p has a meaningful interpretation as the direction of the signal represented by d, and we can use the scalar $p \cdot d = Re[s^* \cdot z]$ in further processing, such as making a detection decision. For example, it is well known that $Re[s^* \cdot z]$ ($=p \cdot d$) compared to a threshold is the optimal structure for detecting s against a background of white Gaussian noise when the initial signal offset phase is known. This has the geometric interpretation of projecting the data onto the steering direction.

However, the initial phase offset is often unknown and in many problems is modeled as a uniformly distributed random variable on $[0,2\pi]$.[5] Thus, it is not helpful to project d_p onto the vector p, because in this case, the direction of d_p within the p,p' plane is randomly distributed with respect to the direction of p. In other words, without knowledge of ϕ, the direction of p has no meaningful relation to the direction of the signal to which we are trying to match. In this instance, our geometric diagram suggests that instead of using the spinor $s^* \cdot z$, which has a magnitude and an angle measured relative to p, we use only its magnitude $\rho_z = |s^* \cdot z| = |d|\cos\psi_p = |d_p|$ in making processing decisions. Thus, we use the magnitude of the vector obtained by projecting our data onto the subspace i_p, rather than the magnitude of the vector obtained by projecting our data onto the specific vector p. For example, it is well known that when the initial phase of the signal is uniformly distributed on $[0,2\pi]$, $|s^* \cdot z|$ compared to a threshold is the optimal structure for detecting s against a background of white Gaussian noise. This is seen to have a geometric interpretation analogous to the earlier case. Namely, the optimal detector in white Gaussian noise projects the data onto the subspace i_p, which may be thought of as a steering subspace or a steering bivector, instead of a steering vector.

The preceding geometric considerations show that

$$|s^* \cdot z| = |d|\cos\psi_p \tag{5.21}$$

[5] It may not always be desirable to model an unknown phase as a uniformly distributed random variable, although in radar problems this is generally the approach that is adopted. For different perspectives on this problem, examine Jaynes (1973), Gray and Addison (2010), and Gray and Alouani (2014).

has a meaningful geometric interpretation as the magnitude of the real $2N$-dimensional vector $\boldsymbol{d}_p$ obtained by orthogonally projecting the real $2N$-dimensional vector $\boldsymbol{d}$ through the angle ψ_p onto the $\boldsymbol{p},\boldsymbol{p}'$ plane.

B. SIGNAL DETECTION

In accordance with the orthogonality principle (Poor, 1988), the vector $\boldsymbol{d}_p$ in (5.10) has an interpretation as a linear estimate of the signal. In effect, it is the best linear estimate of a signal that has been corrupted by white Gaussian noise. If we know the initial phase offset ϕ of the signal, we can use $\boldsymbol{d}_p = (\boldsymbol{p} \cdot \boldsymbol{d})\boldsymbol{p}$, i.e., the orthogonal projection of $\boldsymbol{d}$ onto $\boldsymbol{p}$, which in this case is the direction of the signal. If we do not know ϕ or if we model it as uniformly distributed on $[0,2\pi]$, then we can use $\boldsymbol{d}_p = (\boldsymbol{p} \cdot \boldsymbol{d})\boldsymbol{p} + (\boldsymbol{p}' \cdot \boldsymbol{d})\boldsymbol{p}'$, i.e., the orthogonal projection of $\boldsymbol{d}$ onto the $\boldsymbol{p},\boldsymbol{p}'$ plane, which is the subspace containing the signal. In either case, $\boldsymbol{d}_p$ is an estimate of the signal; thus, $\boldsymbol{d}_p / |\boldsymbol{d}_p|$ represents an estimate of the direction of the signal. We use this observation later to study a basic detection problem.

Consider a detection problem in the form of a simple binary hypothesis test:

$$\begin{aligned} H_0 &: \quad \boldsymbol{z} = \boldsymbol{n} \\ H_1 &: \quad \boldsymbol{z} = \boldsymbol{s} + \boldsymbol{n} \end{aligned} \tag{5.22}$$

where $\boldsymbol{n}$ is a real m-dimensional noise vector and $\boldsymbol{s}$ is a known real m-dimensional signal vector. Assume

$$\begin{aligned} E\left[\boldsymbol{n}\right] &= 0 \\ Cov\left[\boldsymbol{n}\right] &= I \end{aligned} \tag{5.23}$$

where I is an $m \times m$ identity matrix.[6] In general, the solution to this binary hypothesis test is to separate the m-dimensional space into two regions via a separating hypersurface (Fig. 26).

If the observed vector $\boldsymbol{z}$ falls into region 0, no detection is declared, whereas if the observed vector $\boldsymbol{z}$ falls into region 1, a signal detection is declared. This is a general formulation of this simple detection problem, and the task is to define the two decision regions. It is well known that if the multivariate statistics of the m-dimensional vector $\boldsymbol{z}$ are known under both hypotheses, the decision regions may be determined implicitly according to various optimality criteria by comparing the likelihood ratio to a threshold. However, in many problems of interest, the required multivariate statistics are not known; therefore, the likelihood ratio is not available to define the decision regions. Thus, other approaches to defining the decision regions are of interest.

In the following presentation, we look at the problem of defining the decision regions by defining the separating surface directly in the m-dimensional space. We make no claims to optimality but instead are guided by intuitively reasonable observations about the geometry of the decision regions.

1) *A General Detection Structure*: To motivate our definition of the separating surface, we assume the separating hypersurface in the m-dimensional space divides the space into two open regions. In this regard, we are motivated by the observation that the optimal detector in Gaussian noise separates the space via a hyperplane into two open regions, and we choose a separating hypersurface in our more general problem analogously.

[6] If the noise is correlated, the observed data vector $\boldsymbol{z}$ can be whitened and the signal vector $\boldsymbol{s}$ can be prewhitened.

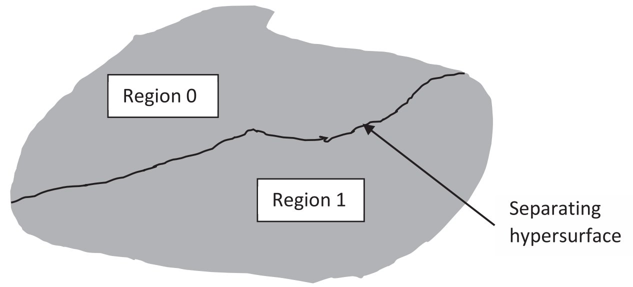


Fig. 26.
General solution for binary hypothesis test.

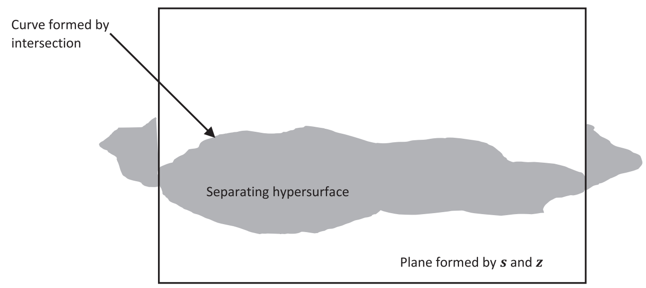


Fig. 27.
Hyperplane separating hypothesis test regions.

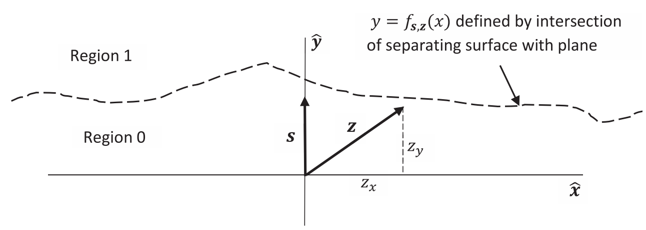


Fig. 28.
Curve defining hypothesis test regions.

First, for any particular decision, we have two vectors available: the signal vector $\boldsymbol{s}$ and the observed data vector $\boldsymbol{z}$ (Fig. 27). Taken together, these two vectors define a plane in our m-dimensional space. We assume this plane intersects with the separating hypersurface to form a curve, as shown in Fig. 27.

For each observed $\boldsymbol{z}$, we obtain in general a different plane and hence a different intersecting curve. Because this curve is in the plane defined by $\boldsymbol{s}$ and $\boldsymbol{z}$, we may write a parametric equation for it as

$$y = f_{s,z}\left(x\right) \tag{5.24}$$

where we have defined our coordinate system so that y is parallel to $\boldsymbol{s}$—and thus x is perpendicular to $\boldsymbol{s}$—with both y and x lying in the $\boldsymbol{s},\boldsymbol{z}$ plane. We have shown the function f as depending on $\boldsymbol{s}$ and $\boldsymbol{z}$, because its form depends on the coordinates, and we have chosen coordinates that lie in the $\boldsymbol{s},\boldsymbol{z}$ plane.

For any particular detection decision, we therefore have the diagram in Fig. 28. Within this plane, we have $\boldsymbol{z} = z_x\hat{\boldsymbol{x}} + z_y\hat{\boldsymbol{y}}$, where $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$ form unit basis vectors. Furthermore, we have $\hat{\boldsymbol{y}} = \hat{\boldsymbol{s}}$, where the unit vector $\hat{\boldsymbol{s}}$ is pointing in the direction of the signal vector $\boldsymbol{s}$. It is evident from this diagram that we declare a detection whenever

$$z_y \geq f_{s,z}\left(z_x\right). \tag{5.25}$$

In other words, whenever the component of the observed vector $\boldsymbol{z}$ parallel to the signal vector $\boldsymbol{s}$ exceeds some function of the component of $\boldsymbol{z}$ perpendicular to $\boldsymbol{s}$, we declare a detection. Now, we have for the component parallel to the signal vector

$$z_y = \hat{\boldsymbol{s}} \cdot \boldsymbol{z} \tag{5.26}$$

which is simply the projection of $\boldsymbol{z}$ onto the direction of the signal $\boldsymbol{s}$. We also see that z_x, the component of $\boldsymbol{z}$ that is perpendicular to $\boldsymbol{s}$, is given by

$$z_x = \pm\sqrt{\boldsymbol{z}\cdot\boldsymbol{z} - \left(\hat{\boldsymbol{s}}\cdot\boldsymbol{z}\right)^2}. \tag{5.27}$$

Equations (5.25)–(5.27) lead to the following detection structure, obtained purely by geometric considerations:

$$\hat{\boldsymbol{s}}\cdot\boldsymbol{z} \gtrless f_{s,z}\left(\pm\sqrt{\boldsymbol{z}\cdot\boldsymbol{z} - \left(\hat{\boldsymbol{s}}\cdot\boldsymbol{z}\right)^2}\right) \tag{5.28}$$

We expect this general detection structure to be complicated in the optimal case. However, the geometric formulation presented here has the advantage of showing that the quantities $\hat{\boldsymbol{s}}\cdot\boldsymbol{z}$ and $\boldsymbol{z}\cdot\boldsymbol{z}$ play fundamental roles in the solution to our simple binary hypothesis test. Furthermore, (5.28) shows that the general detector may be viewed as the Gaussian detector $\hat{\boldsymbol{s}}\cdot\boldsymbol{z}$ compared to a threshold that varies with the data in accordance with the function $f_{s,z}\left(\pm\sqrt{\boldsymbol{z}\cdot\boldsymbol{z} - \left(\hat{\boldsymbol{s}}\cdot\boldsymbol{z}\right)^2}\right)$.

As written in (5.28), the detection structure is too general to be useful. To proceed, although the form of $f_{s,z}(\cdot)$ in general depends on the observed vector $\boldsymbol{z}$ as discussed earlier, unless we have an *a priori* reason to know the nature of this dependence, we assume $f_{s,z}(\cdot)$ has the same form for all observed vectors $\boldsymbol{z}$. Thus, we assume the form of the intersection of the separating hypersurface with the $\boldsymbol{s}$,$\boldsymbol{z}$ plane does not vary as the plane changes. This imposes symmetry on the separating hypersurface around the steering vector $\hat{\boldsymbol{s}}$. With these assumptions, our detector becomes

$$\hat{\boldsymbol{s}}\cdot\boldsymbol{z} \gtrless f_s\left(\sqrt{\boldsymbol{z}\cdot\boldsymbol{z} - \left(\hat{\boldsymbol{s}}\cdot\boldsymbol{z}\right)^2}\right). \tag{5.29}$$

This detection structure is general and depends on the choice of $f_s(\cdot)$, which may be different for each signal vector $\boldsymbol{s}$. Rather than trying to determine $f_s(\cdot)$ from multivariate statistics and the likelihood ratio, however, we choose this function directly. Before we choose $f_s(\cdot)$, however, we must first extend these ideas to include in-phase and quadrature signals.

To extend these ideas to complex-valued data, we use our geometric interpretation of in-phase and quadrature signals as discussed earlier. If we know the initial signal phase, the ideas about the earlier detector apply by simply setting $m = 2N$, representing our data as $\boldsymbol{d}$, and letting our steering vector be the appropriately chosen $\boldsymbol{p}$ that incorporates the initial target phase ϕ. However, as discussed earlier, we often do not know the initial signal phase. In this case, instead of projecting our data $\boldsymbol{d}$ onto a steering vector, we project it onto the steering bivector i_p; this, as we discussed earlier, results in an estimate of the direction of the signal.

Thus, our geometric approach leads to the following detection structure:

$$\begin{matrix} \boldsymbol{p}\cdot\boldsymbol{d} \\ \text{– or –} \\ \left|\boldsymbol{d}\right|\cos\psi_p \end{matrix} \gtrless f_s\left(\left|\boldsymbol{d}\right|\sin\psi_p\right) \tag{5.30}$$

In this expression, we have used

$$Re\left[\boldsymbol{s}^*\cdot\boldsymbol{z}\right] = \boldsymbol{p}\cdot\boldsymbol{d} \tag{5.31a}$$

$$\left|\boldsymbol{s}^*\cdot\boldsymbol{z}\right| = \left|\boldsymbol{d}\right|\cos\psi_p = \frac{\boldsymbol{d}_p}{\left|\boldsymbol{d}_p\right|}\cdot\boldsymbol{d} = \tilde{\boldsymbol{p}}\cdot\boldsymbol{d} \tag{5.31b}$$

where $\tilde{\boldsymbol{p}} = \dfrac{\boldsymbol{d}_p}{\left|\boldsymbol{d}_p\right|}$ may be viewed as an estimate of the signal direction. In what follows, we assume that the offset phase is random and use the lower choice in (5.30) as our detector. In effect, we make a detection decision based on a comparison of how close the data vector $\boldsymbol{d}$ is to the steering bivector i_p—i.e., how small the angle ψ_p is between them—or equivalently how close $\boldsymbol{d}$ is to the estimated signal direction $\boldsymbol{d}_p / |\boldsymbol{d}_p|$. In addition, we have

$$\left|\boldsymbol{d}\right|\sin\psi_p = \sqrt{\left|\boldsymbol{d}\right|^2 - \left(\left|\boldsymbol{d}\right|\cos\psi_p\right)^2}. \tag{5.32}$$

Our general detector may therefore be written

$$\left|\boldsymbol{s}^*\cdot\boldsymbol{z}\right| \gtrless f_s\left(\sqrt{\left|\boldsymbol{z}\right|^2 - \left|\boldsymbol{s}^*\cdot\boldsymbol{z}\right|^2}\right) \tag{5.33}$$

where we have used $|\boldsymbol{s}| = |\boldsymbol{p}| = 1$ and $|\boldsymbol{d}| = |\boldsymbol{z}|$.

2) *Specific Detection Structures*: We determine the precise requirements on how close the data vector $\boldsymbol{d}$ must be to the steering bivector i_p through choice of the function $f_s(x)$. As an example, let

$$f_s\left(x\right) = \left(a_s + b_s\left(x^2\right)^{c_s}\right)^{\frac{1}{2c_s}} \tag{5.34}$$

where a_s, b_s, $c_s > 0$. This leads to

$$\frac{\left|\boldsymbol{s}^*\cdot\boldsymbol{z}\right|^{2c_s} - a_s}{\left[\left|\boldsymbol{z}\right|^2 - \left|\boldsymbol{s}^*\cdot\boldsymbol{z}\right|^2\right]^{c_s}} \gtrless b_s. \tag{5.35}$$

First, this structure requires $\left|\boldsymbol{s}^{*}\cdot\boldsymbol{z}\right|^{2c_s}\geq a_s$; otherwise, no detection can be declared. Hence, the parameter $a_s^{1/2c_s}$ may be thought of as setting a minimum threshold required for detection. Once this condition is satisfied, the factor $\left(\left|\boldsymbol{z}\right|^{2}-\left|\boldsymbol{s}^{*}\cdot\boldsymbol{z}\right|^{2}\right)^{c_s}$ then acts to emphasize or deemphasize the amount by which $\left|\boldsymbol{s}^{*}\cdot\boldsymbol{z}\right|^{2c_s}$ exceeds a_s before comparison with b_s. When $\left|\boldsymbol{z}\right|^{2}-\left|\boldsymbol{s}^{*}\cdot\boldsymbol{z}\right|^{2}=\left|\boldsymbol{z}\right|^{2}\sin^{2}\psi_p\ll 1$, i.e., when $\boldsymbol{z}$ is sufficiently close to the steering bivector, then $\left|\boldsymbol{s}^{*}\cdot\boldsymbol{z}\right|^{2c_s}-a_s$ is greatly amplified (the degree of amplification is controlled by the value of c_s). Data vectors that are closer to the steering bivector thus have a better chance of resulting in a detection. Intuitively, this makes sense. If the received signal includes a sinusoidal signal plus noise, and if the steering bivector corresponds to the plane where the vector associated with this sinusoidal signal resides, then the noise will push the vector associated with the received signal out of the plane (in the absence of noise, this vector would be in the plane). When the noise is small, one expects the vector to be close to the plane, whereas when the noise is large, the effect may be to push the vector farther outside the plane.

For the choice $c_s = 1$, this detector may be rewritten as the equivalent detector

$$\left|\boldsymbol{s}^{*}\cdot\boldsymbol{z}\right|^{2}\underset{<}{>}\frac{a_s}{1+b_s}+\frac{b_s}{1+b_s}\left|\boldsymbol{z}\right|^{2} \tag{5.36}$$

which is the detector given by Sangston et al. (2012) for operation in Student-t distributed compound-Gaussian clutter. For this detector, the decision region is defined by

$$f_s\left(z_x\right)=\sqrt{a_s+b_s z_x^{2}}. \tag{5.37}$$

When $a_s = 0$, this relation becomes

$$f_s\left(z_x\right)=\sqrt{b_s}\left|z_x\right| \tag{5.38}$$

which is the decision region for the generalized likelihood ratio test (GLRT), to be discussed further later. Other choices for c_s lead to other detectors for operation in the more general class of compound-Gaussian clutter models (Sangston and Gerlach, 1994; Sangston et al., 1999). Each optimal detector in the class of compound-Gaussian models has the form of the general detector in (5.30).

For the choice $c_s=\frac{1}{2}$, the detector in (5.35) becomes

$$\frac{\left|\boldsymbol{s}^{*}\cdot\boldsymbol{z}\right|-a_s}{\sqrt{\left|\boldsymbol{z}\right|^{2}-\left|\boldsymbol{s}^{*}\cdot\boldsymbol{z}\right|^{2}}}\underset{<}{>}b_s. \tag{5.39}$$

In this case, we have the diagram for the decision region shown in Fig. 29. The y-axis represents either the plane i_p or the estimated steering direction $\tilde{\boldsymbol{p}}=\boldsymbol{d}_p/\left|\boldsymbol{d}_p\right|$.

When $b_s = 0$, this detector becomes

$$\left|\boldsymbol{s}^{*}\cdot\boldsymbol{z}\right|\underset{<}{>}a_s \tag{5.40}$$

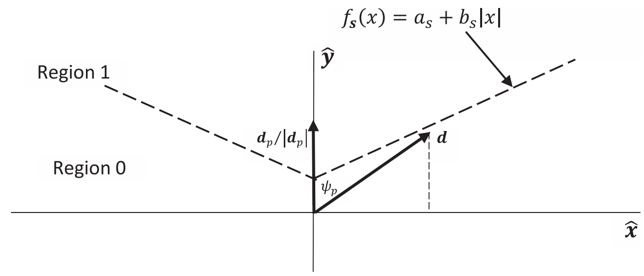


Fig. 29.
Specific curve defining detection processor.

which follows because the relation

$$\left|\boldsymbol{z}\right|^{2}\geq\left|\boldsymbol{s}^{*}\cdot\boldsymbol{z}\right|^{2} \tag{5.41}$$

is always true (recall we assume $|\boldsymbol{s}| = 1$) and thus the detector exceeds a zero threshold only when $|\boldsymbol{s}^{*}\cdot\boldsymbol{z}|$ exceeds a_s. The detector in (5.40) is the optimal Gaussian detector when the initial phase of the signal is uniformly distributed on $[0,2\pi]$. When $a_s = 0$, the detector in (5.39) may be rewritten as

$$\frac{\left|\boldsymbol{s}^{*}\cdot\boldsymbol{z}\right|}{\left|\boldsymbol{z}\right|}\underset{<}{>}\frac{b_s}{\sqrt{b_s^{2}+1}} \tag{5.42}$$

which is the well-known GLRT (Korado, 1968; Picinbono and Vezzosi, 1970; Scharf and Lytle, 1971; Conte et al., 1995). Because $\frac{\left|\boldsymbol{s}^{*}\cdot\boldsymbol{z}\right|}{\left|\boldsymbol{z}\right|}=\cos\psi_p$, in this formulation, the GLRT explicitly uses the cosine of the angle between the data and the steering bivector in making a detection decision. Moreover, the minimum threshold required for detection in the GLRT is 0—e.g., if the data vector is in the subspace defined by the bivector i_p, then $\cos\psi_p = 1$ and a detection is declared regardless of the size of the data vector.

The detector in (5.39) is a reasonably simple extension of the traditional Gaussian detector. Moreover, by varying a_s and b_s in (5.39), we can transition between the Gaussian detector and the GLRT. Because noise alone can result in a vector that has a component in the plane of the steering bivector, false alarms must be controlled in the GLRT detector through choice of the slope term b_s alone (because $a_s = 0$ in the GLRT). Choosing $a_s > 0$ therefore reduces the number of false alarms resulting from noise components in the plane and thus may allow a reduction in the value of b_s. The question remains as to how to choose these parameters, and this is an area that merits further exploration.

From this geometric perspective, we see that a potentially viable approach to extending signal detection beyond the Gaussian and GLRT detectors would be to choose a detection structure such as that given in (5.39) and then adaptively obtain the parameters a_s and b_s that define the structure. This approach is analogous to—but fundamentally different from—ordinary CFAR processing. In ordinary CFAR processing, we essentially choose the detection structure defined in (5.40) and then determine the single parameter a_s adaptively. The approach here is analogous but incorporates a more general detection structure that is not specifically tied to the

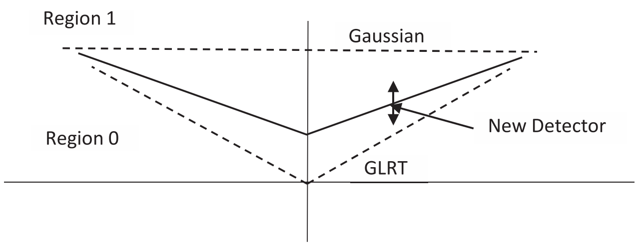


Fig. 30.
Transition between Gaussian detector and GLRT.

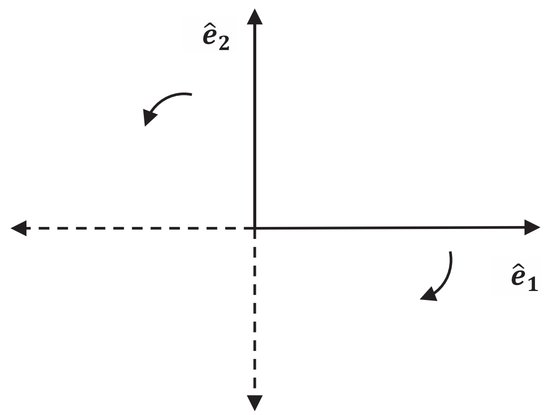


Fig. 31.
Operation of left multiplication by $\boldsymbol{I}_{\hat{e}_1\hat{e}_2}$ in hyperbolic plane.

assumption of Gaussian noise and thus may allow better adaptivity in the presence of more general, non-Gaussian environments. It leads to a diagram as shown in Fig. 30 of the decision regions.

Finally, in the general detector of (5.30), the function $f_s(x)$ can be chosen so that even a small value of $|\boldsymbol{s}^* \cdot \boldsymbol{z}|$ will lead to a detection if the observed data vector is sufficiently close to the steering bivector. For example, the GLRT exhibits this behavior: if the observed vector is in the subspace defined by the steering bivector, then $\cos\psi_p = 1$ and a detection occurs regardless of how small the observed data vector is. The spinor DFT-based detector discussed earlier also exhibits this behavior—there, the instantaneous detectability of the target is determined by the local SNR, i.e., the comparison of the signal power to the power of the noise that accompanies the signal in the DFT. This feature of the detector should be useful in non-Gaussian noise, where small values of the noise occur intermittently. With a traditional Gaussian detector (or the traditional magnitude DFT detector), detections of small- or medium-strength signals can be missed even when those signals are significantly larger than the noise that accompanies them. With the approach to formulating a detector discussed here, small values of noise accompanying a small- or medium-strength signal will likely not cause the resulting data vector to rotate far outside the plane of the steering bivector i_p. In that case, even though the signal is small, it can potentially be detected because the received data vector will likely be close to i_p. This behavior was clearly demonstrated earlier with respect to the spinor DFT detector.

VI. APPLICATION: ROTATIONS IN THE HYPERBOLIC PLANE

As an alternative application of geometric algebra, in this section we change our focus and examine rotations in the hyperbolic plane. To do so, we assume basis vectors $\hat{\boldsymbol{e}}_1$ and $\hat{\boldsymbol{e}}_2$ in a plane satisfying

$$\hat{\boldsymbol{e}}_1^2 = 1 \tag{6.1}$$

$$\hat{\boldsymbol{e}}_2^2 = -1 \tag{6.2}$$

$$\hat{\boldsymbol{e}}_1 \cdot \hat{\boldsymbol{e}}_2 = 0. \tag{6.3}$$

As we discussed earlier, in the hyperbolic plane, the unit vector $I_{\hat{e}_1\hat{e}_2}$ satisfies

$$I_{\hat{e}_1\hat{e}_2}^2 = 1. \tag{6.4}$$

In this case, we may write

$$\boldsymbol{ab} = \boldsymbol{a}\cdot\boldsymbol{b} + \boldsymbol{a}\wedge\boldsymbol{b} = a_1b_1 - a_2b_2 + \left(a_1b_2 - a_2b_1\right)I_{\hat{e}_1\hat{e}_2} \tag{6.5}$$

$$= |\boldsymbol{a}||\boldsymbol{b}|\cosh\theta_{ab} + I_{\hat{e}_1\hat{e}_2}|\boldsymbol{a}||\boldsymbol{b}|\sinh\theta_{ab} \tag{6.6}$$

$$= |\boldsymbol{a}||\boldsymbol{b}|e^{I_{\hat{e}_1\hat{e}_2}\theta_{ab}}. \tag{6.7}$$

In this expression, θ_{ab} defines the hyperbolic angle between the vectors $\boldsymbol{a}$ and $\boldsymbol{b}$, and for simplicity, we have assumed $(a_1b_2 - a_2b_1) > 0$ (if $(a_1b_2 - a_2b_1) < 0$, then $I_{\hat{e}_1\hat{e}_2}$ is replaced by $I_{\hat{e}_2\hat{e}_1}$). As in the Euclidean case, the product $\boldsymbol{ab}$ is a spinor that operates to dilate a vector in the hyperbolic plane and rotate it through the hyperbolic angle θ_{ab} to obtain another vector in the hyperbolic plane.

To better understand these notions, let us examine the effect that $I_{\hat{e}_1\hat{e}_2}$ has on the basis vectors. First, consider

$$I_{\hat{e}_1\hat{e}_2}\hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_1 = -\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 = -\hat{\boldsymbol{e}}_2. \tag{6.8}$$

Similarly, we find

$$I_{\hat{e}_1\hat{e}_2}\hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_2 = -\hat{\boldsymbol{e}}_1. \tag{6.9}$$

In obtaining this last result, we used $\hat{\boldsymbol{e}}_2^2 = -1$. This property of $\hat{\boldsymbol{e}}_2$ leads to behavior in the hyperbolic plane that is fundamentally different from the behavior in the Euclidean plane. We see in Fig. 31 that multiplication from the left by $I_{\hat{e}_1\hat{e}_2}$ rotates each of the basis vectors by 90 deg, but it rotates them in opposite directions.

Let us examine multiplication by $I_{\hat{e}_1\hat{e}_2}$ from the right. We find in a straightforward manner that

$$\hat{\boldsymbol{e}}_1 I_{\hat{e}_1\hat{e}_2} = \hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{e}}_2 \tag{6.10}$$

$$\hat{\boldsymbol{e}}_2 I_{\hat{e}_1\hat{e}_2} = \hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_1\hat{\boldsymbol{e}}_2 = -\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_2\hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{e}}_1. \tag{6.11}$$

This operation also results in 90-deg rotations of the basis vectors—again in opposite directions (Fig. 32).

Let us use these results to examine the effect of the unit spinor $e^{I_{\hat{e}_1\hat{e}_2}\psi}$ on a vector:

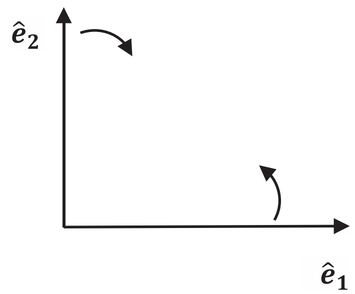


Fig. 32.
Operation of right multiplication by $\boldsymbol{I}_{\hat{e}_1\hat{e}_2}$ in hyperbolic plane.

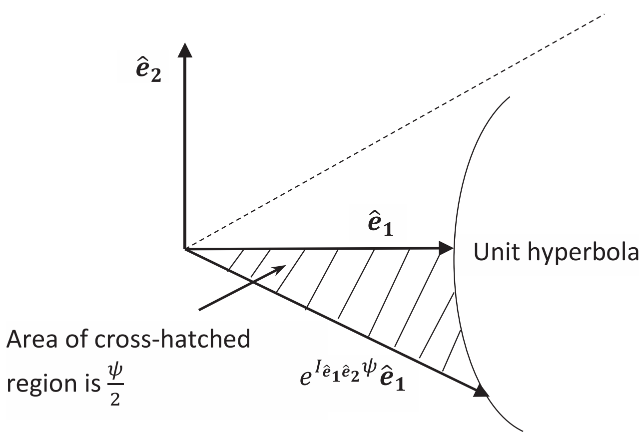


Fig. 33.
Rotation in hyperbolic plane.

$$
\begin{aligned}
e^{I_{\hat{e}_1\hat{e}_2}\psi}\boldsymbol{a} &= \left(\cosh\psi + I_{\hat{e}_1\hat{e}_2}\sinh\psi\right)\left(a_1\hat{\boldsymbol{e}}_1 + a_2\hat{\boldsymbol{e}}_2\right) = a_1\cosh\psi\hat{\boldsymbol{e}}_1 \\
&\quad + a_2\sinh\psi I_{\hat{e}_1\hat{e}_2}\hat{\boldsymbol{e}}_2 + a_2\cosh\psi\hat{\boldsymbol{e}}_2 + a_1\sinh\psi I_{\hat{e}_1\hat{e}_2}\hat{\boldsymbol{e}}_1 \\
&= \left(a_1\cosh\psi - a_2\sinh\psi\right)\hat{\boldsymbol{e}}_1 + \left(a_2\cosh\psi - a_1\sinh\psi\right)\hat{\boldsymbol{e}}_2
\end{aligned}
\tag{6.12}
$$

This is a vector in the hyperbolic plane.

Let us examine the magnitude of the vector we have obtained. It is given by the following:

$$
\begin{aligned}
\left|e^{I_{\hat{e}_1\hat{e}_2}\psi}\boldsymbol{a}\right|^2 &= \left(a_1\cosh\psi - a_2\sinh\psi\right)^2 - \left(a_2\cosh\psi - a_1\sinh\psi\right)^2 \\
&= a_1^2\cosh^2\psi - 2a_1a_2\cosh\psi\sinh\psi + a_2^2\sinh^2\psi \\
&\quad -\left\{a_2^2\cosh^2\psi - 2a_1a_2\cosh\psi\sinh\psi + a_1^2\sinh^2\psi\right\} \\
&= a_1^2\left(\cosh^2\psi - \sinh^2\psi\right) - a_2^2\left(\cosh^2\psi - \sinh^2\psi\right) \\
&= a_1^2 - a_2^2 = \left|\boldsymbol{a}\right|^2
\end{aligned}
\tag{6.13}
$$

Thus, this action does not change the magnitude of the vector on which it operates (where we are defining magnitude to be $|\boldsymbol{a}| = \sqrt{|\boldsymbol{a}\cdot\boldsymbol{a}|}$). This property is consistent with a rotation. For concreteness, let us choose $\boldsymbol{a} = \hat{\boldsymbol{e}}_1$. We then find

$$
e^{I_{\hat{e}_1\hat{e}_2}\psi}\hat{\boldsymbol{e}}_1 = \cosh\psi\hat{\boldsymbol{e}}_1 - \sinh\psi\hat{\boldsymbol{e}}_2. \tag{6.14}
$$

These two vectors are plotted in Fig. 33.

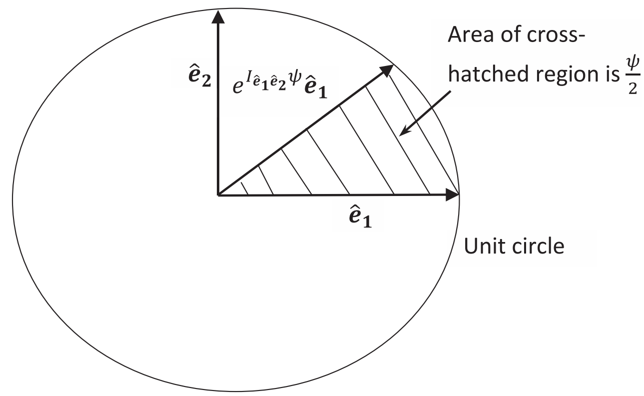


Fig. 34.
Rotation in Euclidean plane.

The tip of the vector moves along a unit hyperbola (defined by $x^2 - y^2 = 1$) as ψ varies. This is analogous to the property that the tip of a rotated unit vector in a Euclidean plane moves along a unit circle (defined by $x^2 + y^2 = 1$). In addition, the area swept out is equal to $\frac{\psi}{2}$, just as with a rotation in the Euclidean plane (Fig. 34). Similar observations hold for the action of $e^{I_{\hat{e}_1\hat{e}_2}\psi}$ on $\hat{\boldsymbol{e}}_2$—and hence on any vector. This behavior justifies calling $e^{I_{\hat{e}_1\hat{e}_2}\psi}\boldsymbol{a}$ a rotation of $\boldsymbol{a}$ and calling ψ the angle through which $\boldsymbol{a}$ is rotated.

The general effect of a spinor in the hyperbolic plane is to rotate and dilate a vector into another vector in the same plane—where "rotate" refers to hyperbolic rotation. Let us write a general spinor as $\rho e^{I_{\hat{e}_1\hat{e}_2}\psi}$. We have, from our earlier results,

$$
\rho e^{I_{\hat{e}_1\hat{e}_2}\psi}\boldsymbol{a} = \rho\left(a_1\cosh\psi - a_2\sinh\psi\right)\hat{\boldsymbol{e}}_1 + \rho\left(a_2\cosh\psi - a_1\sinh\psi\right)\hat{\boldsymbol{e}}_2. \tag{6.15}
$$

Using simple hyperbolic trigonometric identities allows us to rewrite this result as

$$
\rho e^{I_{\hat{e}_1\hat{e}_2}\psi}\boldsymbol{a} = \rho\frac{\left(a_1 - a_2\tanh\psi\right)}{\sqrt{1-\tanh^2\psi}}\hat{\boldsymbol{e}}_1 + \rho\frac{\left(a_2 - a_1\tanh\psi\right)}{\sqrt{1-\tanh^2\psi}}\hat{\boldsymbol{e}}_2. \tag{6.16}
$$

By setting $\rho = 1$ and

$$
\tanh\psi = \frac{v}{c} \tag{6.17}
$$

where $v \leq c$ is a velocity and c is the speed of light, and by associating $\hat{\boldsymbol{e}}_1$ with a spatial dimension in Minkowski space-time and $\hat{\boldsymbol{e}}_2$ with a temporal dimension (or vice versa), we obtain the Lorentz-Fitzgerald transformations used in special relativity. From this point of view, the Lorentz-Fitzgerald transformations are seen to be rotations in the hyperbolic plane. These results can be extended to the full four-dimensional Minkowski space; refer to Hestenes (1966) for a more complete and fascinating discussion of these ideas.

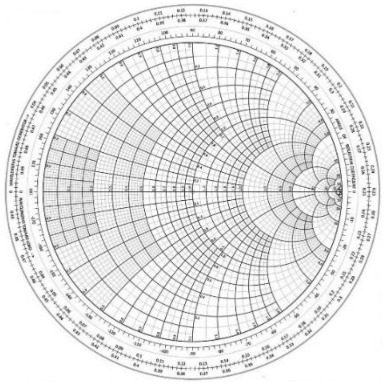


Fig. 35.
Smith chart.

VII. APPLICATION: THE THREE-DIMENSIONAL SMITH CHART

As a final application of the utility of geometric algebra in electrical engineering, we consider a stereographic projection of the Smith chart onto a three-dimensional Euclidean sphere, which we call the Smith sphere, given by Muller et al. (2011). The Smith sphere enables us to extend Smith chart analysis to both active and passive resistance loads and thus holds promise as a useful tool in microwave engineering. As we will see, motions along the various arcs of the Smith chart become motions along the circles on the surface of the Smith sphere. As a result, these motions are amenable to descriptions as rotations. The author is not an expert in microwave theory but has included this section (in conjunction with Section VI) with the aim of motivating the idea that geometric algebra may provide some benefit in application to microwave circuit problems. In addition to the work of Muller et al. (2011), refer to the works of Bolinder (1957–1959, 1981, 1982, 1986, 1987), who has given a stereographic projection of the Smith chart onto a three-dimensional hyperboloid and has used an insertion power loss approach in the context of geometric algebra to study microwave circuit problems. In Bolinder's work (1957–1959, 1981, 1982, 1986, 1987), the effects of various two-port devices are described as rotations in a hyperbolic space, which as we have seen in Section VI, are readily described in geometric algebra.

The well-known Smith chart is shown in Fig. 35. It comprises circles intersecting with arcs, where the circles are all tangent to the same open circuit point and the arcs all emanate from this same open circuit point. The arcs all orthogonally intersect a bounding circle. It is particularly revealing to view the Smith chart as part of a larger figure extending to the entire complex plane. Fig. 36 is taken from Muller et al. (2011). In this extended figure, the boundary of the Smith chart is shown by the thickened line. The continuous lines show constant resistance circles, whereas the dashed lines show constant reactance circles. This view of the extended Smith chart suggests that it can be further extended by stereographically projecting the plane onto the surface of a unit sphere. This is precisely what Muller et al. (2011) did, as shown by Fig. 37, also taken from that paper.

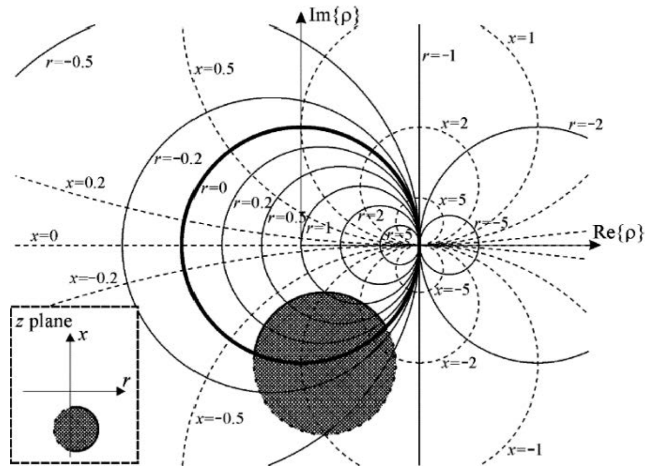


Fig. 36.
Extension of Smith chart to entire plane (from Muller et al., 2011).

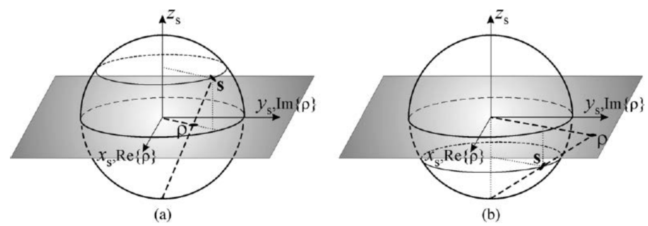


Fig. 37.
Stereographic projection of plane onto surface of unit sphere (from Muller et al., 2011).

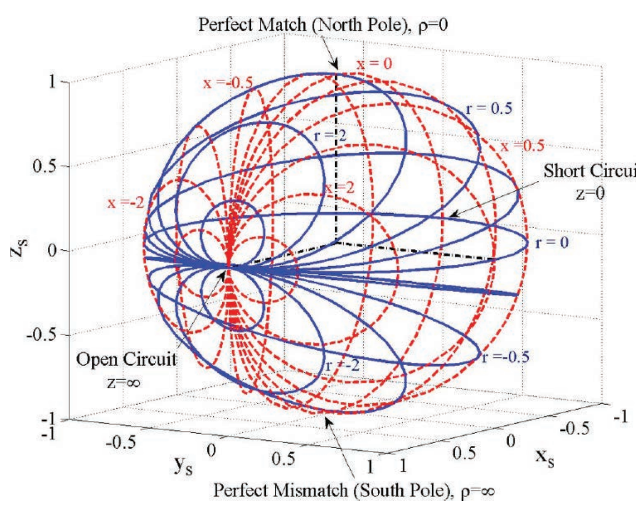


Fig. 38.
Smith sphere (from Muller et al., 2011).

Points on a unit circle in the plane project onto the equator of the unit sphere, whereas points inside the unit circle project onto the upper half of the spherical surface and points on the outside of the unit circle project onto the lower half of the spherical surface. Because points outside the unit circle would correspond to negative resistance loads in the context of the Smith chart, this stereographic projection allows us to incorporate both passive and active loads within the Smith sphere.

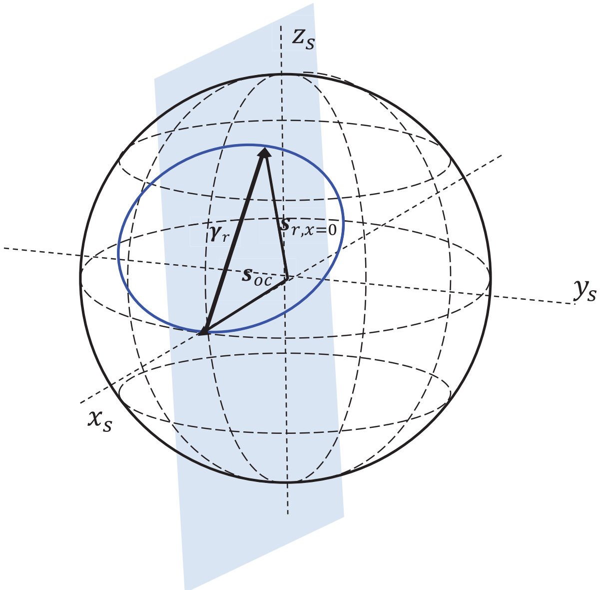


Fig. 39.
Vectors for use with constant-r circle.

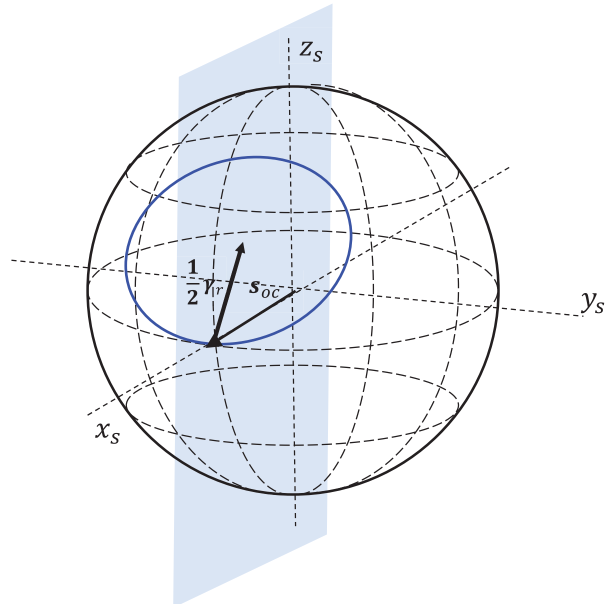


Fig. 40.
Center of constant-r circle.

As discussed by Muller et al. (2011), the point on the complex plane

$$z = r + jx \tag{7.1}$$

where r and x represent load normalized resistance and reactance, respectively, is mapped onto the point on the sphere

$$s = \frac{1}{|z|^2 + 1}\left(|z|^2 - 1, 2x, 2r\right). \tag{7.2}$$

This mapping leads to the representation of the Smith sphere in Fig. 38, taken from Muller et al. (2011). The same constant resistance and constant reactance circles plotted in Fig. 37 are plotted in Fig. 38 for comparison.

As may be seen, the constant resistance and constant reactance curves trace out circles on the surface of the Smith sphere. The Smith chart is generally used by starting from a point on the chart and then moving along either constant resistance circles or constant reactance arcs until a desired point is reached. Thus, it is important to understand how to move along the corresponding circles in the Smith sphere.

Consider first motion along a constant resistance circle (Fig. 39). Let the resistance be r, and consider the following vectors:

$$\boldsymbol{s}_{oc} = (1,0,0) \tag{7.3}$$

$$\boldsymbol{s}_{r,x=0} = \frac{1}{r^2+1}\left(r^2 - 1, 0, 2r\right) \tag{7.4}$$

The vector $\boldsymbol{s}_{oc}$ gives the open circuit point, and the vector $\boldsymbol{s}_{r,x=0}$ describes the point along the constant-r circle where the reactance $x = 0$. This vector lies in the x_s,z_s plane. The vector

$$\boldsymbol{s}_{r,x=0} - \boldsymbol{s}_{oc} = \boldsymbol{\gamma}_r \tag{7.5}$$

then points from the open circuit point to the point described by $\boldsymbol{s}_{r,x=0}$. It also lies in the x_s,z_s plane.

It is apparent from the Smith sphere that this constant-r circle lies in the plane shared by the vector $\boldsymbol{\gamma}_r$ and the unit vector in the y_s direction, which we denote $\hat{\boldsymbol{y}}_s$. Let us define the unit vector:

$$\hat{\boldsymbol{v}}_r = \frac{\boldsymbol{\gamma}_r}{|\boldsymbol{\gamma}_r|} \tag{7.6}$$

With a little algebra, we find

$$\hat{\boldsymbol{v}}_r = \frac{1}{\sqrt{r^2+1}}(-1, 0, r). \tag{7.7}$$

Hence, we may define a unit bivector as follows:

$$\boldsymbol{U}_r = \hat{\boldsymbol{v}}_r \wedge \hat{\boldsymbol{y}}_s \tag{7.8}$$

If we write

$$\hat{\boldsymbol{v}}_r = \frac{1}{\sqrt{r^2+1}}\left(-\hat{\boldsymbol{x}}_s + r\hat{\boldsymbol{z}}_s\right) \tag{7.9}$$

we find

$$\boldsymbol{U}_r = \frac{1}{\sqrt{r^2+1}}\left(\hat{\boldsymbol{y}}_s\hat{\boldsymbol{x}}_s + r\hat{\boldsymbol{z}}_s\hat{\boldsymbol{y}}_s\right). \tag{7.10}$$

When $r = 0$, the bivector $\boldsymbol{U}_{r=0}$ is in the $\hat{\boldsymbol{y}}_s\hat{\boldsymbol{x}}_s$ plane, as expected from Fig. 38.

The center of the constant-r circle (Fig. 40) is given by the coordinates of the vector

$$\boldsymbol{c}_{center,r} = \boldsymbol{s}_{oc} + \frac{1}{2}\boldsymbol{\gamma}_r = \frac{1}{2}\left(\boldsymbol{s}_{oc} + \boldsymbol{s}_{r,x=0}\right) = \left(\frac{r^2}{r^2+1}, 0, r\right). \tag{7.11}$$

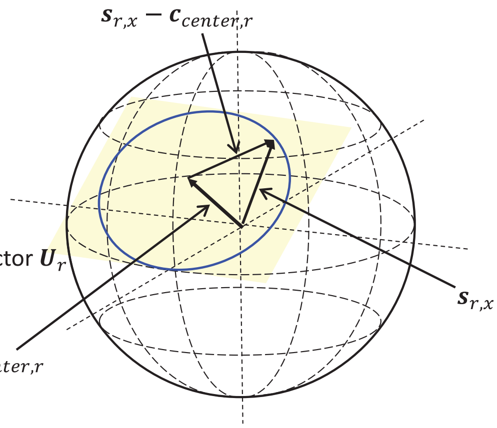


Fig. 41.
Rotation on constant-r circle.

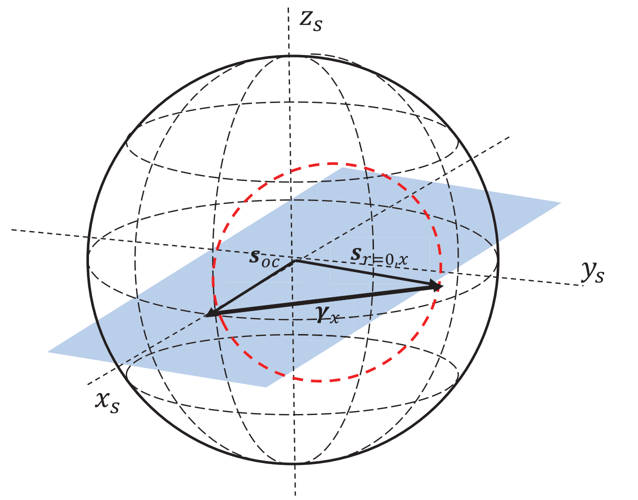


Fig. 42.
Vectors for use with constant-r circle.

Now consider a general point $s_{r,x}$ on this constant-r circle. We may write

$$s_{r,x} = c_{center,r} + \left(s_{r,x} - c_{center,r}\right). \tag{7.12}$$

We have the rotation of a constant-r circle shown in Fig. 41. Observe that $s_{r,x} - c_{center,r}$ lies in the plane of the bivector U_r. We may therefore rotate the point $s_{r,x}$ to a new point along the constant-r circle by rotating the vector $s_{r,x} - c_{center,r}$ in this plane:

$$s_{r,x'} = c_{center,r} + e^{U_r \frac{\psi}{2}}\left(s_{r,x} - c_{center,r}\right)e^{-U_r \frac{\psi}{2}} \tag{7.13}$$

where x' is the new reactance after we have rotated along the constant-r circle through the angle ψ.

Similarly we can consider motion along a constant reactance circle. We follow steps analogous to those taken earlier, except that now we consider the vectors

$$s_{r=0,x} = \frac{1}{x^2+1}\left(x^2-1, 2x, 0\right) \tag{7.14}$$

$$s_{r=0,x} - s_{oc} = \gamma_x \tag{7.15}$$

where $s_{r=0,x}$ is the vector on the constant-x circle with $r = 0$. The vector γ_x points from the open circuit point to $s_{r=0,x}$ and lies in the $x_s y_s$ plane (Fig. 42).

In this case, the constant-x circle lies in the plane shared by the vector γ_x and the unit vector in the z_s direction, which we denote $\hat{z}_s$. Define the unit vector:

$$\hat{v}_x = \frac{\gamma_x}{|\gamma_x|} \tag{7.16}$$

We find

$$\hat{v}_x = \frac{1}{\sqrt{x^2+1}}\left(-1, x, 0\right). \tag{7.17}$$

We define a unit bivector as follows:

$$U_x = \hat{v}_x \wedge \hat{z}_s \tag{7.18}$$

We write

$$\hat{v}_x = \frac{1}{\sqrt{x^2+1}}\left(-\hat{x}_s + x\hat{y}_s\right) \tag{7.19}$$

and obtain

$$U_x = \frac{1}{\sqrt{x^2+1}}\left(\hat{z}_s\hat{x}_s + x\hat{y}_s\hat{z}_s\right). \tag{7.20}$$

When $x = 0$, the bivector $U_{x=0}$ is in the $\hat{z}_s\hat{x}_s$ plane, as expected from Fig. 38.

Now consider a general point $s_{r,x}$ on this constant-x circle:

$$s_{r,x} = c_{center,x} + \left(s_{r,x} - c_{center,x}\right) \tag{7.21}$$

The center of the constant-x circle is given by

$$c_{center,x} = s_{oc} + \frac{1}{2}\gamma_x = \frac{1}{2}\left(s_{oc} + s_{r=0,x}\right) = \left(\frac{x^2}{x^2+1}, x, 0\right). \tag{7.22}$$

We may rotate the point $s_{r,x}$ along the constant-x circle by rotating the vector $s_{r,x} - c_{center,x}$, which lies in the plane defined by the bivector U_x:

$$\boldsymbol{s}_{r',x} = \boldsymbol{c}_{center,x} + e^{\boldsymbol{U}_x \frac{\psi}{2}} \left(\boldsymbol{s}_{r,x} - \boldsymbol{c}_{center,x} \right) e^{-\boldsymbol{U}_x \frac{\psi}{2}} \tag{7.23}$$

where r' is the new resistance after we have rotated along the constant-x circle through the angle ψ.

Combined use of the two preceding results gives an algebraic description of motion along the constant resistance and constant reactance circles of the Smith sphere and gives a mechanism for implementing these motions. In particular, the rotors defined herein should be amenable to computer implementation and should thereby facilitate the use of the Smith sphere. This section, along with Section VI, should suffice to provide entry to the works of Bolinder (1957–1959, 1981, 1982, 1986, 1987) on the use of geometric algebra in a non-Euclidean space to study microwave circuit problems.

VIII. CONCLUSION

Geometric algebra provides geometric interpretations to complex numbers, complex data vectors, and Hermitian inner products of complex vectors. As a result, it should facilitate geometric approaches to processing in-phase and quadrature data and thereby allow geometric intuition to be brought to bear on the development of new signal processing approaches. This article has reviewed the basics of geometric algebra and then shown how it may be applied to various topics that use complex data in electrical engineering, including power concepts, the DFT, a simple binary hypothesis signal detection problem, the hyperbolic plane, and a three-dimensional Smith chart. In each case, the geometric perspective gives insights that can be lost with traditional approaches. For example, with respect to power concepts, the resistance and the reactance concepts arising from the impedance notion each stem from geometrically distinct concepts. Such a distinction may help students more quickly grasp such fundamental concepts.

With respect to the DFT, the geometric perspective leads to a general discrete transform pair, of which the DFT is one example, and suggests a possible new approach to detecting signals. It also suggests that further research be pursued to explore whether other useful discrete transforms may be obtained from the general result by choosing other bases in which to expand a data vector. For example, is it possible to rotate the observed data basis continuously into the Fourier basis so that we can obtain transforms that are intermediate between the data and the DFT in a way that is analogous to the fractional Fourier transform? This seems like an interesting question to explore. Similarly, with respect to the detection problem, the geometric approach yields a general detection structure and suggests that further research be pursued to explore the utility of various specific examples of the general structure. Finally, this article showed how the geometric concepts of geometric algebra may be of interest to engineers working in microwave circuits and suggested several references that may be consulted for further information. ❖

ACKNOWLEDGMENTS

The authors thank the three anonymous referees whose suggestions and insightful questions have greatly improved the quality of this material.

REFERENCES

Geometric algebra has a tremendous body of literature, and the references may be consulted for more extensive treatments of both geometric algebra and related geometric calculus (which has not been mentioned in this tutorial but is fascinating in its own right). Only a small sampling of the growing literature in this area has been provided in the hope of giving entry into the field. There are vast areas of interest that have not been mentioned in this tutorial, such as the application of geometric algebra to electromagnetism and the application of conformal geometric algebra to computer vision. A sampling of papers in these areas is included in the references.

The references are organized into categories (based solely on the author's assessment of them) as an aid to consulting the literature.

BOOKS

Ablamowicz, R., and Sobczyk, G. (Eds.) *Lectures on Clifford (Geometric) Algebras and Applications.* Springer, 2004.

Arthur, J. W. *Understanding Geometric Algebra for Electromagnetic Theory.* IEEE Press, 2011.

Artin, E. *Geometric Algebra.* Wiley, 1957.

Baylis, W. E. *Electrodynamics: A Modern Geometrical Approach.* Birkhauser, 1999.

Baylis, W. E. (Ed.) *Clifford (Geometric) Algebras with Applications in Physics, Mathematics, and Engineering.* Birkhauser, 1996.

Bayro-Corrochano, E., and Scheuermann, G. (Eds.) *Geometric Algebra Computing in Engineering and Computer Science.* Springer, 2010.

Bayro-Corrochano, E., and Sobczyk, G. (Eds.) *Geometric Algebra with Applications in Science and Engineering.* Birkhauser, 2001.

Doran, C., and Lasenby, A. *Geometric Algebra for Physicists.* Cambridge Univ. Press, 2003.

Dorst, L., and Lasenby, J. (Eds.) *Guide to Geometric Algebra in Practice.* Springer-Verlag, 2011.

Dorst, L., Doran, C., and Lasenby, J. (Eds.) *Applications of Geometric Algebra in Computer Science and Engineering.* Springer, 2002.

Hestenes, D. *New Foundations for Classical Mechanics,* 2nd ed. Kluwer Academic, 1999.

Hestenes, D. *Space-Time Algebra.* New York, NY: Gordon and Breach, 1966.

Hestenes, D., and Sobczyk, G. *Clifford Algebra to Geometric Calculus.* D. Reidel Publishing, 1984.

Hladik, J. *Spinors in Physics.* Springer-Verlag, 1999.

Li, H., Olver, P. J., and Sommers, G. (Eds.) *Computer Algebra and Geometric Algebra with Applications.* Springer, 2005.

Lounesto, P. *Clifford Algebras and Spinors.* Cambridge Univ. Press, 2001.

Macdonald, A. *Linear and Geometric Algebra.* 2010. Createspace Independent Publishing Platform, an Amazon Company at www.createspace.com

Macdonald, A. *Vector and Geometric Calculus.* 2012. Createspace Independent Publishing Platform, an Amazon Company at www.createspace.com

Needham, T. *Visual Complex Analysis.* Oxford Univ. Press, 1997.

Perlwass, C. *Geometric Algebra with Applications in Engineering.* Springer, 2009.

Sabbata, V., and Datta, B. K. *Geometric Algebra and Applications to Physics.* CRC Press, 2006.

Snygg, J. *Clifford Algebra: A Computational Tool for Physicists.* Oxford Univ. Press, 1997.

Snygg, J. *A New Approach to Differential Geometry Using Clifford's Geometric Algebra.* Birkhauser, 2010.

Sobczyk, G. *New Foundations in Mathematics: The Geometric Concept of Number.* Birkhauser Mathematics, 2013.

INTRODUCTORY AND/OR GENERAL PAPERS

Baylis, W. E., Huschilt, J., and Wei, J. Why *i*? *American Journal of Physics,* **60**, 9 (Sep. 1992), 788.

Bolinder, E. F. Clifford algebra: What is it? *IEEE Antennas and Propagation Society Newsletter* (Aug. 1987), 18–23.

Buchanan, M. Geometric intuition. *Nature Physics,* **7** (June 2011). Corrected online 28 Oct. 2011.

Carroll, J. E. Traditional vectors as an introduction to geometric algebra. *European Journal of Physics,* **24** (2003).

Chappell, J. M., Drake, S. P., Seidel, C. L., Gunn, L. J., Iqbal, A., Allison, A., and Abbott, D. Geometric algebra for electrical and electronic engineers. *Proceedings of the IEEE,* **102**, 9 (Sep. 2014).

Franchini, S., Vassallo, G., and Sorbello, F. A brief introduction to Clifford algebra. Universita degli Studi di Palermo, Dipartimento di Ingegneria Informatica, Tech Rep., 2/2010, Feb. 2010.

Gull, S. Lasenby, A., and Doran, C. Imaginary numbers are not real—The geometric algebra of spacetime. *Foundations of Physics,* **23**, 9 (1993).

Hestenes, D. Mathematical viruses. In *Clifford Algebras and Their Applications in Mathematical Physics,* A. Micali et al., Eds. Dordrecht, The Netherlands: D. Reidel Publishing, 1992.

Hestenes, D. Oersted Medal lecture 2002: Reforming the mathematical language of physics. *American Journal of Physics,* **71**, 2 (Feb. 2003).

Hestenes, D. Tutorial on geometric calculus. *Advances in Applied Clifford Algebras* (2013).

Hestenes, D. A unified language for mathematics and physics. In *Clifford Algebras and Their Applications in Mathematical Physics,* J. S. R. Chisolm and A. K. Commons, Eds. D. Reidel Publishing, 1986.

Hitzer, E. Introduction to Clifford's geometric algebra. *SICE Journal of Control, Measurement, and System Integration,* **4**, 1 (Jan. 2011).

Macdonald, A. An elementary construction of the geometric algebra. *Advances in Applied Clifford Algebras,* **12**, 1 (2002).

Meisel, L. V. A mathematical formulation of geometric algebra in 3-space. US Army Armament Research, Development and Engineering Center, Tech Rep. ARCCB-TR-95016, Mar. 1995.

Rui, W., and Xuegang, Y. The geometric connections of two kinds of complex number. Presented at the International Conference on Information Technology and Computer Science, 2009.

Sobczyk, G. The hyperbolic number plane. *College Mathematics Journal,* **26**, 4 (Sep. 1995).

Stewart, I. Hermann Grassmann was right. *Nature,* **321** (May 1986).

Vold, T. G. An introduction to geometric algebra with an application in rigid body mechanics, *American Journal of Physics,* **61,** 6 (1993), 491–504.

Vold, T. G. An introduction to geometric calculus and its application to electrodynamics. *American Journal of Physics,* **61**, 6 (1993), 505–513.

MORE ADVANCED PAPERS

Aragon, G., Aragon, J. L., and Rodriguez, M. Clifford algebras and geometric algebra. *Advances in Applied Clifford Algebras,* **7**, 2 (1997), 91–102.

Delanghe, R. Clifford analysis: History and perspective. *Computational Methods and Function Theory,* **1** (2001).

Doran, C., Lasenby, A., and Gull, S. Grassmann mechanics, multivector derivatives and geometric algebra. In *Spinors, Twistors, Clifford Algebras and Quantum Deformations,* Z. Oziewicz, A. Borowiec, and B. Jancewicz, Eds. Kluwer Academic, 1993, p. 215.

Dorst, L. The inner products of geometric algebra. In *Applications of Geometric Algebra in Computer Science and Engineering,* L. Dorst, C. Doran, and J. Lasenby, Eds. Springer, 2002, pp. 35–46.

Dorst, L. Tutorial: Structure-preserving representation of Euclidean motions through conformal geometric algebra. In *Geometric Algebra Computing in Engineering and Computer Science,* E. Bayro-Corrochano and G. Scheuermann, Eds. Springer, 2010, pp. 35–52.

Dress, A., and Havel, T. Distance geometry and geometric algebra. *Foundations of Physics,* **23**, 10 (1993), 1357–1374.

Hestenes, D. The design of linear algebra and geometry. *Acta Applicandae Mathematicae,* **23** (1991).

Hestenes, D. Differential forms in geometric calculus. In *Clifford Algebras and Their Applications in Mathematical Physics,* F. Brackx, et al., Eds. D. Reidel Publishing, 1993.

Hestenes, D. Multivector calculus. *Journal of Mathematical Analysis and Applications,* **24** (1968).

Hestenes, D. Multivector functions. *Journal of Mathematical Analysis and Applications,* **24** (1968).

Hestenes, D. Old wine in new bottles: A new algebraic framework for computational geometry. In *Geometric Algebra with Applications in Science and Engineering,* E. Bayro-Corrochano and G. Sobzcyk, Eds. Birkhauser, 2001.

Hestenes, D. The shape of differential geometry in geometric calculus. In *Guide to Geometric Algebra in Practice,* L. Dorst and J. Lasenby, Eds. Springer-Verlag, 2011.

Hestenes, D. Universal geometric algebra. *Simon Stevin,* **62**, 3–4 (Sep.–Dec. 1988).

Hestenes, D. Vectors, spinors, and complex numbers in classical and quantum physics. *American Journal of Physics,* **39** (1971).

Hestenes, D., and Ziegler, R. Projective geometry with Clifford algebra. *Acta Applicandae Mathematicae,* **23** (1991).

Hitzer, E. Antisymmetric matrices are real bivectors. *Memoirs of the Faculty of Engineering, Fukui University,* **49**, 2 (2001).

Khosravi, M., and Taylor, M. D. The wedge product and analytic geometry. *Mathematical Association of America Monthly,* **115** (Aug.–Sep. 2008).

Lounesto, P. Clifford algebras and Hestenes spinors. *Foundations of Physics,* **23**, 9 (1993), 1203–1237.

Notte-Cuello, E., and Rodrigues, W., Jr. Differential structure of the hyperbolic Clifford algebra. *Advances in Applied Clifford Algebras,* published online Aug. 1, 2014.

Sobczyk, G. Conformal mappings in geometric algebra. *Notices of the American Mathematics Society,* **59**, 2 (Feb. 2012).

Sobczyk, G. The generalized spectral decomposition of a linear operator. *The College Mathematics Journal,* **28**, 1 (Jan. 1997).

Sobczyk, G. Geometric matrix algebra. *Linear Algebra and its Applications,* **429** (2008).

Sobczyk, G. Linear transformations in unitary geometric algebra. *Foundations of Physics,* **23**, 10 (1993).

Sobczyk, G. Unitary geometric algebra. *Advances in Applied Clifford Algebras,* **22** (2012).

Tian, Y. Biquaternions and their complex matrix representations. *Beiträge zur Algebra und Geometrie,* **54** (2013), 575–592.

Wietzke, L., Sommer, G., Schmaltz, C., and Weickert, J. Differential geometry of monogenic signal representations. In *Proceedings of the Robot Vision, Second International Workshop (RoboVis),* G. Sommer, and R. Klette, Eds. Springer, 2008.

ELECTRIC POWER APPLICATIONS

Bolinder, E. F. Cascade synthesis of lumped and distributed networks by means of the Minkowski model of Lorentz space. In *Proceedings of the 11th European Microwave Conference,* Helsinki, Finland, 1982, 25–35.

Bolinder, E. F. Electromagnetic theory and network theory using Clifford algebra. In *Clifford Algebras and Their Applications in Mathematical Physics,* J. S. R. Chisolm and A. K. Commons, Eds. Dordrecht, The Netherlands: D. Reidel Publishing, 1986.

Bolinder, E. F. Geometric-analytic theory of transition in electrical engineering. *Proceedings of the IRE* (June 1959), 1124-1129.

Bolinder, E. F. Impedance and power transformations by the isometric circle method and non-Euclidean hyperbolic geometry. Massachusetts Institute of Technology, Research Laboratory of Electronics, Tech Rep. 312, June 14, 1957.

Bolinder, E. F. A survey of the use of non-Euclidean geometry in electrical engineering. *Journal of the Franklin Institute,* **265**, 3 (Mar. 1958), 169–186.

Bolinder, E. F. Theory of noisy two-port networks. Massachusetts Institute of Technology, Research Laboratory of Electronics, Tech Rep. 344, June 14, 1958.

Bolinder, E. F. Unified microwave theory based on Clifford algebra in Lorentz space. In *Proceedings of the 12th European Microwave Conference,* Amsterdam, the Netherlands, 1981, 93–98.

Castilla, M., and Bravo, J. C. An approach to the multivectorial apparent power in terms of a generalized poynting multivector. *Progress in Electromagnetics Research B,* **15** (2009), 401–422.

Gupta, M. S. Escher's art, Smith chart, and hyperbolic geometry. *IEEE Microwave Magazine* (Oct. 2006).

Lev-Ari, H., and Stankovic, A. M. A Geometric algebra approach to decomposition of apparent power in general polyphase networks. In *Proceedings of the North American Power Symposium,* Starkville, MS, 2009, 1–6.

MacFarlane, A. On the analytical treatment of alternating currents. In *Proceedings of the International Electrical Congress,* Chicago, IL, 1893, 24–32.

Menti, A., Zacharias, T., and Milia-Argitis, J. Geometric algebra: A powerful tool for representing power under nonsinusoidal conditions. *IEEE Transactions on Circuits and Systems,* **54**, 3 (2007), 601–609.

Muller, A. A., Soto, P., Dascalu, D., Neculoiu, D., and Boria, V. E. A 3-D Smith chart based on the Riemann sphere for active and passive microwave circuits. *IEEE Microwave and Wireless Components Letters,* **21**, 6 (June 2011), 286–288.

Petroianu, A. I. A geometric algebra reformulation and interpretation of Steinmetz's symbolic method and his power expression in alternating current electrical circuits. *Electrical Engineering,* **97**, 3 (Sep. 2015), 175–180.

Steinmetz, C. Complex quantities and their use in electrical engineering. In *Proceedings of the International Electrical Congress,* Chicago, IL, 1893, 33–75.

Willems, J. L. Mathematical foundations of the instantaneous power concepts: A geometrical approach. *European Transactions on Electrical Power,* **6**, 5 (Sep./Oct. 1996), 299–304.

SIGNALS, TRANSFORMS, AND SIGNAL PROCESSING APPLICATIONS

Alfsmann, D., Gockler, H., Sangwine, S., and Ell, T. Hypercomplex algebras in digital signal processing: Benefits and drawbacks. In *Proceedings of the 15th European Signal Processing Conference,* Poland, Sep. 3–7, 2007.

Batard, T., and Berthier, M. Spinor Fourier transform for image processing. *IEEE Journal on Selected Topics in Signal Processing,* **7**, 4 (2013), 605–613.

Bayro-Corrochano, E., and Zhang, Y. The motor extended Kalman filter: A geometric approach for rigid motion estimation. *Journal of Mathematical Imaging and Vision,* **13** (2000), 205–228.

Brackx, F., De Knock, B., and De Schepper, H. Hilbert transforms in Clifford analysis. In *Geometric Algebra Computing,* E. Bayro-Corrochano and G. Scheuermann, Eds. Springer, 2010.

Bujack, R., Scheuermann, G., and Hitzer, E. A general geometric Fourier transform convolution theorem. *Advances in Applied Clifford Algebra,* **23** (2013), 15–38.

Bulow, T., and Sommer, G. Hypercomplex signals—A novel extension of the analytic signal to the multidimensional case. *IEEE Transactions on Signal Processing,* **49**, 11 (2002), 2844–2852.

Carrol, J. Complex signals can be real. *European Journal of Physics,* **28** (2007), 1151–1160.

Ebling, J., and Scheuermann, G. Clifford Fourier transform on vector fields. *IEEE Transactions on Visualization and Computer Graphics,* **11**, 4 (2005), 469–479.

Felsberg, M., and Sommer, G. The monogenic signal. *IEEE Transactions on Signal Processing,* **49**, 12 (2001), 3136–3144.

Felsberg, M., and Sommer, G. The multidimensional isotropic generalization of quadrature filters in geometric algebra. In *Algebraic Frames for the Perception-Action Cycle, Lecture Notes in Computer Science,* **1888** (2000), 175–185.

Fontijne, D., and Dorst, L. Modeling 3-D Euclidean geometry. *IEEE Computer Graphics and Applications* (Mar./Apr. 2003), 2–12.

Gebken, C., Perwass, C., and Sommer, G. Parameter estimation from uncertain data in geometric algebra. *Advances in Applied Clifford Algebra,* **18** (2008), 647–664.

Guillemard, M., Iske, A., and Zolzer, U. Clifford algebras and dimensionality reduction for signal separation and classification. Available: http://www.math.uni-hamburg.de/home/guillemard/clifford/.

Hinidama, S., Gamage, U., and Lasenby, J. Optimal estimation and tracking of general rotations using geometric algebra with applications in computer vision. *Proceedings of SPIE,* **4117**, Vision Geometry IX (2000), 261–271.

Hitzer, E. The quest for conformal geometric algebra Fourier transformations: 11th International Conference of Numerical Analysis and Applied Mathematics, *AIP Conference Proceeding,* **1558** (2013), 529–532.

Ilchenko, M., Narytnik, T., and Didkovsky, R. Clifford algebra in multiple-access noise-signal communication systems. *Telecommunications and Radio Engineering,* **72**, 18 (2013), 1651–1663.

Jiang, J., and Zhang, J. Geometric algebra of Euclidean 3-space for electromagnetic vector-sensor array processing, Part 1: Modeling. *IEEE Transactions on Antennas and Propagation,* **58**, 12 (2010), 3961–3973.

Jiang, J., and Zhang, J. A weighted inner product estimator in the geometric algebra of Euclidean 3-space for source localization using an EM vector-Sensor. *Chinese Journal of Aeronautics,* **25** (2012), 83–93.

Kuroe, Y., Nitta, T., and Hitzer, E. Applications of Clifford's geometric algebra. *SICE Journal of Control, Measurement, and System Integration,* **4**, 1 (Jan. 2011), 1–10.

Le Bihan, N., Miron, S., and Mars, J. MUSIC algorithm for vector-sensors array using biquaternions. *IEEE Transactions on Signal Processing,* **55**, 9 (2007), 4523–4533.

Rockwood, A., and Hildebrand, D. Engineering graphics in geometric algebra. In *Geometric Algebra Computing,* E. Bayro-Corrochano and G. Scheuermann, Eds. Springer, 2010.

Rosenhahn, B., and Sommer, G. Pose estimation in conformal geometric algebra. Part 1: The stratification of mathematical spaces. *Journal of Mathematical Imaging and Vision,* **22** (2005), 27–48.

Stanway, J., and Kinsey, J. Sensor alignment using rotors in geometric algebra. In *Proceedings of the 2011 IEEE International Conference on Robotics and Automation,* Shanghai, China, May 9–13, 2011, 994–999.

Wietzke, L., and Sommer, G. The signal multi-vector. *Journal of Mathematical Imaging and Visualization,* **37** (2010), 132–150.

Wu, D., and Wang, Z. Strapdown INS/GPS integrated navigation using geometric algebra. *Advances in Applied Clifford Algebra,* **23** (2013), 767–785.

Wu, D., Wang, Z., and Wang, Q. A rotation vector differential equation in geometric algebra. In *Proceedings of the 6th International Conference on Telecommunication Systems, Services, and Applications,* 2011.

APPLICATIONS TO ELECTROMAGNETISM

Abreu-Blaya, R., Avila-Avila, R., Bory-Reyes, J., and Rodriguez-Dagnino, R. M. 2D quaternionic time-harmonic Maxwell System in elliptic coordinates. *Advances in Applied Clifford Algebras* (Aug. 1, 2014).

Arthur, J. W. The evolution of Maxwell's equations from 1862 to the present day. *IEEE Antennas and Propagation Magazine,* **55**, 3 (2013), 61–81.

Baylis, W. E., Bonenfant, J., Derbyshire, J., and Huschilt, J. Light polarization: A geometric algebra approach. *American Journal of Physics,* **61**, 6 (1993), 534–545.

Chantaveerod, A., and Seagar, A. D. Iterative solutions for electromagnetic fields at perfectly reflective and transmissive interfaces using Clifford algebra and the multidimensional Cauchy integral. *IEEE Transactions on Antennas and Propagation,* **57**, 11 (2009), 3489–3499.

Chantaveerod, A., Seagar, A. D., and Angkaew, T. Calculation of electromagnetic field with integral equation based on Clifford algebra. In *Proceedings of the PIERS,* Aug. 27–30, 2007, Prague, Czech Republic, 62–71.

Hillion, P. Constitutive relations and Clifford algebra in electromagnetism. *Advances in Applied Clifford Algebras,* **5** (1995), 141–158.

Hillion, P. Spinor representation of electromagnetic fields. *Journal of the Optical Society of America,* **66**, 8 (1976), 865.

Jancewicz, B. Trivector Fourier transformation and electromagnetic field. *Journal of Mathematical Physics,* **31**, 8 (1990), 1847–1852.

Klimek, M., Romer, U., Schops, A., and Weiland, T. Space-time discretization of Maxwell's equations in the setting of geometric algebra. In *Proceedings of the 2013 International Symposium on Electromagnetic Theory,* 2013, 1101–1104.

Matos, S. A., Ribeiro, M. A., and Paiva, C. R. Anisotropy without tensors: A novel approach using geometric algebra. *Optics Express,* **15**, 23 (2007), 15 175–151 86.

Puska, P. Clifford algebra and electromagnetic boundary conditions at an interface. *Journal of Electromagnetic Waves and Applications,* **14** (2000), 11–24.

Puska, P. Covariant isotropic constitutive relations in Clifford's geometric algebra. *PIER,* **32** (2001), 413–428.

Reimer, M., and Yevick, D. A Clifford-algebra analysis of polarization-mode dispersion and polarization-dependent loss. *IEEE Photonics Technology Letters,* **18**, 6 (Mar. 2006), 734–736.

Santos, A. G., Sugon, Q. M., Jr., and McNamara, D. J. Polarization ellipse and stokes parameters in geometric algebra. *Journal of the Optical Society of America A,* **29**, 1 (2012), 89–98.

Soliman, G., Yevick, D., and Jessop, P. Geometric algebra description of polarization mode dispersion, polarization-dependent loss, and stokes tensor transformations. *Journal of the Optical Society of America A,* **31**, 9 (2014), 1956–1962.

Sugon, Q. M., Jr., and McNamara, D. J. A geometric algebra reformulation of geometric optics. *American Journal of Physics,* **72** (2004), 92–97.

Vold, T. G. Computing electromagnetic fields by the method of least action. *Advances in Applied Clifford Algebras,* **24** (2014), 275–292.

Zou, L., Lasenby, J., and He, Z. Pattern analysis of conformal array based on geometric algebra. *IET Microwaves, Antennas & Propagation,* **5**, 10 (2011), 1210–1218.

Zou, L., Lasenby, J., and He, Z. Polarization diversity of conformal arrays based on geometric algebra via convex optimization. *IET Radar, Sonar, and Navigations,* **6**, 6 (2012), 417–424.

OTHER REFERENCES

Conte, E., Longo, M., and Ricci, G. Asymptotically optimum radar detection in compound-Gaussian clutter. *IEEE Transactions on Aerospace and Electronic Systems,* **31**, 2 (1995), 617–625.

Gray, J. E., and Addison, S. R. The Rayleigh problem is everywhere. *Proceedings of the SPIE, Radar Sensor Technology XIV,* **7669** (2010), 76690G-1 - 76690G-13.

Gray, J. E., and Alouani, A. T. Characterization of the PDFs of coordinate transformations in tracking. *IEEE Transactions on Aerospace and Electronic Systems,* **50**, 2 (2014), 1479–1492.

Jaynes, E. T. The well-posed problem. *Foundations of Physics,* **3** (1973), 477–493.

Korado, V. A. Optimum detection of signals with random parameters against the background noise of unknown intensity under conditions of constant false alarm probability. *Radio Engineering and Electronic Physics,* **13**, 6 (1968), 969–972.

Picinbono, B., and Vezzosi, G. Detection d'un signal certain dans un bruit non stationnaire and Gaussien. *Annales des Telecommunications,* **25** (1970), 433–439.

Poor, H. V. *An Introduction to Signal Detection and Estimation.* New York, NY: Springer-Verlag, 1988.

Sangston, K. J., and Gerlach, K. R. Coherent detection of radar targets in a non-Gaussian background. *IEEE Transactions on Aerospace and Electronic Systems,* **30**, 2 (Apr. 1994), 330–340.

Sangston, K. J., Gini, F., and Greco, M. V. New results on coherent radar target detection in heavy-tailed compound-Gaussian clutter. *IEEE Transactions on Aerospace and Electronic Systems,* **48**, 1 (Jan. 2012), 64–77.

Sangston, K. J., Gini, F., Greco, M. V., and Farina, A. Structures for radar detection in compound Gaussian clutter. *IEEE Transactions on Aerospace and Electronic Systems,* **35**, 2 (Apr. 1999), 445–458.

Scharf, L., and Lytle, D. Signal detection in Gaussian noise of unknown level: An invariance application. *IEEE Transactions on Information Theory,* **17**, 4 (1971), 404–411.

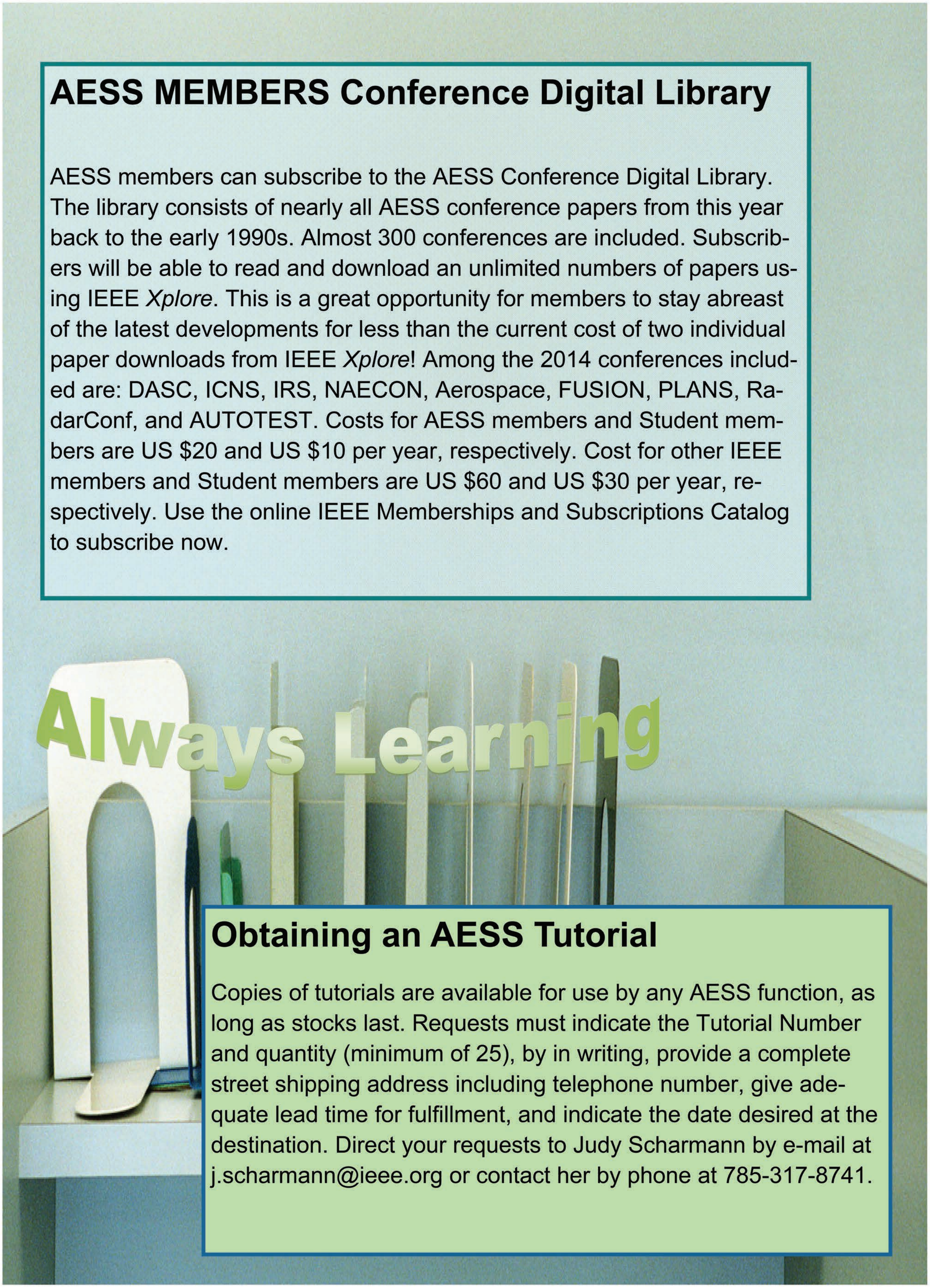
K. James Sangston was born in Uniontown, PA, in 1961. He received the B.S.E. degree in systems science and engineering in 1983 from the Moore School of Electrical Engineering

K. James Sangston

at the University of Pennsylvania, PA. He received the M.S. degree in 1987 and the D.Sc. degree in 1995, both in electrical engineering from George Washington University, Washington, D.C. He received the J.D. degree in 2003 from the College of Law, Georgia State University, Atlanta. He was on the engineering staff of the I.I.T. Research Institute, Annapolis, MD, during 1984–1987; was with the Radar Division of the Naval Research Laboratory, Washington, D.C., during 1987–1995; and was a senior research engineer with the Georgia Tech Research Institute, Atlanta, during 1995–2000. At that time, he returned to law school, and from 2003 to 2013, he practiced law with focus in patent law. He is currently a senior research engineer with the Georgia Tech Research Institute, Atlanta where he continues to pursue his research interests in the areas of radar target detection, radar signal processing, and electromagnetic scattering.

AESS MEMBERS Conference Digital Library

AESS members can subscribe to the AESS Conference Digital Library. The library consists of nearly all AESS conference papers from this year back to the early 1990s. Almost 300 conferences are included. Subscribers will be able to read and download an unlimited numbers of papers using IEEE *Xplore*. This is a great opportunity for members to stay abreast of the latest developments for less than the current cost of two individual paper downloads from IEEE *Xplore*! Among the 2014 conferences included are: DASC, ICNS, IRS, NAECON, Aerospace, FUSION, PLANS, RadarConf, and AUTOTEST. Costs for AESS members and Student members are US $20 and US $10 per year, respectively. Cost for other IEEE members and Student members are US $60 and US $30 per year, respectively. Use the online IEEE Memberships and Subscriptions Catalog to subscribe now.



Obtaining an AESS Tutorial

Copies of tutorials are available for use by any AESS function, as long as stocks last. Requests must indicate the Tutorial Number and quantity (minimum of 25), by in writing, provide a complete street shipping address including telephone number, give adequate lead time for fulfillment, and indicate the date desired at the destination. Direct your requests to Judy Scharmann by e-mail at j.scharmann@ieee.org or contact her by phone at 785-317-8741.